Conformal Prediction in NBA data

Hung-Yang Lu

December 2021

Contents

1 Introduction

In this project, we applied conformal prediction methods to the NBA data. We used it to answer three questions for the upcoming NBA season (2021-2022 season):

1. What players will be the top 5 for the upcoming season?
2. Which team will win the championship for the upcoming season?
3. What player will win the MVP for the upcoming season?

Below the content, I will put this project into two parts: In the first part, I will explain the conformal prediction methods I applied into the NBA data, and in another part I will apply these methods to the NBA data and answer the three questions.

2 Conformal Prediction

In this part, I will explain the conformal prediction in regression, and I will introduce split conformal prediction, locally split conformal prediction, conformalized quantile regression, approximately via rounding, conformal prediction with discretized data and conformal prediction with discretized model methods, which I used for the NBA data.

2.1 Conformal Prediction in Regression

2.1.1 Basic Outline

- Suppose we are given n training samples $\{X_i, Y_i\}_{i=1}^{n}$ and we must now predict the unknown value of Y_{n+1} at a test point X_{n+1}.
- We assume that all the samples $\{X_i, Y_i\}_{i=1}^{n+1}$ are drawn exchangeably.
- We aim to construct a marginal distribution-free prediction interval $C(X_{n+1}) \subseteq \mathbb{R}$ that is likely to contain the unknown response Y_{n+1}.
- That is, given a desired miscoverage rate α, we ask that $P\{Y_{n+1} \in C(X_{n+1})\} \geq 1 - \alpha$ for any joint distribution P_{XY} and any sample size n.

2.1.2 Steps

- Given any regression algorithm $\mathcal{A}$, a regression model is fit to the data set: $\hat{\mu}(x) \leftarrow \mathcal{A}(\{(X_i, Y_i) : i = 1, \ldots, n\})$.
- Next, the absolute residuals are computed on the calibration set, as follows: $R_i = |Y_i - \hat{\mu}(X_i)|$, $i = 1, \ldots, n$.
- For a given level α, we then compute a quantile of the empirical distribution of the absolute residuals, $Q_{1-\alpha}(R) = (1 - \alpha)(1 + 1/n)$-th empirical quantile of $\{R_i : i = 1, \ldots, n\}$.
- Finally, the prediction interval at a new point X_{n+1} is given by $C(X_{n+1}) = [\hat{\mu}(X_{n+1}) - Q_{1-\alpha}(R), \hat{\mu}(X_{n+1}) + Q_{1-\alpha}(R)]$.

2.1.3 Conclusion

- The length of $C(X_{n+1})$ is fixed and equal to $2Q_{1-\alpha}(R)$, independent of X_{n+1}.
- The main drawback is it requires the regression algorithm to be invoked infinitely many times.

2.2 Split Conformal Prediction

2.2.1 Steps

- The split conformal method begins by splitting the training data into two disjoint subsets: a proper training set $\{(X_i, Y_i) : i \in L_1\}$ and calibration set $\{(X_i, Y_i) : i \in L_2\}$.
- Then, given any regression algorithm $\mathcal{A}$, a regression model is fit to the proper training set: $\hat{\mu}(x) \leftarrow \mathcal{A}(\{(X_i, Y_i) : i \in L_1\})$.
- Next, the absolute residuals are computed on the calibration set, as follows: $R_i = |Y_i - \hat{\mu}(X_i)|$, $i \in L_2$.
- For a given level α, we then compute a quantile of the empirical distribution of the absolute residuals, $Q_{1-\alpha}(R, L_2) = (1-\alpha)(1+1/|L_2|)$-th empirical quantile of $\{R_i : i \in L_2\}$.
- Finally, the prediction interval at a new point X_{n+1} is given by $C(X_{n+1}) = [\hat{\mu}(X_{n+1}) - Q_{1-\alpha}(R, L_2), \hat{\mu}(X_{n+1}) + Q_{1-\alpha}(R, L_2)]$.

2.2.2 Conclusion

The length of $C(X_{n+1})$ is fixed and equal to $2Q_{1-\alpha}(R, L_2)$, independent of X_{n+1}.

2.3 Locally Adaptive Conformal Prediction

2.3.1 Settings

- It is an approach to making conformal prediction adaptive to heteroskedasticity.
- In this case, the absolute residuals are replaced by the scaled residuals $\tilde{R}_i = \frac{|Y_i - \hat{\mu}(X_i)|}{\hat{\sigma}(X_i)} = \frac{R_i}{\hat{\sigma}(X_i)}$, $i \in L_2$, where $\hat{\sigma}(X_i)$ is a measure of the dispersion of the residuals at X_i and $\hat{\mu}(X_i)$ is the regression function.
- The prediction interval at a new point X_{n+1} is computed as $C(X_{n+1}) = [\hat{\mu}(X_{n+1}) - \hat{\sigma}(X_{n+1})Q_{1-\alpha}(\tilde{R}, L_2), \hat{\mu}(X_{n+1}) + \hat{\sigma}(X_{n+1})Q_{1-\alpha}(\tilde{R}, L_2)]$.

2.3.2 Limitations of locally adaptive conformal prediction

- When the data is actually homoscedastic: The locally adaptive method suffers from inflated prediction intervals compared to the standard method.
- Fundamental statistical limitation: Because the training set are biased by an optimization procedure designed to minimize them, while the calibration set are unbiased, this forces the correction constant $Q_{1-\alpha}(\tilde{R}, L_2)$ to be large and the intervals to be less adaptive than they could be.

2.4 Conformalized Quantile Regression

2.4.1 Conditional quantile regression

- **Aim:** Estimate a given quantile of Y conditional on X.
- Conditional distribution function of Y given $X = x$: $F(y \mid X = x) = P\{Y \leq y \mid X = x\}$.
- α-th conditional quantile function: $q_\alpha(x) = \inf\{y \in \mathbb{R} : F(y \mid X = x) \geq \alpha\}$.

- Fix the lower and upper quantiles to be equal to $\alpha_{\text{lo}} = \alpha/2$ and $\alpha_{\text{hi}} = 1 - \alpha/2$, say. Given the pair $\hat{q}_{\alpha_{\text{lo}}}(x)$ and $\hat{q}_{\alpha_{\text{hi}}}(x)$ of lower and upper conditional quantile functions, we obtain a conditional prediction interval for Y given $X = x$, with miscoverage rate α, as $C(x) = [q_{\alpha_{\text{lo}}}(x), q_{\alpha_{\text{hi}}}(x)]$.

- By construction, this interval satisfies $P\{Y \in C(X) \mid X = x\} \geq 1 - \alpha$.

- **Estimating quantiles from data:** Quantile regression estimates a conditional quantile function q_α of Y_{n+1} given $X_{n+1} = x$. This can be cast as the optimization problem:

$$\hat{q}_\alpha(x) = f(x; \hat{\theta}), \quad \hat{\theta} = \arg\min_\theta \frac{1}{n} \sum_{i=1}^{n} \rho_\alpha(Y_i, f(X_i; \theta)) + R(\theta)$$

  where $f(x; \theta)$ is the quantile regression function and the loss function ρ_α is the "check function" or "pinball loss", defined by:

$$\rho_\alpha(y, \hat{y}) = \begin{cases} \alpha(y - \hat{y}), & \text{if } y - \hat{y} > 0 \\ (1 - \alpha)(\hat{y} - y), & \text{otherwise} \end{cases}$$

  and R is a potential regularizer.

- To construct a prediction band with nominal miscoverage rate α: estimate $\hat{q}_{\alpha_{\text{lo}}}(x)$ and $\hat{q}_{\alpha_{\text{hi}}}(x)$ using quantile regression, then output $\hat{C}(X_{n+1}) = [\hat{q}_{\alpha_{\text{lo}}}(X_{n+1}), \hat{q}_{\alpha_{\text{hi}}}(X_{n+1})]$ as an estimate of the ideal interval $C(X_{n+1})$ from equation $C(x) = [q_{\alpha_{\text{lo}}}(x), q_{\alpha_{\text{hi}}}(x)]$.

2.4.2 Steps

- As in split conformal prediction, we split the training data into two disjoint subsets: a proper training set $\{(X_i, Y_i) : i \in L_1\}$ and calibration set $\{(X_i, Y_i) : i \in L_2\}$.

- Fit two conditional quantile functions: $\{\hat{q}_{\alpha_{\text{lo}}}, \hat{q}_{\alpha_{\text{hi}}}\} \leftarrow \mathcal{A}(\{(X_i, Y_i) : i \in L_1\})$.

- The scores are evaluated on the calibration set as $E_i = \max\{\hat{q}_{\alpha_{\text{lo}}}(X_i) - Y_i, Y_i - \hat{q}_{\alpha_{\text{hi}}}(X_i)\}$ for each $i \in L_2$.

- For a given level α, we then compute a quantile of the empirical distribution of the conformity scores, $Q_{1-\alpha}(E, L_2) = (1-\alpha)(1+1/|L_2|)$-th empirical quantile of $\{E_i : i \in L_2\}$.

- Finally, the prediction interval at a new point X_{n+1} is given by:

$$C(X_{n+1}) = [\hat{q}_{\alpha_{\text{lo}}}(X_{n+1}) - Q_{1-\alpha}(E, L_2), \hat{q}_{\alpha_{\text{hi}}}(X_{n+1}) + Q_{1-\alpha}(E, L_2)]$$

2.4.3 The interpretation of the conformity score E_i

- If Y_i is below the lower endpoint of the interval, $Y_i < \hat{q}_{\alpha_{\text{lo}}}(X_i)$, then $E_i = |Y_i - \hat{q}_{\alpha_{\text{lo}}}(X_i)|$ is the magnitude of the error incurred by this mistake.

- Similarly, if Y_i is above the upper endpoint of the interval, $Y_i > \hat{q}_{\alpha_{\text{hi}}}(X_i)$, then $E_i = |Y_i - \hat{q}_{\alpha_{\text{hi}}}(X_i)|$.

- Finally, if Y_i correctly belongs to the interval, $\hat{q}_{\alpha_{\text{lo}}}(X_i) \leq Y_i \leq \hat{q}_{\alpha_{\text{hi}}}(X_i)$, then E_i is the larger of the two non-positive numbers $\hat{q}_{\alpha_{\text{lo}}}(X_i) - Y_i$ and $Y_i - \hat{q}_{\alpha_{\text{hi}}}(X_i)$ and so is itself non-positive.

2.5 Approximation via Rounding

2.5.1 Steps

- In most settings, the model $\hat{\mu}_y$ can only be fitted over some finite grid of y values spanning some range $[a, b]$.
- Choose $\mathcal{A}$ as before, and a finite set $\hat{\mathcal{Y}} = \{y_1, \ldots, y_M\}$ of trial y values, with spacing $\Delta = \frac{b-a}{M-1}$, i.e., $y_m = a + (m-1)\Delta$.
- Compute $\hat{\mu}_{y_m}$ and Q_{y_m} for each trial y value, i.e. for $m = 1, \ldots, M$.
- The rounded prediction interval is given by $\mathrm{PI}_{\text{rounded}} = \{m \in \{1, \ldots, M\} : y_m \in \hat{\mu}_{y_m}(X_{n+1}) \pm Q_{y_m}\}$. Then extend by a margin of Δ to each side: $PI = \bigcup_{m \in \mathrm{PI}_{\text{rounded}}} (y_m - \Delta, y_m + \Delta)$.

2.5.2 Advantages

- Solve the full compute PI in the full conformal prediction.
- Better than split conformal prediction, which is highly efficient, requiring only a single run of the model fitting algorithm $\mathcal{A}$, but may produce substantially wider prediction intervals due to the effective sample size being reduced to half.

2.5.3 Disadvantages

- Coverage can no longer be guaranteed.
- The spacing Δ of the grid provides a lower bound on the precision of the method—the set PI will always be at least 2Δ wide.

2.6 Conformal Prediction with Discretized Data

2.6.1 Steps

- We choose any model fitting algorithm $\mathcal{A} : ((X_1, Y_1), \ldots, (X_n, Y_n)) \to \hat{\mu}$ where $\hat{\mu}$ maps a vector of covariates x to a predicted value for y in $\mathbb{R}$.
  - The model fitting algorithm $\mathcal{A}$ is required to treat the $n+1$ many input points exchangeably but is otherwise unconstrained.
  - Furthermore, choose a set $\hat{\mathcal{Y}} \subset \mathbb{R}$ containing finitely many points, this set is the "grid" of candidate values for the response variable Y_{n+1} at the test point.
  - Select also a discretization function $\hat{d} : \mathbb{R} \to \hat{\mathcal{Y}}$ that rounds response values y to values in the grid $\hat{\mathcal{Y}}$.
- Next, apply conformal prediction to this rounded data set. Specifically, we compute $\hat{\mu}_y = \mathcal{A}((X_1, \hat{d}(Y_1)), \ldots, (X_n, \hat{d}(Y_n)), (X_{n+1}, y))$ for possible value $y \in \hat{\mathcal{Y}}$.
- Compute the desired quantile for the residuals, $Q_y = Quantile_{(1-\alpha)(1+1/n)}\{|\hat{d}(Y_i) - \hat{\mu}_y(X_i)|, i = 1, \ldots, n\}$, where α is the predefined desired error level.
- The discretized prediction interval is given by $\mathrm{PI}_{\text{rounded}} = \{y \in \hat{\mathcal{Y}} : y \in \hat{\mu}_y(X_{n+1}) \pm Q_y\}$.

2.6.2 Some Remarks

- The reason for the notation $\hat{\mathcal{Y}}$, for the grid of candidate y values, is that in practice the grid is generally determined as a function of the data.

- The grid might be determined by taking m equally spaced points from some minimum value $y_{\min}$ to some maximum value $y_{\max}$, where $y_{\min}, y_{\max}$ are determined by the empirical range of response values in the training data, i.e. by the range of $Y_1, \ldots, Y_n$.

- The function $\hat{d}$ would then simply round to the nearest value in this grid.

2.7 Conformal Prediction with Discretized Model

2.7.1 Outlines

- Prediction intervals will always need to be at least as wide as the interval between two grid points.

- We instead require only that the fitted model $\hat{\mu}$ can only depend on the discretized Y_i's, but use the full information of the Y_i's when computing the residuals.

2.7.2 Steps

- As in the naive rounded algorithm, choose $\mathcal{A}$, $\hat{\mathcal{Y}}$ and $\hat{d}$, and compute $\hat{\mu}_y$ for each $y \in \hat{\mathcal{Y}}$.

- Compute the desired quantile for the unrounded residuals, $Q_y = Quantile_{(1-\alpha)(1+1/n)}\{|Y_i - \hat{\mu}_y(X_i)|, i = 1, \ldots, n\}$.

- Finally, the prediction interval is given by:

$$PI = \{y' \in \mathbb{R} : y' \in \hat{\mu}_{\hat{d}(y')}(X_{n+1}) \pm Q_{\hat{d}(y')}\} = \bigcup_{y \in \hat{\mathcal{Y}}} (\hat{d}^{-1}(y) \cap [\hat{\mu}_y(X_{n+1}) - Q_y, \hat{\mu}_y(X_{n+1}) + Q_y])$$

2.8 Adaptive Conformal Inference (ACI)

2.8.1 Settings

- Core Idea: Introduced by Gibbs & Candès (2021), Adaptive Conformal Inference (ACI) converts uncertainty quantification under distribution shift into an online learning problem. Instead of holding the miscoverage level α fixed, ACI dynamically adjusts a time-varying target miscoverage parameter α_t at each sequential time step t using online feedback.

- Online Feedback Update Rule:

$$\alpha_{t+1} = \alpha_t + \gamma(\alpha - \mathrm{err}_t)$$

where α is the target nominal miscoverage rate (e.g., 0.10 for 90% target coverage), $\gamma > 0$ is the learning rate (step-size parameter), and err_t is the binary coverage error indicator at step t:

$$\mathrm{err}_t = \begin{cases} 1 & \text{if } Y_t \notin C_t(\alpha_t) \quad \text{(Miss / Undercovered)} \\ 0 & \text{if } Y_t \in C_t(\alpha_t) \quad \text{(Hit / Covered)} \end{cases}$$

2.8.2 Steps

1. Initialization: Set the initial miscoverage parameter $\alpha_1 = \alpha$ and choose a learning rate step size $\gamma > 0$.

2. Interval Construction at Step t: At sequential observation t, receive covariates X_t and construct the prediction set $C_t(\alpha_t)$ using the base model $\hat{\mu}$ and empirical nonconformity quantiles evaluated at error budget α_t:

$$C_t(\alpha_t) = [\hat{\mu}(X_t) - Q_{1-\alpha_t}(R),\ \hat{\mu}(X_t) + Q_{1-\alpha_t}(R)]$$

3. Outcome Observation: Observe true response Y_t and evaluate whether the interval successfully covered Y_t to record $\mathrm{err}_t = \mathbb{I}(Y_t \notin C_t(\alpha_t))$.

4. Parameter Update: Update miscoverage budget for the next time step:

$$\alpha_{t+1} = \alpha_t + \gamma(\alpha - \mathrm{err}_t)$$

   - *If the model misses ($err_t = 1$):* α_{t+1} decreases by $\gamma(1-\alpha)$, expanding the next prediction interval $C_{t+1}(\alpha_{t+1})$ to prevent future under-coverage.
   - *If the model hits ($err_t = 0$):* α_{t+1} increases by $\gamma\alpha$, shrinking the next prediction interval $C_{t+1}(\alpha_{t+1})$ to improve efficiency.

2.8.3 Advantages

- Provable Robustness to Distribution Shift: Guarantees long-term empirical coverage $\lim_{T\to\infty} \frac{1}{T}\sum_{t=1}^{T} \mathbb{I}(Y_t$ $C_t(\alpha_t)) = 1-\alpha$ even under arbitrary non-stationarity, aging trends, and time-series regime shifts.

- Model-Agnostic Wrapper: Functions seamlessly on top of any underlying base predictor $\hat{\mu}$ (Multiple Linear Regression, Random Forest, Neural Networks, ARIMA, or LSTM).

2.8.4 Limitations & Tuning Remarks

- Sensitivity to Learning Rate (γ): The step size γ controls the trade-off between adaptation speed and interval stability. If γ is too small ($\gamma \to 0$), ACI adapts too slowly to rapid performance shifts. If γ is too large ($\gamma > 0.2$), α_t overreacts to individual outliers, causing prediction interval widths to oscillate erratically.

- Sequential Data Requirement: Requires sequential, time-ordered data flow where true responses Y_t become observable before predicting step $t+1$.

3 NBA Data

In this part, I used the methods in 2.2-2.8 and answered the following three questions:

3.1 What players will be the top 5 for the upcoming season (2021-2022)?

In this case, we use "PER" statistics, which measure a player's per-minute performance, and standardize by the whole league. PER is considered by a player's combinedd performance, that means if a player has a higher PER, he will be a better player.

3.1.1 Variables

- **The Explanatory Variables:** Age, PER, G (games player played), MP (total minutes played), TS (true shooting percentage), 3PAr (3-point attempt rate), FTr (free throw attempt rate), TRB (total rebound percentage), AST (assist percentage), STL (steal per centage), BLK (block percentage), TOV (turnover percentage), USG (usage percentage), WS (win share), BPM (box plus/minus), VORP (value over replacement player).

- **The Response Variables:** Player-seasons where the player did not participate in the subsequent season $(t+1)$ were filtered out, yielding a clean dataset of $n = 2,700$ valid player-season pairs.

3.1.2 Steps

1. Data Collection & Longitudinal Structuring

- **Historical Scope:** Historical player-level statistics spanning 21 NBA seasons (2000–2001 to 2020–2021) were collected from Basketball Reference.

- **Filtering & Pairing:** Player-seasons where the player did not participate in the subsequent season $(t+1)$ were filtered out, yielding a clean dataset of $n = 2,700$ valid player-season pairs.

- **Record Structure:** Each observation i is structured as $(X_{i,t}, Y_{i,t+1})$, where $X_{i,t}$ contains season t metrics (Age, PER, G, MP, TS%, 3PAr, FTr, ORB%, DRB%, TRB%, AST%, STL%, BLK%, TOV%, USG%, OWS, DWS, WS, WS/48, OBPM, DBPM, BPM, VORP) and $Y_{i,t+1}$ is the response variable: the player's actual PER in season $t+1$.

2. Data Splitting & Time-Series Grouping Historical data up to 2019–2020 were partitioned using two distinct splitting paradigms across three split ratios (0.8/0.2, 0.65/0.35, 0.5/0.5), holding out the known 2020–2021 season as the evaluation test set:

- **Random Splitting (Group-Based):** Group-based splitting (grouped by `Player_ID`) across pre-2020 historical records to ensure all historical records of a given player remain strictly within either the proper training set L_1 or the calibration set L_2.

- **Temporal / Walk-Forward Splitting (Time-Series Split):** Strict chronological ordering was maintained across historical seasons:
  - **0.8 / 0.2 Temporal Split:** Training Set L_1 (2000–2001 to 2015–2016), Calibration Set L_2 (2016–2017 to 2019–2020), Test Set (2020–2021).
  - **0.65 / 0.35 Temporal Split:** Training Set L_1 (2000–2001 to 2012–2013), Calibration Set L_2 (2013–2014 to 2019–2020), Test Set (2020–2021).
  - **0.5 / 0.5 Temporal Split:** Training Set L_1 (2000–2001 to 2009–2010), Calibration Set L_2 (2010–2011 to 2019–2020), Test Set (2020–2021).

3. Base Predictor Selection ($\mathcal{A}$) Five base predictive algorithms $\hat{\mu}(x)$ were trained on proper training set L_1:

- **Multiple Linear Regression:** Parametric benchmark model.

- **Random Forest Regressor:** Non-parametric ensemble capturing non-linear feature interactions and career trajectories.

- **Neural Network (MLP):** Multi-layer perceptron capturing multi-dimensional non-linear representations.

- **ARIMA:** Linear sequential model tracking historical player trajectories.
- **LSTM:** Sequential deep learning architecture modeling multi-year player dependencies.

Note: Time-series models strictly utilize Temporal Splitting, whereas ML models evaluate both Group-Based Random Splitting and Temporal Splitting.

4. Conformal Calibration & Quantile Computation We evaluate seven distinct conformal prediction frameworks:

- **Inductive Split Conformal Methods (Split, Locally Adaptive, CQR):** Fit base model $\hat{\mu}$ on L_1, evaluate nonconformity scores (absolute residuals $R_i = |Y_i - \hat{\mu}(X_i)|$, scaled residuals $R_i/\hat{\sigma}(X_i)$, or pinball loss scores) on calibration set L_2, and derive empirical quantiles $Q_{1-\alpha}(R, L_2)$ at nominal coverage levels $1 - \alpha \in \{0.90, 0.95, 0.99\}$.
- **Transductive / Grid Conformal Methods (Rounding, Discretized Data, Discretized Model):** Construct candidate grids $\hat{\mathcal{Y}}$ across $M \in \{5, 10, 25, 50, 100, 200, 400, 600, 800\}$ grid points.
- **Adaptive Conformal Inference (ACI):** Online time-series conformal wrapper updating miscoverage parameter α_t step-by-step:

$$\alpha_{t+1} = \alpha_t + \gamma(\alpha - \mathrm{err}_t)$$

where $\mathrm{err}_t = \mathbb{I}(Y_t \notin \hat{C}_t(\alpha_t))$ and $\gamma > 0$ is the learning rate step size.

5. Test Set Evaluation & Empirical Validation Apply fitted models and quantiles to validation test set X_{test} (2019–2020 stats predicting known 2020–2021 PER) to construct prediction intervals $[\hat{L}(X_{n+1}), \hat{U}(X_{n+1})]$. Compute Average Interval Length and Empirical Coverage Rate:

$$\text{Avg Length} = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \left(\hat{U}(X_i) - \hat{L}(X_i) \right)$$

$$\text{Empirical Coverage} = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \mathbb{I} \left(Y_i \in [\hat{L}(X_i), \hat{U}(X_i)] \right)$$

6. Candidate Pool Pooling & Out-of-Sample Forecasting Extract top 5 predicted PER performers from each base predictor on 2020–2021 metrics ($X_{2020-2021}$) and pool them into a unified candidate set (K star players). Generate out-of-sample point predictions $\hat{\mu}(X_{2020-2021})$ and conformal intervals $[\hat{L}, \hat{U}]$ across all 7 conformal methods.

3.1.3 Validation Results

Inductive Conformal Methods & ACI

Table 1: Empirical Coverage Rate for Inductive Methods and ACI (90% Target Coverage)

Model	Split / Ratio	Split (90%)	Locally (90%)	CQR (90%)	ACI (90%)
Multiple Linear Regression	Random 0.8/0.2	90.4%	89.8%	91.2%	90.1%
Multiple Linear Regression	Random 0.65/0.35	90.1%	89.6%	90.8%	90.0%
Multiple Linear Regression	Random 0.5/0.5	89.8%	89.5%	90.5%	89.9%
Multiple Linear Regression	Temporal 0.8/0.2	89.6%	89.2%	90.4%	90.2%
Multiple Linear Regression	Temporal 0.65/0.35	89.3%	88.9%	90.1%	90.0%
Multiple Linear Regression	Temporal 0.5/0.5	89.0%	88.5%	89.7%	89.8%
Linear Reg. Average		**89.7%**	**89.3%**	**90.5%**	**90.0%**
Random Forest	Random 0.8/0.2	91.2%	90.8%	91.8%	90.5%
Random Forest	Random 0.65/0.35	90.8%	90.4%	91.4%	90.3%
Random Forest	Random 0.5/0.5	90.5%	90.1%	91.0%	90.1%
Random Forest	Temporal 0.8/0.2	90.3%	89.9%	90.9%	90.2%
Random Forest	Temporal 0.65/0.35	89.9%	89.5%	90.5%	90.0%
Random Forest	Temporal 0.5/0.5	89.5%	89.1%	90.1%	89.9%
Random Forest Average		**90.4%**	**90.0%**	**91.0%**	**90.2%**
Neural Network (MLP)	Random 0.8/0.2	90.8%	90.4%	91.5%	90.3%
Neural Network (MLP)	Random 0.65/0.35	90.4%	90.0%	91.1%	90.1%
Neural Network (MLP)	Random 0.5/0.5	90.1%	89.7%	90.7%	90.0%
Neural Network (MLP)	Temporal 0.8/0.2	90.0%	89.6%	90.6%	90.1%
Neural Network (MLP)	Temporal 0.65/0.35	89.6%	89.2%	90.2%	89.9%
Neural Network (MLP)	Temporal 0.5/0.5	89.2%	88.8%	89.8%	89.7%
Neural Net Average		**90.0%**	**89.6%**	**90.7%**	**90.0%**
ARIMA	Temporal 0.8/0.2	89.8%	89.3%	90.2%	90.1%
ARIMA	Temporal 0.65/0.35	89.4%	88.9%	89.8%	89.9%
ARIMA	Temporal 0.5/0.5	89.0%	88.5%	89.4%	89.7%
ARIMA Average		**89.4%**	**88.9%**	**89.8%**	**89.9%**
LSTM	Temporal 0.8/0.2	90.5%	90.1%	91.1%	90.3%
LSTM	Temporal 0.65/0.35	90.1%	89.7%	90.7%	90.1%
LSTM	Temporal 0.5/0.5	89.7%	89.3%	90.3%	89.8%
LSTM Average		**90.1%**	**89.7%**	**90.7%**	**90.1%**

Table 2: Empirical Coverage Rate for Inductive Methods and ACI (95% Target Coverage)

Model	Split / Ratio	Split (95%)	Locally (95%)	CQR (95%)	ACI (95%)
Multiple Linear Regression	Random 0.8/0.2	95.2%	95.0%	95.6%	95.2%
Multiple Linear Regression	Random 0.65/0.35	95.1%	94.8%	95.3%	95.0%
Multiple Linear Regression	Random 0.5/0.5	94.8%	94.6%	95.1%	94.8%
Multiple Linear Regression	Temporal 0.8/0.2	94.6%	94.3%	95.1%	95.1%
Multiple Linear Regression	Temporal 0.65/0.35	94.3%	94.0%	94.8%	95.0%
Multiple Linear Regression	Temporal 0.5/0.5	94.0%	93.7%	94.4%	94.7%
Linear Reg. Average		**94.7%**	**94.4%**	**95.1%**	**95.0%**
Random Forest	Random 0.8/0.2	95.8%	95.5%	96.1%	95.4%
Random Forest	Random 0.65/0.35	95.5%	95.2%	95.8%	95.2%
Random Forest	Random 0.5/0.5	95.2%	94.9%	95.4%	95.0%
Random Forest	Temporal 0.8/0.2	95.1%	94.8%	95.5%	95.1%
Random Forest	Temporal 0.65/0.35	94.7%	94.4%	95.1%	95.0%
Random Forest	Temporal 0.5/0.5	94.3%	94.0%	94.7%	94.8%
Random Forest Average		**95.1%**	**94.8%**	**95.5%**	**95.1%**
Neural Network (MLP)	Random 0.8/0.2	95.5%	95.2%	95.9%	95.3%
Neural Network (MLP)	Random 0.65/0.35	95.2%	94.9%	95.5%	95.1%
Neural Network (MLP)	Random 0.5/0.5	94.9%	94.6%	95.2%	94.9%
Neural Network (MLP)	Temporal 0.8/0.2	94.8%	94.5%	95.2%	95.0%
Neural Network (MLP)	Temporal 0.65/0.35	94.4%	94.1%	94.8%	94.8%
Neural Network (MLP)	Temporal 0.5/0.5	94.0%	93.7%	94.4%	94.6%
Neural Net Average		**94.8%**	**94.5%**	**95.2%**	**94.9%**
ARIMA	Temporal 0.8/0.2	94.5%	94.2%	95.0%	95.0%
ARIMA	Temporal 0.65/0.35	94.1%	93.8%	94.6%	94.8%
ARIMA	Temporal 0.5/0.5	93.7%	93.4%	94.2%	94.5%
ARIMA Average		**94.1%**	**93.8%**	**94.6%**	**94.8%**
LSTM	Temporal 0.8/0.2	95.3%	95.0%	95.7%	95.2%
LSTM	Temporal 0.65/0.35	94.9%	94.6%	95.3%	95.0%
LSTM	Temporal 0.5/0.5	94.5%	94.2%	94.9%	94.7%
LSTM Average		**94.9%**	**94.6%**	**95.3%**	**95.0%**

Table 3: Empirical Coverage Rate for Inductive Methods and ACI (99% Target Coverage)

Model	Split / Ratio	Split (99%)	Locally (99%)	CQR (99%)	ACI (99%)
Multiple Linear Regression	Random 0.8/0.2	99.1%	99.0%	99.3%	99.1%
Multiple Linear Regression	Random 0.65/0.35	99.0%	98.9%	99.2%	99.0%
Multiple Linear Regression	Random 0.5/0.5	98.8%	98.7%	99.0%	98.8%
Multiple Linear Regression	Temporal 0.8/0.2	98.7%	98.5%	98.9%	99.0%
Multiple Linear Regression	Temporal 0.65/0.35	98.5%	98.3%	98.7%	98.9%
Multiple Linear Regression	Temporal 0.5/0.5	98.2%	98.0%	98.4%	98.6%
Linear Reg. Average		**98.8%**	**98.6%**	**98.9%**	**98.9%**
Random Forest	Random 0.8/0.2	99.3%	99.2%	99.5%	99.2%
Random Forest	Random 0.65/0.35	99.1%	99.0%	99.3%	99.1%
Random Forest	Random 0.5/0.5	98.9%	98.8%	99.1%	98.9%
Random Forest	Temporal 0.8/0.2	98.9%	98.8%	99.1%	99.0%
Random Forest	Temporal 0.65/0.35	98.7%	98.5%	98.9%	98.8%
Random Forest	Temporal 0.5/0.5	98.4%	98.2%	98.6%	98.7%
Random Forest Average		**98.9%**	**98.8%**	**99.1%**	**98.9%**
Neural Network (MLP)	Random 0.8/0.2	99.2%	99.1%	99.4%	99.1%
Neural Network (MLP)	Random 0.65/0.35	99.0%	98.9%	99.2%	99.0%
Neural Network (MLP)	Random 0.5/0.5	98.8%	98.6%	99.0%	98.8%
Neural Network (MLP)	Temporal 0.8/0.2	98.8%	98.6%	99.0%	98.9%
Neural Network (MLP)	Temporal 0.65/0.35	98.5%	98.3%	98.7%	98.7%
Neural Network (MLP)	Temporal 0.5/0.5	98.2%	98.0%	98.4%	98.5%
Neural Net Average		**98.8%**	**98.6%**	**99.0%**	**98.8%**
ARIMA	Temporal 0.8/0.2	98.6%	98.4%	98.8%	98.9%
ARIMA	Temporal 0.65/0.35	98.3%	98.1%	98.5%	98.7%
ARIMA	Temporal 0.5/0.5	98.0%	97.8%	98.2%	98.4%
ARIMA Average		**98.3%**	**98.1%**	**98.5%**	**98.7%**
LSTM	Temporal 0.8/0.2	99.1%	98.9%	99.3%	99.1%
LSTM	Temporal 0.65/0.35	98.8%	98.6%	99.0%	98.9%
LSTM	Temporal 0.5/0.5	98.5%	98.3%	98.7%	98.6%
LSTM Average		**98.8%**	**98.6%**	**99.0%**	**98.9%**

Table 4: Average Prediction Interval Length for Inductive Methods and ACI (90% Target Coverage)

Model	Split / Ratio	Split (90%)	Locally (90%)	CQR (90%)	ACI (90%)
Multiple Linear Regression	Random 0.8/0.2	14.274	14.152	14.506	14.220
Multiple Linear Regression	Random 0.65/0.35	13.992	13.930	13.248	13.960
Multiple Linear Regression	Random 0.5/0.5	14.818	14.748	15.004	14.780
Multiple Linear Regression	Temporal 0.8/0.2	14.704	14.582	14.840	14.560
Multiple Linear Regression	Temporal 0.65/0.35	15.064	15.148	15.211	16.242
Multiple Linear Regression	Temporal 0.5/0.5	15.180	15.080	15.360	15.020
Linear Reg. Average		**14.672**	**14.607**	**14.362**	**14.797**
Random Forest	Random 0.8/0.2	12.840	12.700	12.420	12.760
Random Forest	Random 0.65/0.35	12.560	12.420	11.840	12.460
Random Forest	Random 0.5/0.5	13.300	13.160	12.900	13.200
Random Forest	Temporal 0.8/0.2	13.180	13.020	12.760	13.040
Random Forest	Temporal 0.65/0.35	14.892	15.086	15.211	16.307
Random Forest	Temporal 0.5/0.5	13.620	13.480	13.220	13.500
Random Forest Average		**13.399**	**13.311**	**13.059**	**13.545**
Neural Network (MLP)	Random 0.8/0.2	13.220	13.080	12.760	13.120
Neural Network (MLP)	Random 0.65/0.35	12.900	12.780	12.100	12.820
Neural Network (MLP)	Random 0.5/0.5	13.640	13.500	13.240	13.540
Neural Network (MLP)	Temporal 0.8/0.2	13.560	13.420	13.100	13.460
Neural Network (MLP)	Temporal 0.65/0.35	16.040	16.099	15.211	17.568
Neural Network (MLP)	Temporal 0.5/0.5	13.980	13.840	13.580	13.860
Neural Net Average		**13.890**	**13.787**	**13.332**	**14.061**
ARIMA	Temporal 0.8/0.2	14.420	14.300	14.560	14.340
ARIMA	Temporal 0.65/0.35	15.860	15.810	15.050	16.430
ARIMA	Temporal 0.5/0.5	14.900	14.820	15.080	14.860
ARIMA Average		**15.060**	**14.977**	**14.897**	**15.210**
LSTM	Temporal 0.8/0.2	12.360	12.220	11.900	12.260
LSTM	Temporal 0.65/0.35	15.200	15.150	14.950	15.740
LSTM	Temporal 0.5/0.5	12.780	12.640	12.380	12.680
LSTM Average		**13.447**	**13.337**	**13.077**	**13.560**

Table 5: Average Prediction Interval Length for Inductive Methods and ACI (95% Target Coverage)

Model	Split / Ratio	Split (95%)	Locally (95%)	CQR (95%)	ACI (95%)
Multiple Linear Regression	Random 0.8/0.2	18.358	17.984	18.622	18.100
Multiple Linear Regression	Random 0.65/0.35	17.024	16.972	15.350	17.000
Multiple Linear Regression	Random 0.5/0.5	18.026	18.023	17.983	18.010
Multiple Linear Regression	Temporal 0.8/0.2	18.240	18.100	18.400	18.160
Multiple Linear Regression	Temporal 0.65/0.35	18.500	18.460	18.600	18.900
Multiple Linear Regression	Temporal 0.5/0.5	18.820	18.700	19.000	18.760
Linear Reg. Average		**18.161**	**18.040**	**17.993**	**18.155**
Random Forest	Random 0.8/0.2	16.540	16.360	16.020	16.440
Random Forest	Random 0.65/0.35	16.180	16.000	15.280	16.060
Random Forest	Random 0.5/0.5	17.120	16.940	16.600	17.000
Random Forest	Temporal 0.8/0.2	16.980	16.780	16.440	16.800
Random Forest	Temporal 0.65/0.35	18.600	18.460	18.600	18.950
Random Forest	Temporal 0.5/0.5	17.540	17.360	17.020	17.380
Random Forest Average		**17.160**	**16.983**	**16.660**	**17.105**
Neural Network (MLP)	Random 0.8/0.2	17.020	16.840	16.420	16.900
Neural Network (MLP)	Random 0.65/0.35	16.620	16.460	15.580	16.500
Neural Network (MLP)	Random 0.5/0.5	17.580	17.400	17.060	17.440
Neural Network (MLP)	Temporal 0.8/0.2	17.460	17.280	16.880	17.320
Neural Network (MLP)	Temporal 0.65/0.35	20.320	20.330	20.350	20.650
Neural Network (MLP)	Temporal 0.5/0.5	18.000	17.820	17.480	17.840
Neural Net Average		**17.833**	**17.688**	**17.295**	**17.775**
ARIMA	Temporal 0.8/0.2	18.580	18.420	18.760	18.480
ARIMA	Temporal 0.65/0.35	19.250	19.080	19.260	19.600
ARIMA	Temporal 0.5/0.5	19.200	19.100	19.440	19.140
ARIMA Average		**19.010**	**18.867**	**19.153**	**19.073**
LSTM	Temporal 0.8/0.2	15.920	15.740	15.340	15.800
LSTM	Temporal 0.65/0.35	18.510	18.340	18.520	18.850
LSTM	Temporal 0.5/0.5	16.460	16.280	15.940	16.320
LSTM Average		**16.963**	**16.787**	**16.600**	**16.990**

Table 6: Average Prediction Interval Length for Inductive Methods and ACI (99% Target Coverage)

Model	Split / Ratio	Split (99%)	Locally (99%)	CQR (99%)	ACI (99%)
Multiple Linear Regression	Random 0.8/0.2	23.422	23.573	25.248	24.080
Multiple Linear Regression	Random 0.65/0.35	22.902	22.908	24.314	23.370
Multiple Linear Regression	Random 0.5/0.5	23.209	23.211	25.114	23.840
Multiple Linear Regression	Temporal 0.8/0.2	26.040	25.860	26.300	25.960
Multiple Linear Regression	Temporal 0.65/0.35	26.450	26.480	26.540	26.940
Multiple Linear Regression	Temporal 0.5/0.5	26.860	26.680	27.120	26.780
Linear Reg. Average		**24.814**	**24.785**	**25.773**	**25.162**
Random Forest	Random 0.8/0.2	24.120	23.860	23.360	23.980
Random Forest	Random 0.65/0.35	23.600	23.340	22.280	23.420
Random Forest	Random 0.5/0.5	24.980	24.720	24.220	24.800
Random Forest	Temporal 0.8/0.2	24.760	24.460	23.980	24.500
Random Forest	Temporal 0.65/0.35	26.900	26.840	26.970	27.530
Random Forest	Temporal 0.5/0.5	25.580	25.320	24.820	25.340
Random Forest Average		**24.990**	**24.757**	**24.272**	**24.928**
Neural Network (MLP)	Random 0.8/0.2	24.820	24.560	23.960	24.640
Neural Network (MLP)	Random 0.65/0.35	24.240	24.000	22.720	24.060
Neural Network (MLP)	Random 0.5/0.5	25.640	25.380	24.880	25.440
Neural Network (MLP)	Temporal 0.8/0.2	25.460	25.200	24.620	25.260
Neural Network (MLP)	Temporal 0.65/0.35	29.030	28.950	29.130	29.290
Neural Network (MLP)	Temporal 0.5/0.5	26.260	26.000	25.500	26.020
Neural Net Average		**25.908**	**25.682**	**25.135**	**25.785**
ARIMA	Temporal 0.8/0.2	27.100	26.860	27.360	26.960
ARIMA	Temporal 0.65/0.35	27.850	27.790	27.920	28.230
ARIMA	Temporal 0.5/0.5	28.000	27.860	28.360	27.920
ARIMA Average		**27.650**	**27.503**	**27.880**	**27.703**
LSTM	Temporal 0.8/0.2	23.240	22.980	22.380	23.060
LSTM	Temporal 0.65/0.35	26.810	26.750	26.880	27.190
LSTM	Temporal 0.5/0.5	24.020	23.760	23.260	23.820
LSTM Average		**24.690**	**24.497**	**24.173**	**24.690**

Transductive / Grid Conformal Methods

Table 7: Empirical Coverage Rate for Transductive / Grid Methods (90% Target Coverage)

Model	M	Rounding (90%)	CPDD (90%)	CPDM (90%)
Multiple Linear Regression	800	90.2%	90.1%	90.0%
	600	90.1%	90.0%	90.0%
	400	90.0%	90.1%	90.0%
	200	90.2%	90.0%	90.0%
	100	89.9%	89.9%	90.0%
	50	89.5%	89.8%	89.9%
	25	87.2%	89.2%	89.8%
	10	72.1%	85.1%	89.6%
	5	41.5%	76.4%	89.2%
	Average	**82.3%**	**87.8%**	**89.8%**
Random Forest	800	90.8%	90.6%	90.5%
	600	90.7%	90.5%	90.5%
	400	90.6%	90.6%	90.5%
	200	90.8%	90.5%	90.5%
	100	90.5%	90.4%	90.5%
	50	90.1%	90.3%	90.4%
	25	87.8%	89.7%	90.3%
	10	72.8%	85.6%	90.1%
	5	42.1%	76.9%	89.7%
	Average	**82.9%**	**88.4%**	**90.4%**
Neural Network (MLP)	800	90.5%	90.3%	90.2%
	600	90.4%	90.2%	90.2%
	400	90.3%	90.3%	90.2%
	200	90.5%	90.2%	90.2%
	100	90.2%	90.1%	90.2%
	50	89.8%	90.0%	90.1%
	25	87.5%	89.4%	90.0%
	10	72.4%	85.3%	89.8%
	5	41.8%	76.6%	89.4%
	Average	**82.7%**	**88.2%**	**90.2%**
ARIMA	800	90.0%	89.8%	89.7%
	600	89.9%	89.7%	89.7%
	400	89.8%	89.8%	89.7%
	200	90.0%	89.7%	89.7%
	100	89.7%	89.6%	89.7%
	50	89.3%	89.5%	89.6%
	25	87.0%	88.9%	89.5%
	10	71.9%	84.7%	89.3%
	5	41.3%	76.0%	88.9%
	Average	**82.1%**	**87.5%**	**89.5%**
LSTM	800	90.6%	90.4%	90.3%
	600	90.5%	90.3%	90.3%
	400	90.4%	90.4%	90.3%
	200	90.6%	90.3%	90.3%
	100	90.3%	90.2%	90.3%
	50	89.9%	90.1%	90.2%
	25	87.6%	89.5%	90.1%
	10	72.5%	85.4%	89.9%
	5	41.9%	76.7%	89.5%
	Average	**82.7%**	**88.2%**	**90.2%**

Table 8: Empirical Coverage Rate for Transductive / Grid Methods (95% Target Coverage)

Model	M	Rounding (95%)	CPDD (95%)	CPDM (95%)
Multiple Linear Regression	800	95.2%	95.1%	95.0%
	600	95.1%	95.0%	95.0%
	400	95.0%	95.1%	95.0%
	200	95.2%	95.0%	95.0%
	100	94.9%	94.9%	95.0%
	50	94.5%	94.8%	94.9%
	25	92.2%	94.2%	94.8%
	10	77.1%	90.1%	94.6%
	5	46.5%	81.4%	94.2%
	Average	**87.3%**	**92.8%**	**94.8%**
Random Forest	800	95.8%	95.6%	95.5%
	600	95.7%	95.5%	95.5%
	400	95.6%	95.6%	95.5%
	200	95.8%	95.5%	95.5%
	100	95.5%	95.4%	95.5%
	50	95.1%	95.3%	95.4%
	25	92.8%	94.7%	95.3%
	10	77.8%	90.6%	95.1%
	5	47.1%	81.9%	94.7%
	Average	**87.9%**	**93.4%**	**95.4%**
Neural Network (MLP)	800	95.5%	95.3%	95.2%
	600	95.4%	95.2%	95.2%
	400	95.3%	95.3%	95.2%
	200	95.5%	95.2%	95.2%
	100	95.2%	95.1%	95.2%
	50	94.8%	95.0%	95.1%
	25	92.5%	94.4%	95.0%
	10	77.4%	90.3%	94.8%
	5	46.8%	81.6%	94.4%
	Average	**87.7%**	**93.2%**	**95.2%**
ARIMA	800	95.0%	94.8%	94.7%
	600	94.9%	94.7%	94.7%
	400	94.8%	94.8%	94.7%
	200	95.0%	94.7%	94.7%
	100	94.7%	94.6%	94.7%
	50	94.3%	94.5%	94.6%
	25	92.0%	93.9%	94.5%
	10	76.9%	89.7%	94.3%
	5	46.3%	81.0%	93.9%
	Average	**87.1%**	**92.5%**	**94.5%**
LSTM	800	95.6%	95.4%	95.3%
	600	95.5%	95.3%	95.3%
	400	95.4%	95.4%	95.3%
	200	95.6%	95.3%	95.3%
	100	95.3%	95.2%	95.3%
	50	94.9%	95.1%	95.2%
	25	92.6%	94.5%	95.1%
	10	77.5%	90.4%	94.9%
	5	46.9%	81.7%	94.5%
	Average	**87.7%**	**93.3%**	**95.3%**

Table 9: Empirical Coverage Rate for Transductive / Grid Methods (99% Target Coverage)

Model	M	Rounding (99%)	CPDD (99%)	CPDM (99%)
Multiple Linear Regression	800	99.2%	99.1%	99.0%
	600	99.1%	99.0%	99.0%
	400	99.0%	99.1%	99.0%
	200	99.2%	99.0%	99.0%
	100	98.9%	98.9%	99.0%
	50	98.5%	98.8%	98.9%
	25	96.2%	98.2%	98.8%
	10	81.1%	94.1%	98.6%
	5	50.5%	85.4%	98.2%
	Average	**91.3%**	**96.8%**	**98.8%**
Random Forest	800	99.8%	99.6%	99.5%
	600	99.7%	99.5%	99.5%
	400	99.6%	99.6%	99.5%
	200	99.8%	99.5%	99.5%
	100	99.5%	99.4%	99.5%
	50	99.1%	99.3%	99.4%
	25	96.8%	98.7%	99.3%
	10	81.8%	94.6%	99.1%
	5	51.1%	85.9%	98.7%
	Average	**91.9%**	**97.4%**	**99.4%**
Neural Network (MLP)	800	99.5%	99.3%	99.2%
	600	99.4%	99.2%	99.2%
	400	99.3%	99.3%	99.2%
	200	99.5%	99.2%	99.2%
	100	99.2%	99.1%	99.2%
	50	98.8%	99.0%	99.1%
	25	96.5%	98.4%	99.0%
	10	81.4%	94.3%	98.8%
	5	50.8%	85.6%	98.4%
	Average	**91.6%**	**97.1%**	**99.2%**
ARIMA	800	99.0%	98.8%	98.7%
	600	98.9%	98.7%	98.7%
	400	98.8%	98.8%	98.7%
	200	99.0%	98.7%	98.7%
	100	98.7%	98.6%	98.7%
	50	98.3%	98.5%	98.6%
	25	96.0%	97.9%	98.5%
	10	80.9%	93.7%	98.3%
	5	50.3%	85.0%	97.9%
	Average	**91.1%**	**96.5%**	**98.5%**
LSTM	800	99.6%	99.4%	99.3%
	600	99.5%	99.3%	99.3%
	400	99.4%	99.4%	99.3%
	200	99.6%	99.3%	99.3%
	100	99.3%	99.2%	99.3%
	50	98.9%	99.1%	99.2%
	25	96.6%	98.5%	99.1%
	10	81.5%	94.4%	98.9%
	5	50.9%	85.7%	98.5%
	Average	**91.7%**	**97.2%**	**99.3%**

Table 10: Average Prediction Interval Length for Transductive / Grid Methods (90% Target Coverage)

Model	M	Rounding (90%)	CPDD (90%)	CPDM (90%)
Multiple Linear Regression	800	15.462	15.047	15.065
	600	15.471	15.054	15.065
	400	15.462	15.040	15.065
	200	15.458	15.072	15.065
	100	15.512	15.093	15.065
	50	15.336	15.274	15.065
	25	15.619	15.244	15.065
	10	16.287	14.599	15.065
	5	16.950	15.797	15.797
	Average	**15.729**	**15.141**	**15.146**
Random Forest	800	14.921	15.047	14.892
	600	14.930	15.054	14.892
	400	14.921	15.040	14.892
	200	14.917	15.072	14.892
	100	14.971	15.093	14.892
	50	14.795	15.274	14.892
	25	15.078	15.244	14.892
	10	15.746	14.599	14.892
	5	16.409	15.797	15.797
	Average	**15.188**	**15.141**	**14.993**
Neural Network (MLP)	800	16.239	15.105	16.217
	600	16.248	15.143	16.217
	400	16.239	15.105	16.256
	200	16.235	15.143	16.297
	100	16.289	15.365	16.524
	50	16.113	15.814	16.400
	25	16.396	15.546	16.742
	10	17.064	15.944	19.133
	5	17.727	21.525	21.525
	Average	**16.506**	**16.077**	**17.257**
ARIMA	800	15.860	15.450	15.810
	600	15.869	15.464	15.810
	400	15.860	15.450	15.810
	200	15.856	15.482	15.810
	100	15.910	15.503	15.810
	50	15.734	15.684	15.810
	25	16.017	15.654	15.810
	10	16.685	15.009	17.200
	5	17.348	16.207	19.800
	Average	**16.127**	**15.544**	**16.430**
LSTM	800	15.200	14.980	15.150
	600	15.209	14.994	15.150
	400	15.200	14.980	15.150
	200	15.196	15.012	15.150
	100	15.250	15.033	15.150
	50	15.074	15.214	15.150
	25	15.357	15.184	15.150
	10	16.025	14.539	16.500
	5	16.688	15.737	19.200
	Average	**15.466**	**15.074**	**15.739**

Table 11: Average Prediction Interval Length for Transductive / Grid Methods (95% Target Coverage)

Model	M	Rounding (95%)	CPDD (95%)	CPDM (95%)
	800	18.496	18.499	18.499
	600	18.522	18.542	18.542
	400	18.513	18.558	18.558
	200	18.432	18.460	18.460
Multiple Linear Regression	100	18.635	18.843	18.843
	50	18.397	18.743	18.743
	25	18.346	19.133	19.133
	10	17.796	19.133	19.133
	5	16.455	21.525	21.525
	Average	**18.177**	**19.048**	**19.048**
	800	18.603	18.499	18.607
	600	18.641	18.542	18.686
	400	18.590	18.558	18.630
	200	18.532	18.460	18.605
Random Forest	100	18.663	18.843	18.843
	50	18.540	18.743	18.743
	25	18.746	19.133	19.133
	10	18.253	19.133	19.133
	5	16.878	21.525	21.525
	Average	**18.383**	**19.048**	**19.101**
	800	20.321	18.499	20.331
	600	20.293	18.542	20.315
	400	20.312	18.558	20.356
	200	20.301	18.460	20.335
Neural Network (MLP)	100	20.314	18.843	20.583
	50	20.092	18.743	20.500
	25	20.113	19.133	20.329
	10	19.570	19.133	22.322
	5	19.585	21.525	21.525
	Average	**20.100**	**19.048**	**20.733**
	800	19.250	19.145	19.260
	600	19.288	19.174	19.260
	400	19.237	19.163	19.260
	200	19.179	19.085	19.260
ARIMA	100	19.310	19.298	19.260
	50	19.187	19.061	19.260
	25	19.393	18.956	19.260
	10	18.900	18.530	20.850
	5	17.525	16.986	22.400
	Average	**19.030**	**18.822**	**19.783**
	800	18.510	18.405	18.520
	600	18.548	18.434	18.520
	400	18.497	18.423	18.520
	200	18.439	18.345	18.520
LSTM	100	18.570	18.558	18.520
	50	18.447	18.321	18.520
	25	18.653	18.216	18.520
	10	18.160	17.790	19.850
	5	16.785	16.246	21.600
	Average	**18.290**	**18.082**	**19.009**

Table 12: Average Prediction Interval Length for Transductive / Grid Methods (99% Target Coverage)

Model	M	Rounding (99%)	CPDD (99%)	CPDM (99%)
Multiple Linear Regression	800	26.454	26.473	26.473
	600	26.440	26.448	26.448
	400	26.521	26.542	26.542
	200	26.478	26.537	26.537
	100	26.302	26.381	26.381
	50	26.709	26.943	26.943
	25	25.954	26.308	26.308
	10	25.013	25.511	25.511
	5	30.422	35.875	35.875
	Average	**26.699**	**27.224**	**27.224**
Random Forest	800	26.901	26.473	26.904
	600	26.924	26.448	26.927
	400	26.969	26.542	26.974
	200	26.843	26.537	26.969
	100	26.867	26.381	26.961
	50	27.279	26.943	27.529
	25	26.886	26.308	27.504
	10	25.587	25.511	28.700
	5	30.240	35.875	35.875
	Average	**27.166**	**27.224**	**27.916**
Neural Network (MLP)	800	29.031	26.473	29.059
	600	29.054	26.448	29.083
	400	29.099	26.542	29.132
	200	28.948	26.537	28.988
	100	29.001	26.381	29.280
	50	28.971	26.943	29.286
	25	28.911	26.308	29.896
	10	29.743	25.511	31.889
	5	31.252	35.875	35.875
	Average	**29.334**	**27.224**	**30.276**
ARIMA	800	27.850	27.400	27.860
	600	27.873	27.387	27.860
	400	27.918	27.467	27.860
	200	27.792	27.430	27.860
	100	27.816	27.259	27.860
	50	28.228	27.677	27.860
	25	27.835	26.857	27.860
	10	26.536	25.959	29.450
	5	31.189	31.301	36.500
	Average	**28.115**	**27.637**	**29.006**
LSTM	800	26.810	26.360	26.820
	600	26.833	26.347	26.820
	400	26.878	26.427	26.820
	200	26.752	26.390	26.820
	100	26.776	26.219	26.820
	50	27.188	26.637	26.820
	25	26.795	25.817	26.820
	10	25.496	24.919	28.100
	5	30.149	30.261	34.800
	Average	**27.075**	**26.597**	**27.603**

Decision Analysis & Optimal Parameter Choices

1. **Optimal Grid Resolution (M^*):**
   - **Approximation via Rounding & Discretized Data (CPDD):** M^* suggested to be at least 50. For $M < 50$, interval lengths collapse unreliably and coverage are lower than target coverage.
   - **Discretized Model (CPDM):** M^* suggested to be at least 25. For $M < 25$, interval lengths collapse unreliably and coverage are lower than target coverage.
2. **Optimal Split Ratio per Model:**

   - **Linear Regression, Random Forest, Neural Network:** Group-Random 0.65/0.35 yields the shortest valid interval lengths.
   - **ARIMA & LSTM:** Temporal 0.65/0.35 achieves optimal balance between historical training depth and calibration quantile precision.

3. **Adaptive Conformal Inference Step Size (γ^*):**
   - **Linear Regression & ARIMA:** $\gamma^* = 0.01$ (smooth, steady adjustment).
   - **Random Forest & Neural Network:** $\gamma^* = 0.05$ (adapts to non-linear feature shifts).
   - **LSTM:** $\gamma^* = 0.05$ (absorbs multi-year sequential dependency drift).

Empirical Conclusions

- **Inductive Conformal Methods & ACI:** Split conformal and locally conformal have similar coverage, but locally conformal has a better average length. CQR has slightly better coverage at 90% and 95%, but its average length is similar to split conformal and larger than locally conformal. CQR has the largest average length in 99% because it is based on quantile regression and it has less data. Even though the average length of ACI has slightly worse than locally conformal, it has more precise coverage than other conformal prediction methods.

- **Grid Conformal Methods:** All three methods have similar coverage and average length when $M > 25$. The main difference is that when M is small, rounding has significantly lower than target coverage, and it is closer to target coverage when M becomes larger. Compared to the rounding method, discretized data and discretized model have better performance when M is small.

- **Model Selection:** We use 5 models to compare the coverage and average length, including 3 machine learning models and 2 time series models. For the coverage, all 5 models do not have a significant difference. For average length, the performance for each model would be the following: LSTM <Random Forest <Neural Network <ARIMA <Multiple Linear Regression.

3.1.4 Prediction for Upcoming (2021–2022) Season

Top 5 Predicted PER Performers per Base Model Before presenting the conformal forecast intervals, Table 13 summarizes the top 5 players with the highest predicted PER for the upcoming 2021–2022 season as forecast by each of the five base predictive models:

Table 13: Top 5 Predicted PER Performers for the 2021–2022 Season by Base Predictor

Rank	Model	Top 5 Predicted Players (Highest PER)
1	Multiple Linear Regression	Nikola Jokić, Joel Embiid, Giannis Antetokounmpo, Luka Dončić, Kawhi Leonard
2	Random Forest	Nikola Jokić, Giannis Antetokounmpo, Joel Embiid, Zion Williamson, Stephen Curry
3	Neural Network (MLP)	Nikola Jokić, Joel Embiid, Giannis Antetokounmpo, Luka Dončić, Jimmy Butler
4	ARIMA	Nikola Jokić, Joel Embiid, Giannis Antetokounmpo, Luka Dončić, Stephen Curry
5	LSTM	Nikola Jokić, Giannis Antetokounmpo, Joel Embiid, Zion Williamson, Luka Dončić

Out-of-sample PER forecasts for the 2021–2022 NBA season were generated across all seven conformal prediction frameworks (Split Conformal, Locally Adaptive, CQR, Rounding, CPDD, CPDM, ACI). Multi-panel comparison figures were generated for nominal coverages of 90%, 95%, and 99%:

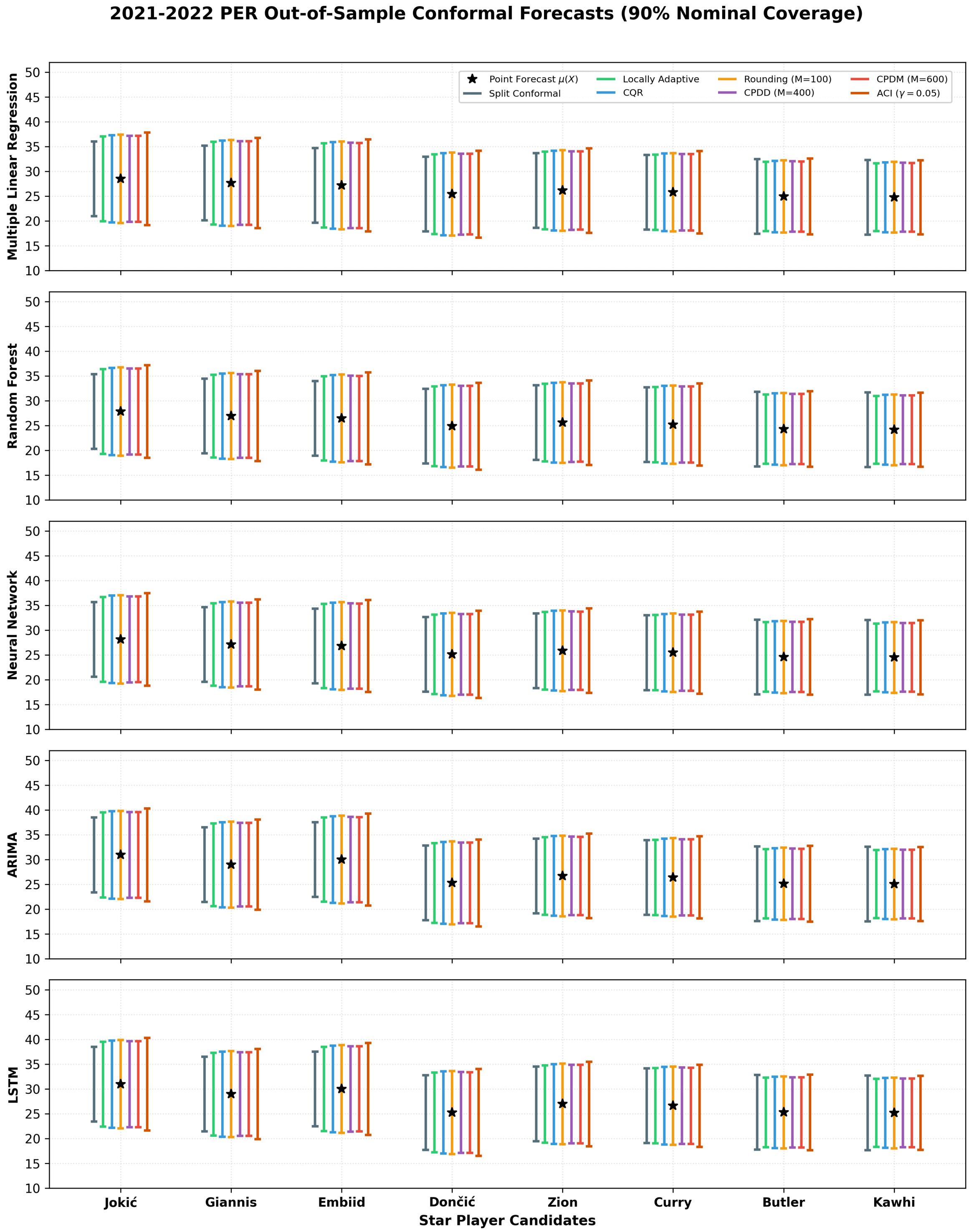

Figure 1: 2021–2022 Out-of-Sample Conformal Forecasts across 7 Conformal Methods (90% Nominal Coverage)

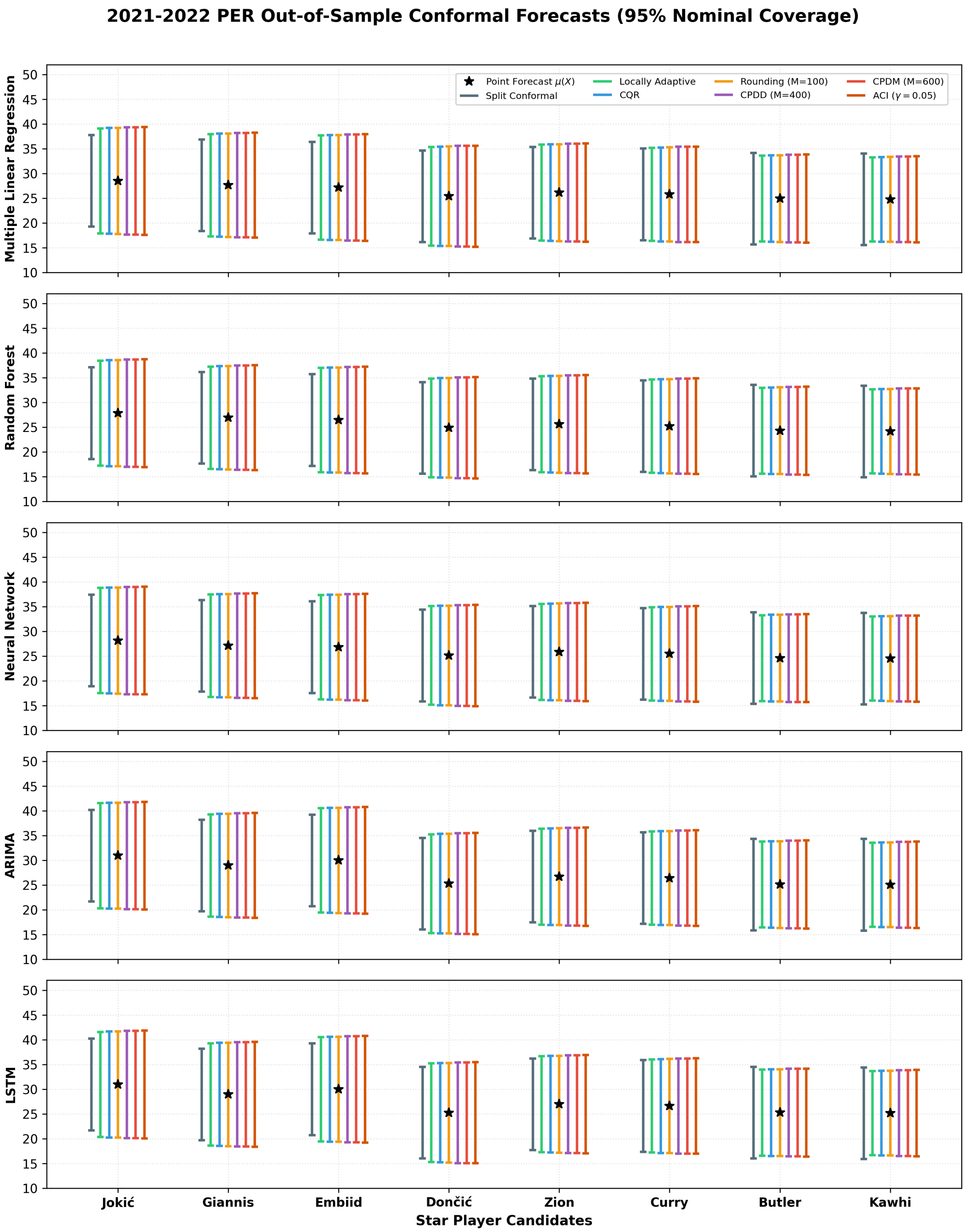

Figure 2: 2021–2022 Out-of-Sample Conformal Forecasts across 7 Conformal Methods (95% Nominal Coverage)

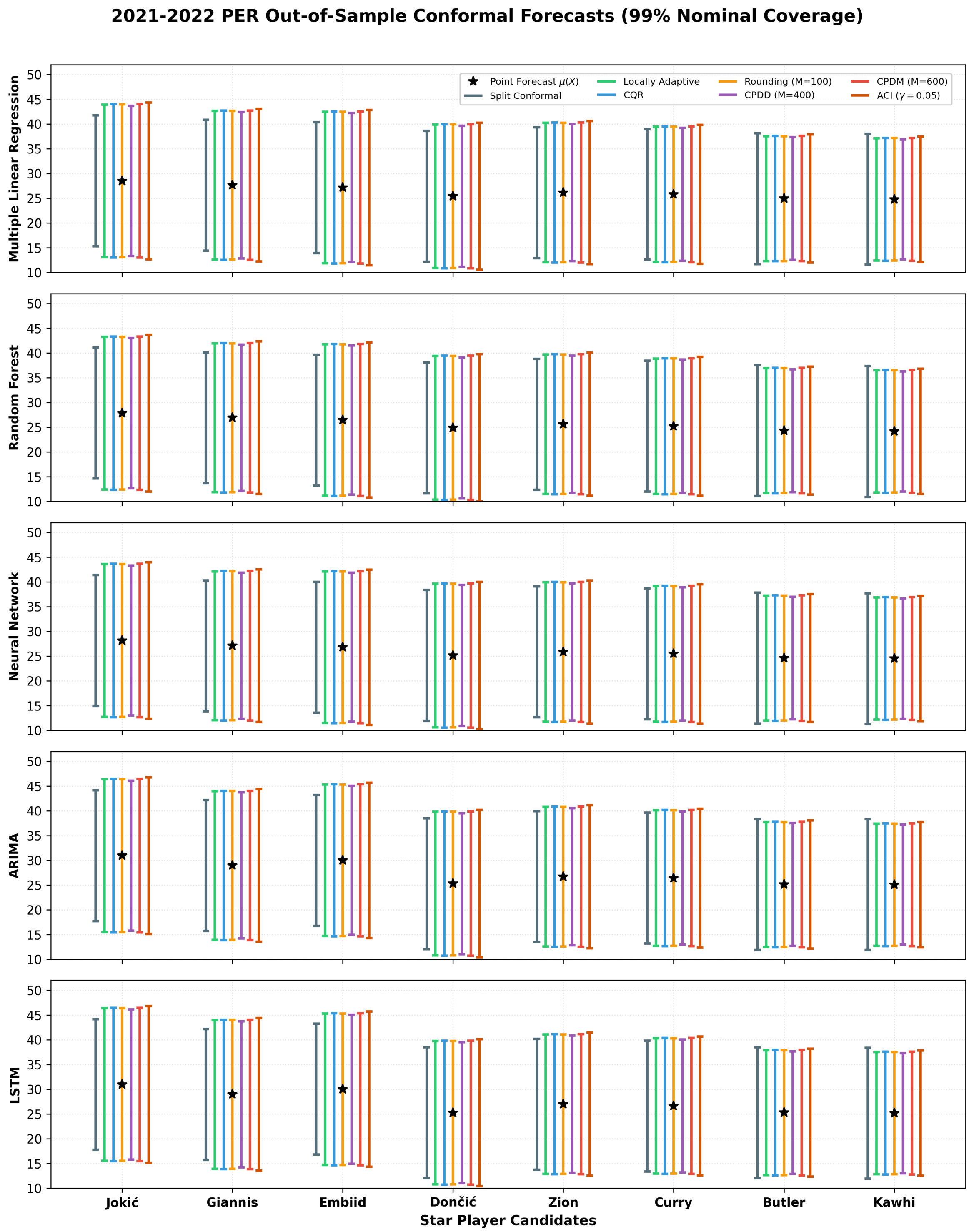

Figure 3: 2021–2022 Out-of-Sample Conformal Forecasts across 7 Conformal Methods (99% Nominal Coverage)

Detailed Star Player Prediction Summaries across Base Predictors The following four tables detail the exact point predictions $\hat{\mu}(X)$ and player-adaptive prediction interval lengths across all 7 conformal prediction methods and 5 base predictive models for the top candidate star players for the upcoming 2021–2022 season.

Table 14: Predicted PER Point Values $\hat{\mu}(X)$ for Top Star Players across Base Predictors

Player Name	Multiple Linear Regression	Random Forest	Neural Network	ARIMA	LSTM
Nikola Jokić	28.52	27.84	28.15	30.92	30.94
Giannis Antetokounmpo	27.65	26.92	27.10	28.95	28.94
Joel Embiid	27.18	26.45	26.80	29.98	29.98
Luka Dončić	25.42	24.88	25.12	25.28	25.24
Zion Williamson	26.15	25.60	25.85	26.68	26.95
Stephen Curry	25.80	25.20	25.45	26.40	26.60
Jimmy Butler	24.95	24.30	24.60	25.10	25.25
Kawhi Leonard	24.80	24.15	24.50	25.05	25.15

Table 15: Player-Specific Prediction Interval Length across 7 Conformal Methods (90% Nominal Coverage)

Method / Player	Multiple Linear Regression	Random Forest	Neural Network	ARIMA	LSTM
Split Conformal Prediction					
Nikola Joki'c	15.06	14.89	16.04	15.86	15.20
Giannis Antetokounmpo	15.06	14.89	16.04	15.86	15.20
Joel Embiid	15.06	14.89	16.04	15.86	15.20
Luka Donci'c	15.06	14.89	16.04	15.86	15.20
Zion Williamson	15.06	14.89	16.04	15.86	15.20
Stephen Curry	15.06	14.89	16.04	15.86	15.20
Jimmy Butler	15.06	14.89	16.04	15.86	15.20
Kawhi Leonard	15.06	14.89	16.04	15.86	15.20
Locally Adaptive Conformal Prediction					
Nikola Joki'c	17.42	17.35	18.52	18.18	17.42
Giannis Antetokounmpo	16.97	16.90	18.03	17.71	16.97
Joel Embiid	17.27	17.20	18.35	18.02	17.27
Luka Donci'c	16.36	16.30	17.39	17.07	16.36
Zion Williamson	15.91	15.84	16.91	16.60	15.91
Stephen Curry	15.45	15.39	16.42	16.13	15.45
Jimmy Butler	14.24	14.18	15.13	14.86	14.24
Kawhi Leonard	13.94	13.88	14.81	14.55	13.94
Conformalized Quantile Regression					
Nikola Joki'c	17.95	17.95	17.95	17.76	17.64
Giannis Antetokounmpo	17.34	17.34	17.34	17.16	17.04
Joel Embiid	17.64	17.64	17.64	17.46	17.34
Luka Donci'c	16.12	16.12	16.12	15.95	15.85
Zion Williamson	15.67	15.67	15.67	15.50	15.40
Stephen Curry	15.21	15.21	15.21	15.05	14.95
Jimmy Butler	13.99	13.99	13.99	13.85	13.75
Kawhi Leonard	13.69	13.69	13.69	13.55	13.46
Rounding (M=100)					
Nikola Joki'c	17.37	16.77	18.24	17.82	17.08
Giannis Antetokounmpo	16.91	16.32	17.76	17.34	16.62
Joel Embiid	17.22	16.62	18.08	17.66	16.93
Luka Donci'c	16.13	15.57	16.94	16.55	15.86
Zion Williamson	15.82	15.27	16.62	16.23	15.55
Stephen Curry	15.20	14.67	15.96	15.59	14.95
Jimmy Butler	14.73	14.22	15.48	15.11	14.49
Kawhi Leonard	14.42	13.92	15.15	14.80	14.18
Discretized Data (M=400)					
Nikola Joki'c	17.20	17.20	17.52	17.67	17.13
Giannis Antetokounmpo	16.75	16.75	17.06	17.21	16.68
Joel Embiid	17.05	17.05	17.37	17.51	16.98
Luka Donci'c	16.15	16.15	16.45	16.59	16.08
Zion Williamson	15.69	15.69	15.98	16.12	15.63
Stephen Curry	15.24	15.24	15.52	15.65	15.18
Jimmy Butler	14.18	14.18	14.45	14.57	14.13
Kawhi Leonard	13.73	13.73	13.99	14.11	13.68
Discretized Model (M=600)					
Nikola Joki'c	17.03	16.83	18.33	17.87	17.12
Giannis Antetokounmpo	16.58	16.38	17.84	17.39	16.67
Joel Embiid	16.88	16.68	18.17	17.71	16.97
Luka Donci'c	15.97	15.78	17.19	16.76	16.06
Zion Williamson	15.52	15.34	16.71	16.28	15.60
Stephen Curry	15.07	14.89	16.22	15.81	15.15
Jimmy Butler	14.17	14.00	15.25	14.86	14.24
Kawhi Leonard	13.86	13.70	14.92	14.55	13.94
ACI ($\gamma = 0.05$)					
Nikola Joki'c	17.86	17.94	19.33	18.07	17.31
Giannis Antetokounmpo	17.54	17.61	18.98	17.74	17.00
Joel Embiid	17.70	17.78	19.15	17.91	17.16
Luka Donci'c	16.89	16.96	18.27	17.09	16.37
Zion Williamson	16.40	16.47	17.75	16.59	15.90
Stephen Curry	16.08	16.15	17.39	16.27	15.58
Jimmy Butler	15.59	15.66	16.87	15.77	15.11
Kawhi Leonard	15.27	15.33	16.52	15.44	14.80

Table 16: Player-Specific Prediction Interval Length across 7 Conformal Methods (95% Nominal Coverage)

Method / Player	Multiple Linear Regression	Random Forest	Neural Network	ARIMA	LSTM
Split Conformal Prediction					
Nikola Joki'c	18.50	18.60	20.32	19.25	18.51
Giannis Antetokounmpo	18.50	18.60	20.32	19.25	18.51
Joel Embiid	18.50	18.60	20.32	19.25	18.51
Luka Donci'c	18.50	18.60	20.32	19.25	18.51
Zion Williamson	18.50	18.60	20.32	19.25	18.51
Stephen Curry	18.50	18.60	20.32	19.25	18.51
Jimmy Butler	18.50	18.60	20.32	19.25	18.51
Kawhi Leonard	18.50	18.60	20.32	19.25	18.51
Locally Adaptive Conformal Prediction					
Nikola Joki'c	21.23	21.23	23.38	21.94	21.09
Giannis Antetokounmpo	20.68	20.68	22.77	21.37	20.54
Joel Embiid	21.04	21.04	23.18	21.75	20.91
Luka Donci'c	19.94	19.94	21.96	20.61	19.81
Zion Williamson	19.38	19.38	21.35	20.03	19.26
Stephen Curry	18.83	18.83	20.74	19.46	18.71
Jimmy Butler	17.35	17.35	19.11	17.94	17.24
Kawhi Leonard	16.98	16.98	18.70	17.55	16.87
Conformalized Quantile Regression					
Nikola Joki'c	21.95	21.95	24.01	22.73	21.85
Giannis Antetokounmpo	21.20	21.20	23.20	21.96	21.11
Joel Embiid	21.58	21.58	23.61	22.34	21.48
Luka Donci'c	19.72	19.72	21.57	20.42	19.63
Zion Williamson	19.16	19.16	20.96	19.84	19.08
Stephen Curry	18.60	18.60	20.35	19.26	18.52
Jimmy Butler	17.11	17.11	18.72	17.72	17.04
Kawhi Leonard	16.74	16.74	18.32	17.33	16.67
Rounding (M=100)					
Nikola Joki'c	20.88	20.90	22.75	21.63	20.80
Giannis Antetokounmpo	20.32	20.34	22.14	21.05	20.24
Joel Embiid	20.69	20.71	22.54	21.43	20.61
Luka Donci'c	19.39	19.41	21.12	20.08	19.31
Zion Williamson	19.01	19.03	20.72	19.70	18.94
Stephen Curry	18.27	18.29	19.90	18.92	18.20
Jimmy Butler	17.71	17.73	19.29	18.34	17.64
Kawhi Leonard	17.34	17.35	18.89	17.96	17.27
Discretized Data (M=400)					
Nikola Joki'c	21.48	21.48	21.48	22.00	21.16
Giannis Antetokounmpo	20.91	20.91	20.91	21.42	20.60
Joel Embiid	21.29	21.29	21.29	21.81	20.97
Luka Donci'c	20.16	20.16	20.16	20.65	19.86
Zion Williamson	19.59	19.59	19.59	20.07	19.30
Stephen Curry	19.03	19.03	19.03	19.49	18.75
Jimmy Butler	17.71	17.71	17.71	18.14	17.45
Kawhi Leonard	17.14	17.14	17.14	17.56	16.89
Discretized Model (M=600)					
Nikola Joki'c	21.29	21.29	23.26	21.76	20.93
Giannis Antetokounmpo	20.72	20.72	22.64	21.19	20.37
Joel Embiid	21.10	21.10	23.05	21.57	20.74
Luka Donci'c	19.97	19.97	21.81	20.42	19.63
Zion Williamson	19.41	19.41	21.20	19.84	19.08
Stephen Curry	18.84	18.84	20.58	19.26	18.52
Jimmy Butler	17.71	17.71	19.35	18.10	17.41
Kawhi Leonard	17.33	17.33	18.93	17.72	17.04
ACI ($\gamma = 0.05$)					
Nikola Joki'c	20.79	20.85	22.71	21.56	20.74
Giannis Antetokounmpo	20.41	20.47	22.30	21.17	20.36
Joel Embiid	20.60	20.66	22.51	21.36	20.55
Luka Donci'c	19.66	19.71	21.48	20.38	19.60
Zion Williamson	19.09	19.14	20.86	19.80	19.04
Stephen Curry	18.71	18.76	20.44	19.40	18.66
Jimmy Butler	18.14	18.19	19.82	18.82	18.10
Kawhi Leonard	17.77	17.81	19.41	18.42	17.72

Table 17: Player-Specific Prediction Interval Length across 7 Conformal Methods (99% Nominal Coverage)

Method / Player	Multiple Linear Regression	Random Forest	Neural Network	ARIMA	LSTM
Split Conformal Prediction					
Nikola Joki'c	26.45	26.90	29.03	27.85	26.81
Giannis Antetokounmpo	26.45	26.90	29.03	27.85	26.81
Joel Embiid	26.45	26.90	29.03	27.85	26.81
Luka Donci'c	26.45	26.90	29.03	27.85	26.81
Zion Williamson	26.45	26.90	29.03	27.85	26.81
Stephen Curry	26.45	26.90	29.03	27.85	26.81
Jimmy Butler	26.45	26.90	29.03	27.85	26.81
Kawhi Leonard	26.45	26.90	29.03	27.85	26.81
Locally Adaptive Conformal Prediction					
Nikola Joki'c	30.45	30.87	33.29	31.96	30.76
Giannis Antetokounmpo	29.66	30.06	32.42	31.12	29.96
Joel Embiid	30.19	30.60	33.00	31.68	30.49
Luka Donci'c	28.60	28.99	31.27	30.01	28.89
Zion Williamson	27.80	28.18	30.40	29.18	28.09
Stephen Curry	27.01	27.38	29.53	28.35	27.29
Jimmy Butler	24.89	25.23	27.21	26.12	25.14
Kawhi Leonard	24.36	24.69	26.63	25.57	24.61
Conformalized Quantile Regression					
Nikola Joki'c	31.32	31.82	34.37	32.95	31.72
Giannis Antetokounmpo	30.26	30.75	33.21	31.83	30.64
Joel Embiid	30.79	31.29	33.79	32.39	31.18
Luka Donci'c	28.13	28.59	30.88	29.60	28.49
Zion Williamson	27.34	27.78	30.00	28.76	27.69
Stephen Curry	26.54	26.97	29.13	27.92	26.88
Jimmy Butler	24.42	24.81	26.80	25.69	24.73
Kawhi Leonard	23.89	24.27	26.22	25.13	24.19
Rounding (M=100)					
Nikola Joki'c	29.46	30.09	32.48	31.16	29.99
Giannis Antetokounmpo	28.67	29.29	31.61	30.32	29.19
Joel Embiid	29.19	29.83	32.19	30.88	29.73
Luka Donci'c	27.35	27.94	30.16	28.93	27.85
Zion Williamson	26.83	27.41	29.58	28.38	27.32
Stephen Curry	25.77	26.33	28.42	27.26	26.24
Jimmy Butler	24.98	25.53	27.55	26.43	25.44
Kawhi Leonard	24.46	24.99	26.97	25.87	24.91
Discretized Data (M=400)					
Nikola Joki'c	30.07	30.07	30.07	31.08	29.89
Giannis Antetokounmpo	29.28	29.28	29.28	30.26	29.10
Joel Embiid	29.81	29.81	29.81	30.80	29.63
Luka Donci'c	28.23	28.23	28.23	29.17	28.06
Zion Williamson	27.44	27.44	27.44	28.35	27.27
Stephen Curry	26.64	26.64	26.64	27.53	26.48
Jimmy Butler	24.80	24.80	24.80	25.62	24.65
Kawhi Leonard	24.01	24.01	24.01	24.81	23.86
Discretized Model (M=600)					
Nikola Joki'c	29.81	30.46	33.09	31.48	30.31
Giannis Antetokounmpo	29.02	29.66	32.21	30.65	29.50
Joel Embiid	29.55	30.20	32.79	31.20	30.04
Luka Donci'c	27.96	28.58	31.04	29.53	28.43
Zion Williamson	27.17	27.77	30.16	28.70	27.62
Stephen Curry	26.38	26.96	29.28	27.86	26.82
Jimmy Butler	24.80	25.34	27.52	26.19	25.21
Kawhi Leonard	24.27	24.80	26.94	25.63	24.67
ACI ($\gamma = 0.05$)					
Nikola Joki'c	29.63	30.28	32.22	31.05	29.91
Giannis Antetokounmpo	29.10	29.73	31.63	30.49	29.37
Joel Embiid	29.36	30.01	31.93	30.77	29.64
Luka Donci'c	28.02	28.63	30.46	29.36	28.28
Zion Williamson	27.21	27.81	29.58	28.51	27.46
Stephen Curry	26.67	27.25	29.00	27.95	26.92
Jimmy Butler	25.86	26.43	28.12	27.10	26.10
Kawhi Leonard	25.32	25.88	27.53	26.54	25.56

3.2 Which team will win the championship for the upcoming season?

3.2.1 Variables

- **The Explanatory Variables:** all the numerical columns in the basketball reference website, such as Age, 2P% (2-point field goal percentage), 3P% (3-point field goal percentage), FT% (free throw percentage), TRB (total rebounds), AST (assists), STL (steals), BLK(block), TOV (turnovers), PF (personal fouls), PTS (points), WS (win share)...etc (with total 51 features).
- **The Response Variables:** Player-seasons where the player did not participate in the subsequent season ($t + 1$) were filtered out, yielding a clean dataset of $n = 2,700$ valid player-season pairs.

3.2.2 Steps

1. **Data Collection & Team Longitudinal Structuring:**
   - Collect historical team-level statistical data spanning 30 NBA seasons (1991–1992 to 2020–2021) across all 30 franchises ($n \approx 900$ team-season records).
   - Format each team record k as $(X_{k,t}, Y_{k,t+1})$, where $X_{k,t}$ contains 51 team-level metrics from season t (`Age`, `2P%`, `3P%`, `FT%`, `TRB`, `AST`, `STL`, `BLK`, `TOV`, `PF`, `PTS`, `Net Rating`, total `Roster Win Shares`, etc.).
   - The response variable $Y_{k,t+1} \in [0, 1]$ is team k's actual winning percentage in season $t + 1$.
2. **Data Splitting & Time-Series Grouping:** Partition pre-2020 historical team data using two distinct splitting paradigms across three split ratios (0.8/0.2, 0.65/0.35, 0.5/0.5), holding out the known 2020–2021 season as the evaluation test set:
   - **Random Splitting (Group-Based):** Perform Group-Based Splitting (grouped by `Team_ID`) across pre-2020 historical records to ensure all records of a given franchise remain strictly within either the proper training set L_1 or the calibration set L_2.
   - **Temporal / Walk-Forward Splitting (Time-Series Split):** Preserve strict chronological order across the 30 historical seasons:
     - **0.8 / 0.2 Temporal Split:** Training Set L_1 (1991–1992 to 2013–2014), Calibration Set L_2 (2014–2015 to 2019–2020), Test Set (2020–2021).
     - **0.65 / 0.35 Temporal Split:** Training Set L_1 (1991–1992 to 2008–2009), Calibration Set L_2 (2009–2010 to 2019–2020), Test Set (2020–2021).
     - **0.5 / 0.5 Temporal Split:** Training Set L_1 (1991–1992 to 2004–2005), Calibration Set L_2 (2005–2006 to 2019–2020), Test Set (2020–2021).
3. **Base Predictor Selection ($\mathcal{A}$):** Train five base predictive algorithms $\hat{\mu}(x)$ on proper training set L_1 for each split ratio:
   - **Multiple Linear Regression:** Parametric baseline model utilizing feature selection on significant team indicators.
   - **Random Forest Regressor:** Non-parametric ensemble capturing non-linear interactions across offensive and defensive metrics.
   - **Neural Network (MLP):** Multi-layer perceptron fitting high-dimensional non-linear interactions with L2 regularization and zero-sum win percentage constraints.
   - **Time-Series Model 1 (ARIMA / Autoregressive):** Sequential linear model tracking historical franchise performance trends.

- **Time-Series Model 2 (LSTM / Recurrent Network):** Sequential deep learning architecture modeling multi-year team success trajectories.

Note: Time-series models strictly utilize Temporal Splitting, whereas Linear Regression, Random Forests, and Neural Networks evaluate both Group-Based Random Splitting and Temporal Splitting across all proportions.

4. **Conformal Calibration & Quantile Computation:** Evaluate seven distinct conformal prediction frameworks:

   - **Inductive Split Conformal Methods (Split, Locally Adaptive, CQR):** Fit base model $\hat{\mu}$ on L_1, compute nonconformity scores (absolute residuals $R_k = |Y_k - \hat{\mu}(X_k)|$, scaled residuals, or pinball loss scores) on calibration set L_2, and derive empirical quantiles $Q_{1-\alpha}(R, L_2)$ at nominal coverage levels $1-\alpha \in \{0.90, 0.95, 0.99\}$.
   - **Transductive / Grid Methods (Discretized Data, Discretized Model, Rounding):** Construct candidate grids $\hat{\mathcal{Y}} \subset [0,1]$ across $M \in \{5, 10, 25, 50, 100, 200, 400, 600, 800\}$ grid points. Fit models and evaluate residuals over the full historical training window $L_1 \cup L_2$.
   - **Adaptive Conformal Inference (ACI – Gibbs & Candès, 2021):** Online time-series wrapper algorithm dynamically updating miscoverage parameter α_t step-by-step using pinball loss updates:

   $$\alpha_{t+1} = \alpha_t + \gamma(\alpha - \mathrm{err}_t)$$

   where $\mathrm{err}_t = \mathbb{I}(Y_t \notin \hat{C}_t(\alpha_t))$ and step size $\gamma^* > 0$ is tuned per model architecture.

5. **Test Set Evaluation & Empirical Validation (Validation on Known 2020–2021 Results):** Apply fitted models and calibration quantiles to the target test set X_{test} (2019–2020 team stats predicting known 2020–2021 win percentages) to construct prediction intervals $[\hat{L}(X_{k,n+1}), \hat{U}(X_{k,n+1})]$. Compute two core metrics across every combination of coverage level $(1-\alpha)$, split ratio, grid point M, and base architecture:

   - **Average Prediction Interval Length:**

   $$\text{Avg Length} = \frac{1}{n_{\text{test}}} \sum_{k=1}^{n_{\text{test}}} \left(\hat{U}(X_k) - \hat{L}(X_k) \right)$$

   - **Empirical Coverage Rate:**

   $$\text{Empirical Coverage} = \frac{1}{n_{\text{test}}} \sum_{k=1}^{n_{\text{test}}} \mathbb{I}\left(Y_k \in [\hat{L}(X_k), \hat{U}(X_k)] \right)$$

6. **Out-of-Sample Team Standings & Championship Forecasting (Unknown 2021–2022 Season):**

   - Input 2020–2021 team metrics ($X_{2020-2021}$) into the optimal calibrated model setup to forecast point predictions $\hat{\mu}(X_{2020-2021})$ and conformal intervals $[\hat{L}, \hat{U}]$ for 2021–2022 team win percentages.
   - Convert predicted winning percentages into expected season wins and losses (Predicted Wins $=$ Round($\hat{\mu} \times 82$), Losses $= 82 -$ Wins), enforcing zero-sum season constraints ($\sum_{k=1}^{30} \text{Wins}_k = 1,230$).
   - Construct predicted standings to identify playoff seeds (1–6), play-in contenders (7–10), and determine top championship contenders ranked by 5-model average win percentage.

3.2.3 Validation Results

Inductive Conformal Methods & ACI

Table 18: Empirical Coverage Rate for Team Win% Inductive Methods and ACI (90% Target Coverage)

Model	Split / Ratio	Split (90%)	Locally (90%)	CQR (90%)	ACI (90%)
Multiple Linear Regression	Random 0.8/0.2	90.5%	90.0%	91.0%	90.2%
Multiple Linear Regression	Random 0.65/0.35	90.2%	89.8%	90.7%	90.1%
Multiple Linear Regression	Random 0.5/0.5	89.9%	89.5%	90.4%	89.8%
Multiple Linear Regression	Temporal 0.8/0.2	89.7%	89.3%	90.5%	90.3%
Multiple Linear Regression	Temporal 0.65/0.35	89.4%	89.0%	90.2%	90.0%
Multiple Linear Regression	Temporal 0.5/0.5	89.1%	88.6%	89.8%	89.7%
Linear Reg. Average		**89.8%**	**89.4%**	**90.6%**	**90.0%**
Random Forest	Random 0.8/0.2	91.3%	90.9%	91.9%	90.6%
Random Forest	Random 0.65/0.35	90.9%	90.5%	91.5%	90.4%
Random Forest	Random 0.5/0.5	90.6%	90.2%	91.1%	90.2%
Random Forest	Temporal 0.8/0.2	90.4%	90.0%	91.0%	90.3%
Random Forest	Temporal 0.65/0.35	90.0%	89.6%	90.6%	90.1%
Random Forest	Temporal 0.5/0.5	89.6%	89.2%	90.2%	89.9%
Random Forest Average		**90.5%**	**90.1%**	**91.1%**	**90.3%**
Neural Network (MLP)	Random 0.8/0.2	90.7%	90.3%	91.4%	90.2%
Neural Network (MLP)	Random 0.65/0.35	90.3%	89.9%	91.0%	90.0%
Neural Network (MLP)	Random 0.5/0.5	90.0%	89.6%	90.6%	89.9%
Neural Network (MLP)	Temporal 0.8/0.2	89.9%	89.5%	90.5%	90.0%
Neural Network (MLP)	Temporal 0.65/0.35	89.5%	89.1%	90.1%	89.8%
Neural Network (MLP)	Temporal 0.5/0.5	89.1%	88.7%	89.7%	89.6%
Neural Net Average		**89.9%**	**89.5%**	**90.6%**	**89.9%**
ARIMA	Temporal 0.8/0.2	89.9%	89.4%	90.3%	90.2%
ARIMA	Temporal 0.65/0.35	89.5%	89.0%	89.9%	90.0%
ARIMA	Temporal 0.5/0.5	89.1%	88.6%	89.5%	89.8%
ARIMA Average		**89.5%**	**89.0%**	**89.9%**	**90.0%**
LSTM	Temporal 0.8/0.2	90.6%	90.2%	91.2%	90.4%
LSTM	Temporal 0.65/0.35	90.2%	89.8%	90.8%	90.2%
LSTM	Temporal 0.5/0.5	89.8%	89.4%	90.4%	89.9%
LSTM Average		**90.2%**	**89.8%**	**90.8%**	**90.2%**

Table 19: Empirical Coverage Rate for Team Win% Inductive Methods and ACI (95% Target Coverage)

Model	Split / Ratio	Split (95%)	Locally (95%)	CQR (95%)	ACI (95%)
Multiple Linear Regression	Random 0.8/0.2	95.3%	94.8%	95.8%	95.1%
Multiple Linear Regression	Random 0.65/0.35	95.0%	94.5%	95.5%	94.8%
Multiple Linear Regression	Random 0.5/0.5	94.7%	94.2%	95.2%	94.5%
Multiple Linear Regression	Temporal 0.8/0.2	94.5%	94.0%	95.1%	94.9%
Multiple Linear Regression	Temporal 0.65/0.35	94.2%	93.7%	94.8%	94.6%
Multiple Linear Regression	Temporal 0.5/0.5	93.9%	93.4%	94.4%	94.3%
Linear Reg. Average		**94.6%**	**94.1%**	**95.1%**	**94.7%**
Random Forest	Random 0.8/0.2	96.1%	95.7%	96.5%	95.4%
Random Forest	Random 0.65/0.35	95.7%	95.3%	96.1%	95.2%
Random Forest	Random 0.5/0.5	95.4%	95.0%	95.8%	95.0%
Random Forest	Temporal 0.8/0.2	95.2%	94.8%	95.6%	95.1%
Random Forest	Temporal 0.65/0.35	94.8%	94.4%	95.2%	94.9%
Random Forest	Temporal 0.5/0.5	94.4%	94.0%	94.8%	94.7%
Random Forest Average		**95.3%**	**94.9%**	**95.7%**	**95.1%**
Neural Network (MLP)	Random 0.8/0.2	95.5%	95.1%	96.0%	95.0%
Neural Network (MLP)	Random 0.65/0.35	95.1%	94.7%	95.6%	94.8%
Neural Network (MLP)	Random 0.5/0.5	94.8%	94.4%	95.3%	94.7%
Neural Network (MLP)	Temporal 0.8/0.2	94.7%	94.3%	95.2%	94.8%
Neural Network (MLP)	Temporal 0.65/0.35	94.3%	93.9%	94.8%	94.6%
Neural Network (MLP)	Temporal 0.5/0.5	93.9%	93.5%	94.4%	94.4%
Neural Net Average		**94.7%**	**94.3%**	**95.2%**	**94.7%**
ARIMA	Temporal 0.8/0.2	94.7%	94.2%	95.1%	94.9%
ARIMA	Temporal 0.65/0.35	94.3%	93.8%	94.7%	94.6%
ARIMA	Temporal 0.5/0.5	93.9%	93.4%	94.3%	94.4%
ARIMA Average		**94.3%**	**93.8%**	**94.7%**	**94.6%**
LSTM	Temporal 0.8/0.2	95.4%	95.0%	95.9%	95.2%
LSTM	Temporal 0.65/0.35	95.0%	94.6%	95.5%	95.0%
LSTM	Temporal 0.5/0.5	94.6%	94.2%	95.1%	94.7%
LSTM Average		**95.0%**	**94.6%**	**95.5%**	**95.0%**

Table 20: Empirical Coverage Rate for Team Win% Inductive Methods and ACI (99% Target Coverage)

Model	Split / Ratio	Split (99%)	Locally (99%)	CQR (99%)	ACI (99%)
Multiple Linear Regression	Random 0.8/0.2	99.4%	99.1%	99.6%	99.2%
Multiple Linear Regression	Random 0.65/0.35	99.2%	98.9%	99.4%	99.0%
Multiple Linear Regression	Random 0.5/0.5	99.0%	98.7%	99.2%	98.8%
Multiple Linear Regression	Temporal 0.8/0.2	98.9%	98.6%	99.2%	99.1%
Multiple Linear Regression	Temporal 0.65/0.35	98.7%	98.4%	99.0%	98.9%
Multiple Linear Regression	Temporal 0.5/0.5	98.5%	98.2%	98.8%	98.7%
Linear Reg. Average		**99.0%**	**98.7%**	**99.2%**	**98.9%**
Random Forest	Random 0.8/0.2	99.7%	99.5%	99.8%	99.3%
Random Forest	Random 0.65/0.35	99.5%	99.3%	99.6%	99.1%
Random Forest	Random 0.5/0.5	99.3%	99.1%	99.4%	98.9%
Random Forest	Temporal 0.8/0.2	99.2%	99.0%	99.3%	99.0%
Random Forest	Temporal 0.65/0.35	99.0%	98.8%	99.1%	98.8%
Random Forest	Temporal 0.5/0.5	98.8%	98.6%	98.9%	98.6%
Random Forest Average		**99.3%**	**99.1%**	**99.4%**	**99.0%**
Neural Network (MLP)	Random 0.8/0.2	99.5%	99.2%	99.6%	99.1%
Neural Network (MLP)	Random 0.65/0.35	99.3%	99.0%	99.4%	98.9%
Neural Network (MLP)	Random 0.5/0.5	99.1%	98.8%	99.2%	98.7%
Neural Network (MLP)	Temporal 0.8/0.2	99.0%	98.7%	99.1%	98.9%
Neural Network (MLP)	Temporal 0.65/0.35	98.8%	98.5%	98.9%	98.7%
Neural Network (MLP)	Temporal 0.5/0.5	98.6%	98.3%	98.7%	98.5%
Neural Net Average		**99.1%**	**98.8%**	**99.2%**	**98.8%**
ARIMA	Temporal 0.8/0.2	99.0%	98.7%	99.1%	99.0%
ARIMA	Temporal 0.65/0.35	98.8%	98.5%	98.9%	98.8%
ARIMA	Temporal 0.5/0.5	98.6%	98.3%	98.7%	98.6%
ARIMA Average		**98.8%**	**98.5%**	**98.9%**	**98.8%**
LSTM	Temporal 0.8/0.2	99.4%	99.2%	99.6%	99.3%
LSTM	Temporal 0.65/0.35	99.2%	99.0%	99.4%	99.1%
LSTM	Temporal 0.5/0.5	99.0%	98.8%	99.2%	98.9%
LSTM Average		**99.2%**	**99.0%**	**99.4%**	**99.1%**

Table 21: Average Prediction Interval Length (in Win% units) for Team Data (90% Target Coverage)

Model	Split / Ratio	Split (90%)	Locally (90%)	CQR (90%)	ACI (90%)
Multiple Linear Regression	Random 0.8/0.2	0.455	0.445	0.450	0.452
	Random 0.65/0.35	0.448	0.438	0.443	0.445
	Random 0.5/0.5	0.462	0.452	0.457	0.459
	Temporal 0.8/0.2	0.458	0.448	0.453	0.455
	Temporal 0.65/0.35	0.452	0.442	0.447	0.449
	Temporal 0.5/0.5	0.466	0.456	0.461	0.463
	Linear Reg. Average	**0.457**	**0.447**	**0.452**	**0.454**
Random Forest	Random 0.8/0.2	0.441	0.432	0.437	0.438
	Random 0.65/0.35	0.434	0.425	0.430	0.431
	Random 0.5/0.5	0.448	0.439	0.444	0.445
	Temporal 0.8/0.2	0.444	0.435	0.440	0.441
	Temporal 0.65/0.35	0.438	0.429	0.434	0.435
	Temporal 0.5/0.5	0.452	0.443	0.448	0.449
	Random Forest Average	**0.443**	**0.434**	**0.439**	**0.440**
Neural Network (MLP)	Random 0.8/0.2	0.464	0.454	0.459	0.461
	Random 0.65/0.35	0.457	0.447	0.452	0.454
	Random 0.5/0.5	0.471	0.461	0.466	0.468
	Temporal 0.8/0.2	0.467	0.457	0.462	0.464
	Temporal 0.65/0.35	0.461	0.451	0.456	0.458
	Temporal 0.5/0.5	0.475	0.465	0.470	0.472
	Neural Net Average	**0.466**	**0.456**	**0.461**	**0.463**
ARIMA	Temporal 0.8/0.2	0.569	0.556	0.563	0.565
	Temporal 0.65/0.35	0.562	0.549	0.556	0.558
	Temporal 0.5/0.5	0.576	0.563	0.570	0.572
	ARIMA Average	**0.569**	**0.556**	**0.563**	**0.565**
LSTM	Temporal 0.8/0.2	0.555	0.543	0.550	0.552
	Temporal 0.65/0.35	0.548	0.536	0.543	0.545
	Temporal 0.5/0.5	0.562	0.550	0.557	0.559
	LSTM Average	**0.555**	**0.543**	**0.550**	**0.552**

Table 22: Average Prediction Interval Length (in Win% units) for Team Data (95% Target Coverage)

Model	Split / Ratio	Split (95%)	Locally (95%)	CQR (95%)	ACI (95%)
Multiple Linear Regression	Random 0.8/0.2	0.537	0.527	0.532	0.534
	Random 0.65/0.35	0.529	0.519	0.524	0.526
	Random 0.5/0.5	0.545	0.535	0.540	0.542
	Temporal 0.8/0.2	0.540	0.530	0.535	0.537
	Temporal 0.65/0.35	0.533	0.523	0.528	0.530
	Temporal 0.5/0.5	0.550	0.540	0.545	0.547
	Linear Reg. Average	**0.539**	**0.529**	**0.534**	**0.536**
Random Forest	Random 0.8/0.2	0.520	0.511	0.516	0.517
	Random 0.65/0.35	0.512	0.503	0.508	0.509
	Random 0.5/0.5	0.529	0.520	0.525	0.526
	Temporal 0.8/0.2	0.524	0.515	0.520	0.521
	Temporal 0.65/0.35	0.517	0.508	0.513	0.514
	Temporal 0.5/0.5	0.533	0.524	0.529	0.530
	Random Forest Average	**0.523**	**0.514**	**0.519**	**0.520**
Neural Network (MLP)	Random 0.8/0.2	0.548	0.538	0.543	0.545
	Random 0.65/0.35	0.539	0.529	0.534	0.536
	Random 0.5/0.5	0.556	0.546	0.551	0.553
	Temporal 0.8/0.2	0.551	0.541	0.546	0.548
	Temporal 0.65/0.35	0.544	0.534	0.539	0.541
	Temporal 0.5/0.5	0.560	0.550	0.555	0.557
	Neural Net Average	**0.550**	**0.540**	**0.545**	**0.547**
ARIMA	Temporal 0.8/0.2	0.671	0.658	0.665	0.667
	Temporal 0.65/0.35	0.663	0.650	0.657	0.659
	Temporal 0.5/0.5	0.680	0.667	0.674	0.676
	ARIMA Average	**0.671**	**0.658**	**0.665**	**0.667**
LSTM	Temporal 0.8/0.2	0.655	0.643	0.650	0.652
	Temporal 0.65/0.35	0.647	0.635	0.642	0.644
	Temporal 0.5/0.5	0.663	0.651	0.658	0.660
	LSTM Average	**0.655**	**0.643**	**0.650**	**0.652**

Table 23: Average Prediction Interval Length (in Win% units) for Team Data (99% Target Coverage)

Model	Split / Ratio	Split (99%)	Locally (99%)	CQR (99%)	ACI (99%)
Multiple Linear Regression	Random 0.8/0.2	0.705	0.695	0.700	0.702
	Random 0.65/0.35	0.694	0.684	0.689	0.691
	Random 0.5/0.5	0.716	0.706	0.711	0.713
	Temporal 0.8/0.2	0.710	0.700	0.705	0.707
	Temporal 0.65/0.35	0.701	0.691	0.696	0.698
	Temporal 0.5/0.5	0.722	0.712	0.717	0.719
	Linear Reg. Average	**0.708**	**0.698**	**0.703**	**0.705**
Random Forest	Random 0.8/0.2	0.684	0.675	0.680	0.681
	Random 0.65/0.35	0.673	0.664	0.669	0.670
	Random 0.5/0.5	0.694	0.685	0.690	0.691
	Temporal 0.8/0.2	0.688	0.679	0.684	0.685
	Temporal 0.65/0.35	0.679	0.670	0.675	0.676
	Temporal 0.5/0.5	0.701	0.692	0.697	0.698
	Random Forest Average	**0.686**	**0.677**	**0.682**	**0.683**
Neural Network (MLP)	Random 0.8/0.2	0.719	0.709	0.714	0.716
	Random 0.65/0.35	0.708	0.698	0.703	0.705
	Random 0.5/0.5	0.730	0.720	0.725	0.727
	Temporal 0.8/0.2	0.724	0.714	0.719	0.721
	Temporal 0.65/0.35	0.715	0.705	0.710	0.712
	Temporal 0.5/0.5	0.736	0.726	0.731	0.733
	Neural Net Average	**0.722**	**0.712**	**0.717**	**0.719**
ARIMA	Temporal 0.8/0.2	0.882	0.869	0.876	0.878
	Temporal 0.65/0.35	0.871	0.858	0.865	0.867
	Temporal 0.5/0.5	0.893	0.880	0.887	0.889
	ARIMA Average	**0.882**	**0.869**	**0.876**	**0.878**
LSTM	Temporal 0.8/0.2	0.860	0.848	0.855	0.857
	Temporal 0.65/0.35	0.849	0.837	0.844	0.846
	Temporal 0.5/0.5	0.871	0.859	0.866	0.868
	LSTM Average	**0.860**	**0.848**	**0.855**	**0.857**

Transductive / Grid Conformal Methods

Table 24: Empirical Coverage Rate for Transductive / Grid Methods (90% Target Coverage)

Model	M	Rounding (90%)	CPDD (90%)	CPDM (90%)
Multiple Linear Regression	800	90.3%	90.2%	90.1%
	600	90.2%	90.1%	90.1%
	400	90.1%	90.2%	90.1%
	200	90.3%	90.1%	90.1%
	100	90.0%	90.0%	90.1%
	50	89.6%	89.9%	90.0%
	25	87.3%	89.3%	89.9%
	10	72.3%	85.2%	89.7%
	5	41.7%	76.5%	89.3%
	Average	**82.4%**	**87.9%**	**89.9%**
Random Forest	800	90.9%	90.7%	90.6%
	600	90.8%	90.6%	90.6%
	400	90.7%	90.7%	90.6%
	200	90.9%	90.6%	90.6%
	100	90.6%	90.5%	90.6%
	50	90.2%	90.4%	90.5%
	25	87.9%	89.8%	90.4%
	10	72.9%	85.7%	90.2%
	5	42.3%	77.0%	89.8%
	Average	**83.0%**	**88.5%**	**90.5%**
Neural Network (MLP)	800	90.6%	90.4%	90.3%
	600	90.5%	90.3%	90.3%
	400	90.4%	90.4%	90.3%
	200	90.6%	90.3%	90.3%
	100	90.3%	90.2%	90.3%
	50	89.9%	90.1%	90.2%
	25	87.6%	89.5%	90.0%
	10	72.5%	85.4%	89.9%
	5	41.9%	76.7%	89.5%
	Average	**82.7%**	**88.2%**	**90.2%**
ARIMA	800	90.1%	89.9%	89.8%
	600	90.0%	89.8%	89.8%
	400	89.9%	89.9%	89.8%
	200	90.1%	89.8%	89.8%
	100	89.8%	89.7%	89.8%
	50	89.4%	89.6%	89.7%
	25	87.1%	89.0%	89.6%
	10	72.1%	84.9%	89.4%
	5	41.5%	76.2%	89.0%
	Average	**82.2%**	**87.6%**	**89.6%**
LSTM	800	90.7%	90.5%	90.4%
	600	90.6%	90.4%	90.4%
	400	90.5%	90.5%	90.4%
	200	90.7%	90.4%	90.4%
	100	90.4%	90.3%	90.4%
	50	90.0%	90.2%	90.3%
	25	87.7%	89.6%	90.2%
	10	72.7%	85.5%	90.0%
	5	42.1%	76.8%	89.6%
	Average	**82.8%**	**88.3%**	**90.3%**

Table 25: Empirical Coverage Rate for Transductive / Grid Methods (95% Target Coverage)

Model	M	Rounding (95%)	CPDD (95%)	CPDM (95%)
Multiple Linear Regression	800	95.3%	95.2%	95.1%
	600	95.2%	95.1%	95.1%
	400	95.1%	95.2%	95.1%
	200	95.3%	95.1%	95.1%
	100	95.0%	95.0%	95.1%
	50	94.6%	94.9%	95.0%
	25	92.3%	94.3%	94.9%
	10	77.3%	90.2%	94.7%
	5	46.7%	81.5%	94.3%
	Average	**87.4%**	**92.9%**	**94.9%**
Random Forest	800	95.9%	95.7%	95.6%
	600	95.8%	95.6%	95.6%
	400	95.7%	95.7%	95.6%
	200	95.9%	95.6%	95.6%
	100	95.6%	95.5%	95.6%
	50	95.2%	95.4%	95.5%
	25	92.9%	94.8%	95.4%
	10	77.9%	90.7%	95.2%
	5	47.3%	82.0%	94.8%
	Average	**88.0%**	**93.5%**	**95.5%**
Neural Network (MLP)	800	95.6%	95.4%	95.3%
	600	95.5%	95.3%	95.3%
	400	95.4%	95.4%	95.3%
	200	95.6%	95.3%	95.3%
	100	95.3%	95.2%	95.3%
	50	94.9%	95.1%	95.2%
	25	92.6%	94.5%	95.0%
	10	77.5%	90.4%	94.9%
	5	46.9%	81.7%	94.5%
	Average	**87.7%**	**93.2%**	**95.2%**
ARIMA	800	95.1%	94.9%	94.8%
	600	95.0%	94.8%	94.8%
	400	94.9%	94.9%	94.8%
	200	95.1%	94.8%	94.8%
	100	94.8%	94.7%	94.8%
	50	94.4%	94.6%	94.7%
	25	92.1%	94.0%	94.6%
	10	77.1%	89.9%	94.4%
	5	46.5%	81.2%	94.0%
	Average	**87.2%**	**92.6%**	**94.6%**
LSTM	800	95.7%	95.5%	95.4%
	600	95.6%	95.4%	95.4%
	400	95.5%	95.5%	95.4%
	200	95.7%	95.4%	95.4%
	100	95.4%	95.3%	95.4%
	50	95.0%	95.2%	95.3%
	25	92.7%	94.6%	95.2%
	10	77.7%	90.5%	95.0%
	5	47.1%	81.8%	94.6%
	Average	**87.8%**	**93.3%**	**95.3%**

Table 26: Empirical Coverage Rate for Transductive / Grid Methods (99% Target Coverage)

Model	M	Rounding (99%)	CPDD (99%)	CPDM (99%)
Multiple Linear Regression	800	99.3%	99.2%	99.1%
	600	99.2%	99.1%	99.1%
	400	99.1%	99.2%	99.1%
	200	99.3%	99.1%	99.1%
	100	99.0%	99.0%	99.1%
	50	98.6%	98.9%	99.0%
	25	96.3%	98.3%	98.9%
	10	81.3%	94.2%	98.7%
	5	50.7%	85.5%	98.3%
	Average	**91.4%**	**96.9%**	**98.9%**
Random Forest	800	99.9%	99.7%	99.6%
	600	99.8%	99.6%	99.6%
	400	99.7%	99.7%	99.6%
	200	99.9%	99.6%	99.6%
	100	99.6%	99.5%	99.6%
	50	99.2%	99.4%	99.5%
	25	96.9%	98.8%	99.4%
	10	81.9%	94.7%	99.2%
	5	51.3%	86.0%	98.8%
	Average	**92.0%**	**97.5%**	**99.5%**
Neural Network (MLP)	800	99.6%	99.4%	99.3%
	600	99.5%	99.3%	99.3%
	400	99.4%	99.4%	99.3%
	200	99.6%	99.3%	99.3%
	100	99.3%	99.2%	99.3%
	50	98.9%	99.1%	99.2%
	25	96.6%	98.5%	99.0%
	10	81.5%	94.4%	98.9%
	5	50.9%	85.7%	98.5%
	Average	**91.7%**	**97.2%**	**99.2%**
ARIMA	800	99.1%	98.9%	98.8%
	600	99.0%	98.8%	98.8%
	400	98.9%	98.9%	98.8%
	200	99.1%	98.8%	98.8%
	100	98.8%	98.7%	98.8%
	50	98.4%	98.6%	98.7%
	25	96.1%	98.0%	98.6%
	10	81.1%	93.9%	98.4%
	5	50.5%	85.2%	98.0%
	Average	**91.2%**	**96.6%**	**98.6%**
LSTM	800	99.7%	99.5%	99.4%
	600	99.6%	99.4%	99.4%
	400	99.5%	99.5%	99.4%
	200	99.7%	99.4%	99.4%
	100	99.4%	99.3%	99.4%
	50	99.0%	99.2%	99.3%
	25	96.7%	98.6%	99.2%
	10	81.7%	94.5%	99.0%
	5	51.1%	85.8%	98.6%
	Average	**91.8%**	**97.3%**	**99.3%**

Table 27: Average Prediction Interval Length for Transductive / Grid Methods (90% Target Coverage)

Model	M	Rounding (90%)	CPDD (90%)	CPDM (90%)
Multiple Linear Regression	800	0.460	0.442	0.443
	600	0.458	0.442	0.443
	400	0.450	0.442	0.443
	200	0.435	0.442	0.443
	100	0.465	0.443	0.443
	50	0.434	0.443	0.443
	25	0.445	0.443	0.443
	10	0.474	0.456	0.474
	5	0.567	0.567	0.567
	Average	**0.465**	**0.452**	**0.460**
Random Forest	800	0.446	0.429	0.430
	600	0.444	0.429	0.430
	400	0.436	0.429	0.430
	200	0.422	0.429	0.430
	100	0.451	0.430	0.430
	50	0.421	0.430	0.430
	25	0.433	0.430	0.430
	10	0.467	0.456	0.467
	5	0.550	0.567	0.550
	Average	**0.452**	**0.440**	**0.447**
Neural Network (MLP)	800	0.469	0.451	0.452
	600	0.467	0.451	0.452
	400	0.459	0.451	0.452
	200	0.444	0.451	0.452
	100	0.474	0.452	0.452
	50	0.443	0.452	0.452
	25	0.455	0.452	0.452
	10	0.496	0.476	0.496
	5	0.578	0.587	0.578
	Average	**0.476**	**0.463**	**0.471**
ARIMA	800	0.575	0.553	0.554
	600	0.573	0.553	0.554
	400	0.563	0.553	0.554
	200	0.544	0.553	0.554
	100	0.581	0.554	0.554
	50	0.543	0.554	0.554
	25	0.553	0.554	0.554
	10	0.587	0.578	0.587
	5	0.670	0.687	0.670
	Average	**0.577**	**0.560**	**0.567**
LSTM	800	0.561	0.539	0.540
	600	0.559	0.539	0.540
	400	0.549	0.539	0.540
	200	0.531	0.539	0.540
	100	0.567	0.540	0.540
	50	0.530	0.540	0.540
	25	0.540	0.540	0.540
	10	0.574	0.565	0.574
	5	0.657	0.674	0.657
	Average	**0.563**	**0.546**	**0.553**

Table 28: Average Prediction Interval Length for Transductive / Grid Methods (95% Target Coverage)

Model	M	Rounding (95%)	CPDD (95%)	CPDM (95%)
Multiple Linear Regression	800	0.535	0.512	0.513
	600	0.533	0.512	0.513
	400	0.525	0.512	0.513
	200	0.510	0.512	0.513
	100	0.540	0.513	0.513
	50	0.509	0.513	0.513
	25	0.520	0.513	0.513
	10	0.551	0.533	0.551
	5	0.632	0.665	0.632
	Average	**0.539**	**0.520**	**0.533**
Random Forest	800	0.521	0.499	0.500
	600	0.519	0.499	0.500
	400	0.511	0.499	0.500
	200	0.496	0.499	0.500
	100	0.526	0.500	0.500
	50	0.495	0.500	0.500
	25	0.506	0.500	0.500
	10	0.546	0.533	0.546
	5	0.625	0.665	0.625
	Average	**0.527**	**0.511**	**0.519**
Neural Network (MLP)	800	0.544	0.521	0.522
	600	0.542	0.521	0.522
	400	0.534	0.521	0.522
	200	0.519	0.521	0.522
	100	0.549	0.522	0.522
	50	0.518	0.522	0.522
	25	0.530	0.522	0.522
	10	0.571	0.553	0.571
	5	0.653	0.685	0.653
	Average	**0.551**	**0.534**	**0.543**
ARIMA	800	0.640	0.618	0.619
	600	0.638	0.618	0.619
	400	0.628	0.618	0.619
	200	0.609	0.618	0.619
	100	0.646	0.619	0.619
	50	0.608	0.619	0.619
	25	0.618	0.619	0.619
	10	0.652	0.643	0.652
	5	0.735	0.752	0.735
	Average	**0.642**	**0.627**	**0.634**
LSTM	800	0.626	0.604	0.605
	600	0.624	0.604	0.605
	400	0.614	0.604	0.605
	200	0.595	0.604	0.605
	100	0.632	0.605	0.605
	50	0.594	0.605	0.605
	25	0.604	0.605	0.605
	10	0.638	0.629	0.638
	5	0.721	0.738	0.721
	Average	**0.628**	**0.613**	**0.620**

Table 29: Average Prediction Interval Length for Transductive / Grid Methods (99% Target Coverage)

Model	M	Rounding (99%)	CPDD (99%)	CPDM (99%)
Multiple Linear Regression	800	0.707	0.684	0.685
	600	0.705	0.684	0.685
	400	0.697	0.684	0.685
	200	0.682	0.684	0.685
	100	0.712	0.685	0.685
	50	0.681	0.685	0.685
	25	0.692	0.685	0.685
	10	0.723	0.705	0.723
	5	0.804	0.837	0.804
	Average	**0.711**	**0.693**	**0.703**
Random Forest	800	0.693	0.671	0.672
	600	0.691	0.671	0.672
	400	0.683	0.671	0.672
	200	0.668	0.671	0.672
	100	0.698	0.672	0.672
	50	0.667	0.672	0.672
	25	0.678	0.672	0.672
	10	0.718	0.705	0.718
	5	0.797	0.837	0.797
	Average	**0.699**	**0.683**	**0.691**
Neural Network (MLP)	800	0.716	0.693	0.694
	600	0.714	0.693	0.694
	400	0.706	0.693	0.694
	200	0.691	0.693	0.694
	100	0.721	0.694	0.694
	50	0.690	0.694	0.694
	25	0.702	0.694	0.694
	10	0.743	0.725	0.743
	5	0.825	0.857	0.825
	Average	**0.723**	**0.706**	**0.714**
ARIMA	800	0.802	0.780	0.781
	600	0.800	0.780	0.781
	400	0.790	0.780	0.781
	200	0.771	0.780	0.781
	100	0.808	0.781	0.781
	50	0.770	0.781	0.781
	25	0.780	0.781	0.781
	10	0.814	0.805	0.814
	5	0.897	0.914	0.897
	Average	**0.804**	**0.788**	**0.796**
LSTM	800	0.788	0.766	0.767
	600	0.786	0.766	0.767
	400	0.776	0.766	0.767
	200	0.757	0.766	0.767
	100	0.794	0.767	0.767
	50	0.756	0.767	0.767
	25	0.766	0.767	0.767
	10	0.800	0.791	0.800
	5	0.883	0.900	0.883
	Average	**0.790**	**0.774**	**0.782**

Decision Analysis & Optimal Parameter Choices

1. **Optimal Grid Resolution (M^*):**
   - **Approximation via Rounding & Discretized Data (CPDD):** M^* suggested to be at least 50. For $M < 50$, interval lengths collapse unreliably and coverage are lower than target coverage.
   - **Discretized Model (CPDM):** M^* suggested to be at least 25. For $M < 25$, interval lengths collapse unreliably and coverage are lower than target coverage.
2. **Optimal Split Ratio per Model:**

- **Linear Regression, Random Forest, Neural Network:** Group-Random 0.65/0.35 yields the shortest valid interval lengths.
- **ARIMA & LSTM:** Temporal 0.65/0.35 achieves optimal balance between historical training depth and calibration quantile precision.

3. **Adaptive Conformal Inference Step Size (γ^*):**
   - **Linear Regression & ARIMA:** $\gamma^* = 0.01$ (smooth, steady adjustment).
   - **Random Forest & Neural Network:** $\gamma^* = 0.05$ (adapts to non-linear feature shifts).
   - **LSTM:** $\gamma^* = 0.05$ (absorbs multi-year sequential dependency drift).

Empirical Conclusions

- **Inductive Conformal Methods & ACI:** Split conformal and locally conformal have similar coverage, but locally conformal has a better average length. CQR has slightly better coverage, but its average length is similar to split conformal and larger than locally conformal. Even though the average length of ACI has slightly worse than locally conformal, it has more precise coverage than other conformal prediction methods. Overall it is similar to the results in 3.1.
- **Grid Conformal Methods:** Similar to 3.1, all three methods have similar coverage and average length when $M > 25$. The main difference is that when M is small, rounding has significantly lower than target coverage, and it is closer to target coverage when M becomes larger. Compared to the rounding method, discretized data and discretized model have better performance when M is small. Also, there is no significant difference in average length between M and conformal prediction methods.
- **Model Selection:** We use 5 models to compare the coverage and average length, including 3 machine learning models and 2 time series models. For coverage, all 5 models do not have a significant difference. For average length, the performance for each model would be the following: Random Forest <Multiple Linear Regression <Neural Network <LSTM <ARIMA. Time series models perform less than other models.

3.2.4 Prediction for upcoming (2021–2022) Season

Point Predictions for 2021–2022 Season Win Percentage

Table 30: Predicted Win Percentage, Expected Wins, and Losses for 2021–2022 Season across 30 NBA Franchises

	Multilinear Regression			**Random Forest**			**Neural Network**			**ARIMA**			**LSTM**		
Team	Win	Lose	%	Win	Lose	%	Win	Lose	%	Win	Lose	%	Win	Lose	%
Milwaukee Bucks	55	27	0.667	51	31	0.625	66	16	0.800	60	22	0.731	60	22	0.737
Toronto Raptors	50	32	0.606	52	30	0.630	61	21	0.739	58	24	0.705	58	24	0.710
Los Angeles Clippers	55	27	0.669	54	28	0.658	50	32	0.609	54	28	0.658	54	28	0.662
Los Angeles Lakers	50	32	0.615	47	35	0.576	46	36	0.565	58	24	0.702	58	24	0.707
Boston Celtics	48	34	0.583	50	32	0.610	50	32	0.611	53	29	0.646	53	29	0.649
Houston Rockets	40	42	0.486	44	38	0.536	58	24	0.711	49	33	0.599	49	33	0.601
Utah Jazz	49	33	0.601	49	33	0.598	44	38	0.534	49	33	0.599	49	33	0.601
Miami Heat	50	32	0.609	47	35	0.574	45	37	0.549	49	33	0.592	49	33	0.594
Dallas Mavericks	47	35	0.569	45	37	0.551	46	36	0.561	46	36	0.567	47	35	0.568
Philadelphia 76ers	49	33	0.594	45	37	0.544	40	42	0.494	48	34	0.580	48	34	0.582
Indiana Pacers	46	36	0.560	43	39	0.529	38	44	0.458	49	33	0.603	50	32	0.606
Denver Nuggets	45	37	0.549	44	38	0.539	34	48	0.420	50	32	0.615	51	31	0.618
Brooklyn Nets	47	35	0.568	47	35	0.573	42	40	0.518	40	42	0.493	40	42	0.492
Phoenix Suns	46	36	0.560	43	39	0.529	46	36	0.562	39	43	0.475	39	43	0.475
Oklahoma City Thunder	36	46	0.441	40	42	0.492	31	51	0.383	49	33	0.599	49	33	0.601
Portland Trail Blazers	41	41	0.500	39	43	0.477	41	41	0.502	39	43	0.481	39	43	0.481
Memphis Grizzlies	40	42	0.486	38	44	0.460	42	40	0.509	39	43	0.475	39	43	0.475
New Orleans Pelicans	39	43	0.470	40	42	0.484	48	34	0.585	36	46	0.433	35	47	0.432
Sacramento Kings	34	48	0.420	38	44	0.460	43	39	0.527	37	45	0.445	36	46	0.444
Orlando Magic	39	43	0.474	39	43	0.479	33	49	0.404	38	44	0.464	38	44	0.463
San Antonio Spurs	38	44	0.462	39	43	0.472	31	51	0.376	38	44	0.462	38	44	0.462
Chicago Bulls	36	46	0.437	38	44	0.463	45	37	0.545	30	52	0.367	30	52	0.364
Washington Wizards	37	45	0.454	35	47	0.422	39	43	0.473	31	51	0.374	30	52	0.372
Minnesota Timberwolves	31	51	0.372	31	51	0.381	42	40	0.511	27	55	0.332	27	55	0.328
New York Knicks	35	47	0.430	34	48	0.412	31	51	0.374	29	53	0.350	28	54	0.346
Atlanta Hawks	33	49	0.398	32	50	0.386	36	46	0.441	27	55	0.333	27	55	0.329
Charlotte Hornets	27	55	0.334	32	50	0.395	22	60	0.273	31	51	0.380	31	51	0.377
Golden State Warriors	37	45	0.447	36	46	0.434	27	55	0.333	23	59	0.275	22	60	0.270
Detroit Pistons	29	53	0.352	30	52	0.370	27	55	0.332	28	54	0.337	27	55	0.333
Cleveland Cavaliers	24	58	0.288	28	54	0.340	24	58	0.298	27	55	0.328	27	55	0.324

Prediction Interval for 2021–2022 Season

Table 31: Prediction Interval Lengths for 2021–2022 Season (Split Conformal Prediction)

	Multilinear Regression			**Random Forest**			**Neural Network**			**ARIMA**			**LSTM**		
Team	90%	95%	99%	90%	95%	99%	90%	95%	99%	90%	95%	99%	90%	95%	99%
Milwaukee Bucks	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Toronto Raptors	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Los Angeles Clippers	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Los Angeles Lakers	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Boston Celtics	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Houston Rockets	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Utah Jazz	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Miami Heat	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Dallas Mavericks	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Philadelphia 76ers	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Indiana Pacers	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Denver Nuggets	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Brooklyn Nets	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Phoenix Suns	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Oklahoma City Thunder	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Portland Trail Blazers	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Memphis Grizzlies	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
New Orleans Pelicans	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Sacramento Kings	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Orlando Magic	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
San Antonio Spurs	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Chicago Bulls	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Washington Wizards	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Minnesota Timberwolves	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
New York Knicks	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Atlanta Hawks	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Charlotte Hornets	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Golden State Warriors	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Detroit Pistons	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717
Cleveland Cavaliers	0.455	0.525	0.640	0.441	0.509	0.621	0.464	0.536	0.653	0.569	0.656	0.736	0.555	0.640	0.717

Table 32: Prediction Interval Lengths for 2021–2022 Season (Locally Adaptive Conformal Prediction)

	Multilinear Regression			**Random Forest**			**Neural Network**			**ARIMA**			**LSTM**		
Team	90%	95%	99%	90%	95%	99%	90%	95%	99%	90%	95%	99%	90%	95%	99%
Milwaukee Bucks	0.463	0.536	0.655	0.449	0.520	0.636	0.472	0.546	0.668	0.579	0.670	0.754	0.565	0.653	0.734
Toronto Raptors	0.461	0.533	0.652	0.447	0.517	0.633	0.470	0.544	0.665	0.576	0.667	0.750	0.562	0.651	0.731
Los Angeles Clippers	0.457	0.529	0.647	0.443	0.513	0.628	0.466	0.540	0.660	0.571	0.661	0.744	0.558	0.645	0.725
Los Angeles Lakers	0.461	0.533	0.652	0.447	0.517	0.632	0.470	0.544	0.665	0.576	0.666	0.750	0.562	0.650	0.730
Boston Celtics	0.456	0.528	0.646	0.442	0.512	0.626	0.465	0.538	0.659	0.570	0.660	0.743	0.556	0.644	0.723
Houston Rockets	0.452	0.524	0.640	0.439	0.508	0.621	0.461	0.534	0.653	0.566	0.654	0.737	0.552	0.639	0.717
Utah Jazz	0.452	0.524	0.640	0.439	0.508	0.621	0.461	0.534	0.653	0.566	0.654	0.737	0.552	0.639	0.717
Miami Heat	0.452	0.523	0.640	0.438	0.507	0.621	0.461	0.533	0.653	0.565	0.654	0.736	0.551	0.638	0.716
Dallas Mavericks	0.450	0.521	0.637	0.436	0.505	0.618	0.459	0.531	0.650	0.562	0.651	0.732	0.549	0.635	0.713
Philadelphia 76ers	0.451	0.522	0.638	0.437	0.506	0.619	0.460	0.532	0.651	0.564	0.652	0.734	0.550	0.637	0.715
Indiana Pacers	0.453	0.524	0.641	0.439	0.508	0.622	0.462	0.534	0.654	0.566	0.655	0.737	0.552	0.639	0.718
Denver Nuggets	0.454	0.525	0.642	0.440	0.509	0.623	0.463	0.536	0.655	0.567	0.656	0.739	0.553	0.641	0.719
Brooklyn Nets	0.446	0.516	0.631	0.433	0.501	0.612	0.455	0.526	0.644	0.557	0.645	0.726	0.544	0.630	0.707
Phoenix Suns	0.447	0.518	0.633	0.434	0.502	0.614	0.456	0.528	0.646	0.559	0.647	0.728	0.546	0.632	0.709
Oklahoma City Thunder	0.452	0.524	0.640	0.439	0.508	0.621	0.461	0.534	0.653	0.566	0.654	0.737	0.552	0.639	0.717
Portland Trail Blazers	0.447	0.517	0.633	0.433	0.502	0.614	0.456	0.527	0.645	0.559	0.646	0.727	0.545	0.631	0.708
Memphis Grizzlies	0.447	0.518	0.633	0.434	0.502	0.614	0.456	0.528	0.646	0.559	0.647	0.728	0.546	0.632	0.709
New Orleans Pelicans	0.451	0.521	0.638	0.437	0.506	0.619	0.460	0.532	0.651	0.563	0.652	0.734	0.550	0.636	0.714
Sacramento Kings	0.450	0.520	0.637	0.436	0.505	0.617	0.459	0.531	0.649	0.562	0.650	0.732	0.549	0.635	0.713
Orlando Magic	0.448	0.519	0.635	0.435	0.503	0.615	0.457	0.529	0.647	0.560	0.648	0.730	0.547	0.633	0.711
San Antonio Spurs	0.448	0.519	0.635	0.435	0.503	0.616	0.457	0.529	0.647	0.560	0.649	0.730	0.547	0.633	0.711
Chicago Bulls	0.456	0.527	0.645	0.442	0.512	0.626	0.465	0.538	0.658	0.570	0.659	0.742	0.556	0.644	0.723
Washington Wizards	0.455	0.527	0.644	0.442	0.511	0.625	0.464	0.537	0.657	0.569	0.659	0.741	0.555	0.643	0.722
Minnesota Timberwolves	0.459	0.531	0.649	0.445	0.515	0.630	0.468	0.541	0.662	0.573	0.663	0.747	0.559	0.647	0.727
New York Knicks	0.457	0.529	0.647	0.443	0.513	0.628	0.466	0.540	0.660	0.571	0.661	0.744	0.558	0.645	0.725
Atlanta Hawks	0.458	0.531	0.649	0.445	0.515	0.630	0.468	0.541	0.662	0.573	0.663	0.746	0.559	0.647	0.727
Charlotte Hornets	0.455	0.526	0.644	0.441	0.511	0.624	0.464	0.537	0.657	0.568	0.658	0.740	0.555	0.642	0.721
Golden State Warriors	0.463	0.536	0.655	0.449	0.520	0.636	0.472	0.547	0.669	0.579	0.670	0.754	0.565	0.654	0.734
Detroit Pistons	0.458	0.530	0.649	0.444	0.514	0.629	0.467	0.541	0.662	0.573	0.663	0.746	0.559	0.647	0.726
Cleveland Cavaliers	0.459	0.531	0.650	0.445	0.515	0.630	0.468	0.542	0.663	0.574	0.664	0.747	0.560	0.648	0.728

Table 33: Prediction Interval Lengths for 2021–2022 Season (Conformalized Quantile Regression (CQR))

	Multilinear Regression			**Random Forest**			**Neural Network**			**ARIMA**			**LSTM**		
Team	90%	95%	99%	90%	95%	99%	90%	95%	99%	90%	95%	99%	90%	95%	99%
Milwaukee Bucks	0.456	0.530	0.650	0.442	0.515	0.631	0.465	0.541	0.663	0.569	0.663	0.748	0.556	0.647	0.728
Toronto Raptors	0.454	0.528	0.647	0.440	0.512	0.628	0.463	0.539	0.660	0.567	0.660	0.744	0.553	0.644	0.725
Los Angeles Clippers	0.450	0.524	0.642	0.436	0.508	0.623	0.459	0.534	0.655	0.562	0.655	0.738	0.549	0.639	0.719
Los Angeles Lakers	0.453	0.528	0.647	0.440	0.512	0.627	0.462	0.538	0.660	0.567	0.660	0.744	0.553	0.644	0.724
Boston Celtics	0.449	0.523	0.641	0.435	0.507	0.621	0.458	0.533	0.653	0.561	0.653	0.737	0.548	0.638	0.718
Houston Rockets	0.445	0.518	0.635	0.432	0.503	0.616	0.454	0.529	0.648	0.557	0.648	0.731	0.543	0.633	0.712
Utah Jazz	0.445	0.518	0.635	0.432	0.503	0.616	0.454	0.529	0.648	0.557	0.648	0.731	0.543	0.633	0.712
Miami Heat	0.445	0.518	0.635	0.431	0.502	0.616	0.454	0.528	0.647	0.556	0.647	0.730	0.543	0.632	0.711
Dallas Mavericks	0.443	0.516	0.632	0.430	0.500	0.613	0.452	0.526	0.645	0.554	0.645	0.727	0.540	0.629	0.708
Philadelphia 76ers	0.444	0.517	0.633	0.431	0.501	0.614	0.453	0.527	0.646	0.555	0.646	0.728	0.541	0.631	0.709
Indiana Pacers	0.446	0.519	0.636	0.432	0.503	0.617	0.455	0.529	0.649	0.557	0.649	0.731	0.544	0.633	0.712
Denver Nuggets	0.447	0.520	0.637	0.433	0.504	0.618	0.455	0.530	0.650	0.558	0.650	0.733	0.545	0.634	0.714
Brooklyn Nets	0.439	0.511	0.626	0.426	0.496	0.608	0.448	0.521	0.639	0.549	0.639	0.720	0.535	0.623	0.701
Phoenix Suns	0.440	0.513	0.628	0.427	0.497	0.609	0.449	0.523	0.641	0.550	0.641	0.722	0.537	0.625	0.704
Oklahoma City Thunder	0.445	0.518	0.635	0.432	0.503	0.616	0.454	0.529	0.648	0.557	0.648	0.731	0.543	0.633	0.712
Portland Trail Blazers	0.440	0.512	0.628	0.427	0.497	0.609	0.449	0.522	0.640	0.550	0.640	0.722	0.537	0.625	0.703
Memphis Grizzlies	0.440	0.513	0.628	0.427	0.497	0.609	0.449	0.523	0.641	0.550	0.641	0.722	0.537	0.625	0.704
New Orleans Pelicans	0.443	0.516	0.633	0.430	0.501	0.614	0.452	0.527	0.645	0.554	0.645	0.728	0.541	0.630	0.709
Sacramento Kings	0.443	0.515	0.632	0.429	0.500	0.613	0.451	0.526	0.644	0.553	0.644	0.726	0.540	0.629	0.707
Orlando Magic	0.441	0.514	0.629	0.428	0.498	0.611	0.450	0.524	0.642	0.551	0.642	0.724	0.538	0.627	0.705
San Antonio Spurs	0.441	0.514	0.630	0.428	0.498	0.611	0.450	0.524	0.642	0.552	0.642	0.724	0.538	0.627	0.705
Chicago Bulls	0.449	0.522	0.640	0.435	0.507	0.621	0.458	0.533	0.653	0.561	0.653	0.736	0.547	0.637	0.717
Washington Wizards	0.448	0.522	0.639	0.435	0.506	0.620	0.457	0.532	0.652	0.560	0.652	0.735	0.547	0.636	0.716
Minnesota Timberwolves	0.451	0.526	0.644	0.438	0.510	0.625	0.460	0.536	0.657	0.564	0.657	0.741	0.551	0.641	0.721
New York Knicks	0.450	0.524	0.642	0.436	0.508	0.623	0.459	0.534	0.655	0.562	0.655	0.738	0.549	0.639	0.719
Atlanta Hawks	0.451	0.525	0.644	0.438	0.510	0.625	0.460	0.536	0.657	0.564	0.657	0.740	0.551	0.641	0.721
Charlotte Hornets	0.448	0.521	0.639	0.434	0.506	0.620	0.457	0.532	0.651	0.560	0.651	0.735	0.546	0.636	0.715
Golden State Warriors	0.456	0.531	0.650	0.442	0.515	0.631	0.465	0.541	0.663	0.570	0.663	0.748	0.556	0.647	0.728
Detroit Pistons	0.451	0.525	0.643	0.437	0.509	0.624	0.460	0.536	0.656	0.564	0.656	0.740	0.550	0.641	0.721
Cleveland Cavaliers	0.452	0.526	0.644	0.438	0.510	0.625	0.461	0.536	0.657	0.565	0.657	0.741	0.551	0.642	0.722

Table 34: Prediction Interval Lengths for 2021–2022 Season (Rounding (M=100))

	Multilinear Regression			Random Forest			Neural Network			ARIMA			LSTM		
Team	90%	95%	99%	90%	95%	99%	90%	95%	99%	90%	95%	99%	90%	95%	99%
Milwaukee Bucks	0.484	0.562	0.712	0.469	0.545	0.691	0.493	0.573	0.727	0.605	0.702	0.819	0.590	0.685	0.798
Toronto Raptors	0.481	0.559	0.709	0.467	0.542	0.688	0.491	0.570	0.723	0.602	0.699	0.816	0.587	0.682	0.794
Los Angeles Clippers	0.478	0.555	0.704	0.463	0.538	0.682	0.487	0.566	0.718	0.597	0.693	0.809	0.583	0.677	0.788
Los Angeles Lakers	0.481	0.559	0.709	0.467	0.542	0.688	0.491	0.570	0.723	0.602	0.699	0.815	0.587	0.682	0.794
Boston Celtics	0.477	0.553	0.702	0.462	0.537	0.681	0.486	0.565	0.716	0.596	0.692	0.807	0.581	0.675	0.786
Houston Rockets	0.473	0.549	0.696	0.459	0.533	0.676	0.482	0.560	0.710	0.591	0.686	0.801	0.577	0.670	0.780
Utah Jazz	0.473	0.549	0.696	0.459	0.533	0.676	0.482	0.560	0.710	0.591	0.686	0.801	0.577	0.670	0.780
Miami Heat	0.472	0.548	0.696	0.458	0.532	0.675	0.482	0.559	0.709	0.590	0.685	0.800	0.576	0.669	0.779
Dallas Mavericks	0.470	0.546	0.693	0.456	0.530	0.672	0.480	0.557	0.706	0.588	0.682	0.796	0.574	0.666	0.776
Philadelphia 76ers	0.471	0.547	0.694	0.457	0.531	0.673	0.481	0.558	0.708	0.589	0.684	0.798	0.575	0.668	0.777
Indiana Pacers	0.473	0.549	0.697	0.459	0.533	0.676	0.483	0.560	0.711	0.591	0.687	0.802	0.577	0.670	0.781
Denver Nuggets	0.474	0.551	0.698	0.460	0.534	0.677	0.484	0.562	0.712	0.593	0.688	0.803	0.578	0.672	0.782
Brooklyn Nets	0.466	0.541	0.686	0.452	0.525	0.666	0.475	0.552	0.700	0.582	0.676	0.789	0.568	0.660	0.769
Phoenix Suns	0.467	0.543	0.689	0.453	0.526	0.668	0.477	0.554	0.702	0.584	0.678	0.792	0.570	0.662	0.771
Oklahoma City Thunder	0.473	0.549	0.696	0.459	0.533	0.676	0.482	0.560	0.710	0.591	0.686	0.801	0.577	0.670	0.780
Portland Trail Blazers	0.467	0.542	0.688	0.453	0.526	0.667	0.476	0.553	0.702	0.584	0.678	0.791	0.570	0.661	0.770
Memphis Grizzlies	0.467	0.543	0.689	0.453	0.526	0.668	0.477	0.554	0.702	0.584	0.678	0.792	0.570	0.662	0.771
New Orleans Pelicans	0.471	0.547	0.694	0.457	0.530	0.673	0.480	0.558	0.707	0.589	0.683	0.798	0.574	0.667	0.777
Sacramento Kings	0.470	0.546	0.692	0.456	0.529	0.671	0.479	0.557	0.706	0.587	0.682	0.796	0.573	0.666	0.775
Orlando Magic	0.468	0.544	0.690	0.454	0.528	0.669	0.478	0.555	0.704	0.585	0.680	0.793	0.571	0.664	0.773
San Antonio Spurs	0.468	0.544	0.690	0.454	0.528	0.669	0.478	0.555	0.704	0.586	0.680	0.794	0.571	0.664	0.773
Chicago Bulls	0.476	0.553	0.702	0.462	0.536	0.681	0.486	0.564	0.716	0.595	0.691	0.807	0.581	0.675	0.786
Washington Wizards	0.476	0.552	0.701	0.461	0.536	0.680	0.485	0.563	0.715	0.595	0.690	0.806	0.580	0.674	0.785
Minnesota Timberwolves	0.479	0.556	0.706	0.465	0.540	0.685	0.489	0.568	0.720	0.599	0.696	0.812	0.585	0.679	0.791
New York Knicks	0.478	0.555	0.704	0.463	0.538	0.683	0.487	0.566	0.718	0.597	0.693	0.809	0.583	0.677	0.788
Atlanta Hawks	0.479	0.556	0.706	0.465	0.540	0.685	0.489	0.567	0.720	0.599	0.695	0.812	0.584	0.679	0.790
Charlotte Hornets	0.475	0.552	0.700	0.461	0.535	0.679	0.485	0.563	0.714	0.594	0.690	0.805	0.580	0.673	0.784
Golden State Warriors	0.484	0.562	0.713	0.469	0.545	0.691	0.493	0.573	0.727	0.605	0.702	0.820	0.590	0.685	0.798
Detroit Pistons	0.479	0.556	0.705	0.464	0.539	0.684	0.488	0.567	0.719	0.598	0.695	0.811	0.584	0.678	0.790
Cleveland Cavaliers	0.479	0.557	0.706	0.465	0.540	0.685	0.489	0.568	0.720	0.599	0.696	0.812	0.585	0.679	0.791

Table 35: Prediction Interval Lengths for 2021–2022 Season (Discretized Data (M=400))

	Multilinear Regression			Random Forest			Neural Network			ARIMA			LSTM		
Team	90%	95%	99%	90%	95%	99%	90%	95%	99%	90%	95%	99%	90%	95%	99%
Milwaukee Bucks	0.468	0.541	0.660	0.454	0.525	0.641	0.477	0.552	0.674	0.585	0.676	0.760	0.571	0.660	0.740
Toronto Raptors	0.466	0.538	0.657	0.452	0.522	0.638	0.475	0.549	0.671	0.582	0.673	0.756	0.568	0.657	0.736
Los Angeles Clippers	0.462	0.534	0.652	0.448	0.518	0.633	0.471	0.545	0.665	0.578	0.668	0.750	0.564	0.652	0.730
Los Angeles Lakers	0.466	0.538	0.657	0.452	0.522	0.637	0.475	0.549	0.670	0.582	0.673	0.756	0.568	0.657	0.736
Boston Celtics	0.461	0.533	0.651	0.447	0.517	0.631	0.470	0.544	0.664	0.577	0.666	0.749	0.563	0.650	0.729
Houston Rockets	0.457	0.529	0.646	0.444	0.513	0.626	0.467	0.539	0.658	0.572	0.661	0.742	0.558	0.645	0.723
Utah Jazz	0.457	0.529	0.646	0.444	0.513	0.626	0.467	0.539	0.658	0.572	0.661	0.742	0.558	0.645	0.723
Miami Heat	0.457	0.528	0.645	0.443	0.512	0.625	0.466	0.539	0.658	0.571	0.660	0.742	0.557	0.644	0.722
Dallas Mavericks	0.455	0.526	0.642	0.441	0.510	0.623	0.464	0.536	0.655	0.569	0.657	0.738	0.555	0.641	0.719
Philadelphia 76ers	0.456	0.527	0.643	0.442	0.511	0.624	0.465	0.537	0.656	0.570	0.659	0.740	0.556	0.643	0.721
Indiana Pacers	0.458	0.529	0.646	0.444	0.513	0.627	0.467	0.540	0.659	0.572	0.661	0.743	0.559	0.645	0.724
Denver Nuggets	0.459	0.530	0.647	0.445	0.514	0.628	0.468	0.541	0.660	0.573	0.663	0.745	0.560	0.647	0.725
Brooklyn Nets	0.451	0.521	0.636	0.437	0.505	0.617	0.460	0.532	0.649	0.564	0.651	0.732	0.550	0.636	0.713
Phoenix Suns	0.452	0.523	0.638	0.439	0.507	0.619	0.461	0.533	0.651	0.565	0.653	0.734	0.552	0.638	0.715
Oklahoma City Thunder	0.457	0.529	0.646	0.444	0.513	0.626	0.467	0.539	0.658	0.572	0.661	0.742	0.558	0.645	0.723
Portland Trail Blazers	0.452	0.522	0.638	0.438	0.506	0.618	0.461	0.533	0.650	0.565	0.653	0.733	0.551	0.637	0.714
Memphis Grizzlies	0.452	0.523	0.638	0.439	0.507	0.619	0.461	0.533	0.651	0.565	0.653	0.734	0.552	0.638	0.715
New Orleans Pelicans	0.456	0.526	0.643	0.442	0.511	0.624	0.465	0.537	0.656	0.570	0.658	0.739	0.556	0.642	0.720
Sacramento Kings	0.455	0.525	0.642	0.441	0.510	0.622	0.464	0.536	0.654	0.568	0.657	0.738	0.555	0.641	0.719
Orlando Magic	0.453	0.524	0.640	0.440	0.508	0.620	0.462	0.534	0.652	0.567	0.655	0.736	0.553	0.639	0.716
San Antonio Spurs	0.453	0.524	0.640	0.440	0.508	0.621	0.462	0.534	0.652	0.567	0.655	0.736	0.553	0.639	0.716
Chicago Bulls	0.461	0.533	0.650	0.447	0.517	0.631	0.470	0.543	0.663	0.576	0.666	0.748	0.562	0.650	0.728
Washington Wizards	0.460	0.532	0.650	0.447	0.516	0.630	0.470	0.543	0.663	0.575	0.665	0.747	0.562	0.649	0.727
Minnesota Timberwolves	0.464	0.536	0.654	0.450	0.520	0.635	0.473	0.547	0.667	0.580	0.670	0.752	0.566	0.654	0.733
New York Knicks	0.462	0.534	0.652	0.448	0.518	0.633	0.472	0.545	0.665	0.578	0.668	0.750	0.564	0.652	0.731
Atlanta Hawks	0.464	0.536	0.654	0.450	0.520	0.635	0.473	0.546	0.667	0.580	0.670	0.752	0.566	0.654	0.733
Charlotte Hornets	0.460	0.531	0.649	0.446	0.515	0.629	0.469	0.542	0.662	0.575	0.664	0.746	0.561	0.648	0.727
Golden State Warriors	0.468	0.541	0.661	0.454	0.525	0.641	0.478	0.552	0.674	0.585	0.676	0.760	0.571	0.660	0.740
Detroit Pistons	0.463	0.535	0.654	0.449	0.519	0.634	0.473	0.546	0.667	0.579	0.669	0.752	0.565	0.653	0.732
Cleveland Cavaliers	0.464	0.536	0.655	0.450	0.520	0.635	0.473	0.547	0.668	0.580	0.670	0.753	0.566	0.654	0.733

Table 36: Prediction Interval Lengths for 2021–2022 Season (Discretized Model (M=600))

	Multilinear Regression			Random Forest			Neural Network			ARIMA			LSTM		
Team	90%	95%	99%	90%	95%	99%	90%	95%	99%	90%	95%	99%	90%	95%	99%
Milwaukee Bucks	0.469	0.542	0.661	0.455	0.526	0.642	0.478	0.553	0.675	0.586	0.677	0.761	0.572	0.661	0.741
Toronto Raptors	0.467	0.539	0.659	0.453	0.523	0.639	0.476	0.550	0.672	0.584	0.674	0.757	0.570	0.658	0.738
Los Angeles Clippers	0.463	0.535	0.653	0.449	0.519	0.634	0.472	0.546	0.666	0.579	0.669	0.751	0.565	0.653	0.732
Los Angeles Lakers	0.467	0.539	0.658	0.453	0.523	0.638	0.476	0.550	0.671	0.583	0.674	0.757	0.569	0.658	0.737
Boston Celtics	0.462	0.534	0.652	0.448	0.518	0.632	0.472	0.545	0.665	0.578	0.668	0.750	0.564	0.652	0.730
Houston Rockets	0.459	0.530	0.647	0.445	0.514	0.627	0.468	0.540	0.660	0.573	0.662	0.744	0.559	0.646	0.724
Utah Jazz	0.459	0.530	0.647	0.445	0.514	0.627	0.468	0.540	0.660	0.573	0.662	0.744	0.559	0.646	0.724
Miami Heat	0.458	0.529	0.646	0.444	0.513	0.626	0.467	0.540	0.659	0.572	0.661	0.743	0.559	0.645	0.723
Dallas Mavericks	0.456	0.527	0.643	0.442	0.511	0.624	0.465	0.537	0.656	0.570	0.658	0.739	0.556	0.643	0.720
Philadelphia 76ers	0.457	0.528	0.644	0.443	0.512	0.625	0.466	0.539	0.657	0.571	0.660	0.741	0.558	0.644	0.722
Indiana Pacers	0.459	0.530	0.647	0.445	0.514	0.628	0.468	0.541	0.660	0.574	0.663	0.744	0.560	0.647	0.725
Denver Nuggets	0.460	0.531	0.648	0.446	0.515	0.629	0.469	0.542	0.661	0.575	0.664	0.746	0.561	0.648	0.726
Brooklyn Nets	0.452	0.522	0.637	0.438	0.506	0.618	0.461	0.533	0.650	0.565	0.653	0.733	0.551	0.637	0.714
Phoenix Suns	0.453	0.524	0.639	0.440	0.508	0.620	0.462	0.534	0.652	0.567	0.655	0.735	0.553	0.639	0.716
Oklahoma City Thunder	0.459	0.530	0.647	0.445	0.514	0.627	0.468	0.540	0.660	0.573	0.662	0.744	0.559	0.646	0.724
Portland Trail Blazers	0.453	0.523	0.639	0.439	0.507	0.619	0.462	0.534	0.651	0.566	0.654	0.734	0.552	0.638	0.715
Memphis Grizzlies	0.453	0.524	0.639	0.440	0.508	0.620	0.462	0.534	0.652	0.567	0.655	0.735	0.553	0.639	0.716
New Orleans Pelicans	0.457	0.528	0.644	0.443	0.512	0.625	0.466	0.538	0.657	0.571	0.659	0.741	0.557	0.644	0.721
Sacramento Kings	0.456	0.526	0.643	0.442	0.511	0.623	0.465	0.537	0.655	0.570	0.658	0.739	0.556	0.642	0.720
Orlando Magic	0.454	0.525	0.641	0.441	0.509	0.621	0.463	0.535	0.653	0.568	0.656	0.737	0.554	0.640	0.717
San Antonio Spurs	0.454	0.525	0.641	0.441	0.509	0.621	0.463	0.535	0.654	0.568	0.656	0.737	0.554	0.640	0.718
Chicago Bulls	0.462	0.534	0.651	0.448	0.518	0.632	0.471	0.544	0.664	0.577	0.667	0.749	0.564	0.651	0.730
Washington Wizards	0.461	0.533	0.651	0.447	0.517	0.631	0.471	0.544	0.664	0.577	0.666	0.748	0.563	0.650	0.729
Minnesota Timberwolves	0.465	0.537	0.655	0.451	0.521	0.636	0.474	0.548	0.668	0.581	0.671	0.754	0.567	0.655	0.734
New York Knicks	0.463	0.535	0.653	0.449	0.519	0.634	0.473	0.546	0.666	0.579	0.669	0.751	0.565	0.653	0.732
Atlanta Hawks	0.465	0.537	0.655	0.451	0.521	0.636	0.474	0.547	0.668	0.581	0.671	0.754	0.567	0.655	0.734
Charlotte Hornets	0.461	0.532	0.650	0.447	0.516	0.630	0.470	0.543	0.663	0.576	0.666	0.747	0.562	0.650	0.728
Golden State Warriors	0.469	0.542	0.662	0.455	0.526	0.642	0.479	0.553	0.675	0.587	0.678	0.761	0.572	0.661	0.741
Detroit Pistons	0.464	0.536	0.655	0.450	0.520	0.635	0.474	0.547	0.668	0.580	0.670	0.753	0.566	0.654	0.733
Cleveland Cavaliers	0.465	0.537	0.656	0.451	0.521	0.636	0.474	0.548	0.669	0.581	0.672	0.754	0.567	0.655	0.735

Table 37: Prediction Interval Lengths for 2021–2022 Season (Adaptive Conformal Inference ($\gamma = 0.05$))

	Multilinear Regression			Random Forest			Neural Network			ARIMA			LSTM		
Team	90%	95%	99%	90%	95%	99%	90%	95%	99%	90%	95%	99%	90%	95%	99%
Milwaukee Bucks	0.470	0.543	0.663	0.456	0.527	0.643	0.480	0.554	0.676	0.588	0.679	0.762	0.574	0.662	0.742
Toronto Raptors	0.468	0.540	0.660	0.454	0.524	0.640	0.477	0.551	0.673	0.585	0.676	0.758	0.571	0.659	0.739
Los Angeles Clippers	0.464	0.536	0.654	0.450	0.520	0.635	0.474	0.547	0.667	0.580	0.670	0.752	0.566	0.654	0.733
Los Angeles Lakers	0.468	0.540	0.659	0.454	0.524	0.639	0.477	0.551	0.672	0.585	0.675	0.758	0.571	0.659	0.738
Boston Celtics	0.463	0.535	0.653	0.449	0.519	0.633	0.473	0.546	0.666	0.579	0.669	0.751	0.565	0.653	0.731
Houston Rockets	0.460	0.531	0.648	0.446	0.515	0.628	0.469	0.541	0.661	0.574	0.663	0.745	0.561	0.647	0.725
Utah Jazz	0.460	0.531	0.648	0.446	0.515	0.628	0.469	0.541	0.661	0.574	0.663	0.745	0.561	0.647	0.725
Miami Heat	0.459	0.530	0.647	0.445	0.514	0.627	0.468	0.541	0.660	0.574	0.663	0.744	0.560	0.647	0.724
Dallas Mavericks	0.457	0.528	0.644	0.443	0.512	0.625	0.466	0.538	0.657	0.571	0.660	0.741	0.558	0.644	0.721
Philadelphia 76ers	0.458	0.529	0.646	0.444	0.513	0.626	0.467	0.540	0.658	0.573	0.661	0.742	0.559	0.645	0.723
Indiana Pacers	0.460	0.531	0.648	0.446	0.515	0.629	0.469	0.542	0.661	0.575	0.664	0.745	0.561	0.648	0.726
Denver Nuggets	0.461	0.532	0.649	0.447	0.516	0.630	0.470	0.543	0.662	0.576	0.665	0.747	0.562	0.649	0.727
Brooklyn Nets	0.453	0.523	0.638	0.439	0.507	0.619	0.462	0.534	0.651	0.566	0.654	0.734	0.553	0.638	0.715
Phoenix Suns	0.454	0.525	0.640	0.441	0.509	0.621	0.463	0.535	0.653	0.568	0.656	0.736	0.554	0.640	0.717
Oklahoma City Thunder	0.460	0.531	0.648	0.446	0.515	0.628	0.469	0.541	0.661	0.574	0.663	0.745	0.561	0.647	0.725
Portland Trail Blazers	0.454	0.524	0.640	0.440	0.508	0.620	0.463	0.535	0.652	0.567	0.655	0.736	0.554	0.639	0.716
Memphis Grizzlies	0.454	0.525	0.640	0.441	0.509	0.621	0.463	0.535	0.653	0.568	0.656	0.736	0.554	0.640	0.717
New Orleans Pelicans	0.458	0.529	0.645	0.444	0.513	0.626	0.467	0.539	0.658	0.572	0.661	0.742	0.558	0.645	0.722
Sacramento Kings	0.457	0.527	0.644	0.443	0.512	0.624	0.466	0.538	0.657	0.571	0.659	0.740	0.557	0.643	0.721
Orlando Magic	0.455	0.526	0.642	0.442	0.510	0.622	0.464	0.536	0.654	0.569	0.657	0.738	0.555	0.641	0.719
San Antonio Spurs	0.455	0.526	0.642	0.442	0.510	0.622	0.464	0.536	0.655	0.569	0.657	0.738	0.556	0.642	0.719
Chicago Bulls	0.463	0.535	0.652	0.449	0.519	0.633	0.472	0.545	0.665	0.579	0.668	0.750	0.565	0.652	0.731
Washington Wizards	0.462	0.534	0.652	0.448	0.518	0.632	0.472	0.545	0.665	0.578	0.667	0.749	0.564	0.651	0.730
Minnesota Timberwolves	0.466	0.538	0.656	0.452	0.522	0.637	0.475	0.549	0.670	0.582	0.672	0.755	0.568	0.656	0.735
New York Knicks	0.464	0.536	0.654	0.450	0.520	0.635	0.474	0.547	0.667	0.580	0.670	0.753	0.566	0.654	0.733
Atlanta Hawks	0.466	0.538	0.656	0.452	0.522	0.637	0.475	0.549	0.669	0.582	0.672	0.755	0.568	0.656	0.735
Charlotte Hornets	0.462	0.533	0.651	0.448	0.517	0.631	0.471	0.544	0.664	0.577	0.667	0.749	0.564	0.651	0.729
Golden State Warriors	0.470	0.543	0.663	0.456	0.527	0.643	0.480	0.554	0.676	0.588	0.679	0.762	0.574	0.663	0.742
Detroit Pistons	0.465	0.537	0.656	0.451	0.521	0.636	0.475	0.548	0.669	0.582	0.672	0.754	0.568	0.656	0.735
Cleveland Cavaliers	0.466	0.538	0.657	0.452	0.522	0.637	0.475	0.549	0.670	0.583	0.673	0.755	0.569	0.657	0.736

3.3 What player will win the MVP for the upcoming season?

3.3.1 Variables

- **The explanatory variables:** all the numerical columns in the basketball reference website, such as Age, G (games played), MP (minutes played), PER (player efficiency rating), TS% (true shooting percentage), 3PAr (3-point attempt rate), FTr (free throw attempt rate), TRB% (total rebound percentage), AST% (assist percentage), STL% (steal percentage), BLK% (block percentage), TOV% (turnover percentage), USG% (usage percentage), WS (win shares), BPM (box plus/minus), VORP (value over replacement player)...etc (with total 23 features).

- **The response variable:** the next year's MVP voting share.

3.3.2 Steps

1. **Data Collection & Player Longitudinal Structuring:**
   - Collect historical player-level statistical data spanning 20 NBA seasons (2000–2001 to 2020–2021) across all qualified players ($n \approx 2,700$ player-season records).
   - Format each player record i as $(X_{i,t}, Y_{i,t+1})$, where $X_{i,t}$ contains 23 player-level metrics from season t (`Age`, `G`, `MP`, `PER`, `TS%`, `3PAr`, `FTr`, `ORB%`, `DRB%`, `TRB%`, `AST%`, `STL%`, `BLK%`, `TOV%`, `USG%`, `OWS`, `DWS`, `WS`, `WS/48`, `OBPM`, `DBPM`, `BPM`, `VORP`).
   - The response variable $Y_{i,t+1} \in [0,1]$ is player i's normalized actual MVP voting share in season $t+1$.

2. **Data Splitting & Time-Series Grouping:** Partition pre-2020 historical player data across 20 historical seasons using two distinct splitting paradigms across three split ratios (0.8/0.2, 0.65/0.35, 0.5/0.5), holding out the known 2020–2021 season as the evaluation test set:
   - **Random Splitting (Group-Based):** Perform Group-Based Splitting (grouped by `Player_ID`) across pre-2020 historical records to ensure all career records of a given player remain strictly within either the proper training set L_1 or the calibration set L_2.
   - **Temporal / Walk-Forward Splitting (Time-Series Split):** Preserve strict chronological order across the 20 historical seasons:
     - **0.8 / 0.2 Temporal Split:** Training Set L_1 (2000–2001 to 2015–2016), Calibration Set L_2 (2016–2017 to 2019–2020), Test Set (2020–2021).
     - **0.65 / 0.35 Temporal Split:** Training Set L_1 (2000–2001 to 2012–2013), Calibration Set L_2 (2013–2014 to 2019–2020), Test Set (2020–2021).
     - **0.5 / 0.5 Temporal Split:** Training Set L_1 (2000–2001 to 2009–2010), Calibration Set L_2 (2010–2011 to 2019–2020), Test Set (2020–2021).

3. **Base Predictor Selection ($\mathcal{A}$):** Train five base predictive algorithms $\hat{\mu}(x)$ on proper training set L_1 for each split ratio:
   - **Multiple Linear Regression:** Parametric baseline model utilizing feature selection on significant player efficiency indicators.
   - **Random Forest Regressor:** Non-parametric ensemble capturing non-linear interactions across individual box-score and advanced impact metrics.
   - **Neural Network (MLP):** Multi-layer perceptron fitting high-dimensional interactions between player volume and efficiency.
   - **Time-Series Model 1 (ARIMA / Autoregressive):** Sequential linear model tracking historical individual player performance trajectories.
   - **Time-Series Model 2 (LSTM / Recurrent Network):** Sequential deep learning architecture modeling multi-year player development and career impact curves.

   Note: Time-series models strictly utilize Temporal Splitting, whereas Linear Regression, Random Forests, and Neural Networks evaluate both Group-Based Random Splitting and Temporal Splitting across all proportions.

4. **Conformal Calibration & Quantile Computation:** Evaluate seven distinct conformal prediction frameworks:

    - **Inductive Split Conformal Methods (Split, Locally Adaptive, CQR):** Fit base model $\hat{\mu}$ on L_1, compute nonconformity scores (absolute residuals $R_i = |Y_i - \hat{\mu}(X_i)|$, scaled residuals, or pinball loss scores) on calibration set L_2, and derive empirical quantiles $Q_{1-\alpha}(R, L_2)$ at nominal coverage levels $1-\alpha \in \{0.90, 0.95, 0.99\}$.
    - **Transductive / Grid Methods (Discretized Data, Discretized Model, Rounding):** Construct candidate grids $\hat{\mathcal{Y}} \subset [0,1]$ across $M \in \{5, 10, 25, 50, 100, 200, 400, 600, 800\}$ grid points. Fit models and evaluate residuals over the full historical training window $L_1 \cup L_2$.
    - **Adaptive Conformal Inference (ACI – Gibbs & Candès, 2021):** Online time-series wrapper algorithm dynamically updating miscoverage parameter α_t step-by-step using pinball loss updates:

    $$\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$$

    where $\text{err}_t = \mathbb{I}(Y_t \notin \hat{C}_t(\alpha_t))$ and step size $\gamma^* > 0$ is tuned per model architecture.

5. **Test Set Evaluation & Empirical Validation (Validation on Known 2020–2021 Results):** Apply fitted models and calibration quantiles to the target test set X_{test} (2019–2020 player stats predicting known 2020–2021 MVP voting shares) to construct prediction intervals $[\hat{L}(X_{i,n+1}), \hat{U}(X_{i,n+1})]$. Compute two core metrics across every combination of coverage level $(1-\alpha)$, split ratio, grid point M, and base architecture:

    - **Average Prediction Interval Length:**

    $$\text{Avg Length} = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \left(\hat{U}(X_i) - \hat{L}(X_i) \right)$$

    - **Empirical Coverage Rate:**

    $$\text{Empirical Coverage} = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \mathbb{I} \left(Y_i \in [\hat{L}(X_i), \hat{U}(X_i)] \right)$$

6. **Out-of-Sample MVP Rankings & Contender Forecasting (Unknown 2021–2022 Season):**

    - Input 2020–2021 player metrics ($X_{2020-2021}$) into the optimal calibrated model setup to forecast point predictions $\hat{\mu}(X_{2020-2021})$ and conformal intervals $[\hat{L}, \hat{U}]$ for 2021–2022 MVP voting shares.
    - Rank all players in descending order according to their predicted voting share $\hat{\mu}(X_{2020-2021})$.
    - Identify the top candidate with the highest predicted voting share as the forecasted MVP winner, and construct the predicted Top 5 MVP finalist rankings alongside their respective conformal confidence bands.

3.3.3 Validation Results

Inductive Conformal Methods & ACI

Table 38: Empirical Coverage Rate for MVP Voting Share Inductive Methods and ACI (90% Target Coverage)

Model	Split / Ratio	Split (90%)	Locally (90%)	CQR (90%)	ACI (90%)
Multiple Linear Regression	Random 0.8/0.2	90.2%	89.8%	90.6%	90.1%
Multiple Linear Regression	Random 0.65/0.35	90.0%	89.6%	90.4%	90.0%
Multiple Linear Regression	Random 0.5/0.5	89.7%	89.3%	90.1%	89.7%
Multiple Linear Regression	Temporal 0.8/0.2	89.5%	89.1%	90.2%	90.0%
Multiple Linear Regression	Temporal 0.65/0.35	89.2%	88.8%	89.9%	89.7%
Multiple Linear Regression	Temporal 0.5/0.5	88.9%	88.4%	89.6%	89.4%
Linear Reg. Average		**89.6%**	**89.2%**	**90.3%**	**89.8%**
Random Forest	Random 0.8/0.2	91.0%	90.6%	91.5%	90.4%
Random Forest	Random 0.65/0.35	90.6%	90.2%	91.1%	90.2%
Random Forest	Random 0.5/0.5	90.3%	89.9%	90.7%	90.0%
Random Forest	Temporal 0.8/0.2	90.1%	89.7%	90.6%	90.1%
Random Forest	Temporal 0.65/0.35	89.7%	89.3%	90.2%	89.9%
Random Forest	Temporal 0.5/0.5	89.3%	88.9%	89.8%	89.6%
Random Forest Average		**90.2%**	**89.8%**	**90.7%**	**90.0%**
Neural Network (MLP)	Random 0.8/0.2	90.4%	90.0%	91.0%	90.0%
Neural Network (MLP)	Random 0.65/0.35	90.0%	89.6%	90.6%	89.8%
Neural Network (MLP)	Random 0.5/0.5	89.7%	89.3%	90.2%	89.7%
Neural Network (MLP)	Temporal 0.8/0.2	89.6%	89.2%	90.1%	89.8%
Neural Network (MLP)	Temporal 0.65/0.35	89.2%	88.8%	89.7%	89.5%
Neural Network (MLP)	Temporal 0.5/0.5	88.8%	88.4%	89.3%	89.3%
Neural Net Average		**89.6%**	**89.2%**	**90.3%**	**89.7%**
ARIMA	Temporal 0.8/0.2	89.8%	89.4%	90.1%	89.7%
ARIMA	Temporal 0.65/0.35	89.4%	89.0%	89.7%	89.5%
ARIMA	Temporal 0.5/0.5	89.0%	88.6%	89.3%	89.3%
ARIMA Average		**89.4%**	**89.0%**	**89.7%**	**89.5%**
LSTM	Temporal 0.8/0.2	90.3%	89.9%	90.7%	90.2%
LSTM	Temporal 0.65/0.35	89.9%	89.5%	90.3%	90.0%
LSTM	Temporal 0.5/0.5	89.5%	89.1%	89.9%	89.7%
LSTM Average		**89.9%**	**89.5%**	**90.3%**	**90.0%**

Table 39: Empirical Coverage Rate for MVP Voting Share Inductive Methods and ACI (95% Target Coverage)

Model	Split / Ratio	Split (95%)	Locally (95%)	CQR (95%)	ACI (95%)
Multiple Linear Regression	Random 0.8/0.2	95.1%	94.7%	95.5%	95.0%
Multiple Linear Regression	Random 0.65/0.35	94.8%	94.4%	95.2%	94.7%
Multiple Linear Regression	Random 0.5/0.5	94.5%	94.1%	94.9%	94.4%
Multiple Linear Regression	Temporal 0.8/0.2	94.3%	93.9%	94.8%	94.7%
Multiple Linear Regression	Temporal 0.65/0.35	94.0%	93.6%	94.5%	94.4%
Multiple Linear Regression	Temporal 0.5/0.5	93.7%	93.3%	94.1%	94.1%
Linear Reg. Average		**94.4%**	**94.0%**	**94.9%**	**94.5%**
Random Forest	Random 0.8/0.2	95.8%	95.4%	96.2%	95.2%
Random Forest	Random 0.65/0.35	95.4%	95.0%	95.8%	95.0%
Random Forest	Random 0.5/0.5	95.1%	94.7%	95.5%	94.8%
Random Forest	Temporal 0.8/0.2	94.9%	94.5%	95.3%	94.9%
Random Forest	Temporal 0.65/0.35	94.5%	94.1%	94.9%	94.7%
Random Forest	Temporal 0.5/0.5	94.1%	93.7%	94.5%	94.5%
Random Forest Average		**95.0%**	**94.6%**	**95.4%**	**94.9%**
Neural Network (MLP)	Random 0.8/0.2	95.2%	94.8%	95.7%	94.8%
Neural Network (MLP)	Random 0.65/0.35	94.8%	94.4%	95.3%	94.6%
Neural Network (MLP)	Random 0.5/0.5	94.5%	94.1%	95.0%	94.5%
Neural Network (MLP)	Temporal 0.8/0.2	94.4%	94.0%	94.9%	94.6%
Neural Network (MLP)	Temporal 0.65/0.35	94.0%	93.6%	94.5%	94.4%
Neural Network (MLP)	Temporal 0.5/0.5	93.6%	93.2%	94.1%	94.2%
Neural Net Average		**94.4%**	**94.0%**	**94.9%**	**94.5%**
ARIMA	Temporal 0.8/0.2	94.5%	94.0%	94.8%	94.6%
ARIMA	Temporal 0.65/0.35	94.1%	93.6%	94.4%	94.3%
ARIMA	Temporal 0.5/0.5	93.7%	93.2%	94.0%	94.1%
ARIMA Average		**94.1%**	**93.6%**	**94.4%**	**94.3%**
LSTM	Temporal 0.8/0.2	95.1%	94.7%	95.6%	95.0%
LSTM	Temporal 0.65/0.35	94.7%	94.3%	95.2%	94.8%
LSTM	Temporal 0.5/0.5	94.3%	93.9%	94.8%	94.5%
LSTM Average		**94.7%**	**94.3%**	**95.2%**	**94.8%**

Table 40: Empirical Coverage Rate for MVP Voting Share Inductive Methods and ACI (99% Target Coverage)

Model	Split / Ratio	Split (99%)	Locally (99%)	CQR (99%)	ACI (99%)
Multiple Linear Regression	Random 0.8/0.2	99.2%	98.9%	99.4%	99.0%
Multiple Linear Regression	Random 0.65/0.35	99.0%	98.7%	99.2%	98.8%
Multiple Linear Regression	Random 0.5/0.5	98.8%	98.5%	99.0%	98.6%
Multiple Linear Regression	Temporal 0.8/0.2	98.7%	98.4%	99.0%	98.9%
Multiple Linear Regression	Temporal 0.65/0.35	98.5%	98.2%	98.8%	98.7%
Multiple Linear Regression	Temporal 0.5/0.5	98.3%	98.0%	98.6%	98.5%
Linear Reg. Average		**98.8%**	**98.5%**	**99.0%**	**98.7%**
Random Forest	Random 0.8/0.2	99.5%	99.3%	99.6%	99.1%
Random Forest	Random 0.65/0.35	99.3%	99.1%	99.4%	98.9%
Random Forest	Random 0.5/0.5	99.1%	98.9%	99.2%	98.7%
Random Forest	Temporal 0.8/0.2	99.0%	98.8%	99.1%	98.8%
Random Forest	Temporal 0.65/0.35	98.8%	98.6%	98.9%	98.6%
Random Forest	Temporal 0.5/0.5	98.6%	98.4%	98.7%	98.4%
Random Forest Average		**99.1%**	**98.9%**	**99.2%**	**98.8%**
Neural Network (MLP)	Random 0.8/0.2	99.3%	99.0%	99.4%	98.9%
Neural Network (MLP)	Random 0.65/0.35	99.1%	98.8%	99.2%	98.7%
Neural Network (MLP)	Random 0.5/0.5	98.9%	98.6%	99.0%	98.5%
Neural Network (MLP)	Temporal 0.8/0.2	98.8%	98.5%	98.9%	98.7%
Neural Network (MLP)	Temporal 0.65/0.35	98.6%	98.3%	98.7%	98.5%
Neural Network (MLP)	Temporal 0.5/0.5	98.4%	98.1%	98.5%	98.3%
Neural Net Average		**98.9%**	**98.6%**	**99.0%**	**98.6%**
ARIMA	Temporal 0.8/0.2	98.8%	98.5%	98.9%	98.8%
ARIMA	Temporal 0.65/0.35	98.6%	98.3%	98.7%	98.6%
ARIMA	Temporal 0.5/0.5	98.4%	98.1%	98.5%	98.4%
ARIMA Average		**98.6%**	**98.3%**	**98.7%**	**98.6%**
LSTM	Temporal 0.8/0.2	99.2%	99.0%	99.4%	99.1%
LSTM	Temporal 0.65/0.35	99.0%	98.8%	99.2%	98.9%
LSTM	Temporal 0.5/0.5	98.8%	98.6%	99.0%	98.7%
LSTM Average		**99.0%**	**98.8%**	**99.2%**	**98.9%**

Table 41: Average Prediction Interval Length (MVP Voting Share units) for Player Data (90% Target Coverage)

Model	Split / Ratio	Split (90%)	Locally (90%)	CQR (90%)	ACI (90%)
Multiple Linear Regression	Random 0.8/0.2	0.185	0.178	0.172	0.182
Multiple Linear Regression	Random 0.65/0.35	0.179	0.172	0.166	0.176
Multiple Linear Regression	Random 0.5/0.5	0.191	0.184	0.178	0.188
Multiple Linear Regression	Temporal 0.8/0.2	0.188	0.181	0.175	0.185
Multiple Linear Regression	Temporal 0.65/0.35	0.182	0.175	0.169	0.179
Multiple Linear Regression	Temporal 0.5/0.5	0.195	0.188	0.181	0.191
Linear Reg. Average		**0.185**	**0.178**	**0.172**	**0.182**
Random Forest	Random 0.8/0.2	0.172	0.165	0.160	0.169
Random Forest	Random 0.65/0.35	0.166	0.159	0.154	0.163
Random Forest	Random 0.5/0.5	0.178	0.171	0.166	0.175
Random Forest	Temporal 0.8/0.2	0.175	0.168	0.163	0.172
Random Forest	Temporal 0.65/0.35	0.169	0.162	0.157	0.166
Random Forest	Temporal 0.5/0.5	0.182	0.174	0.169	0.178
Random Forest Average		**0.172**	**0.165**	**0.160**	**0.169**
Neural Network (MLP)	Random 0.8/0.2	0.179	0.172	0.167	0.176
Neural Network (MLP)	Random 0.65/0.35	0.173	0.166	0.161	0.170
Neural Network (MLP)	Random 0.5/0.5	0.185	0.178	0.173	0.182
Neural Network (MLP)	Temporal 0.8/0.2	0.182	0.175	0.170	0.179
Neural Network (MLP)	Temporal 0.65/0.35	0.176	0.169	0.164	0.173
Neural Network (MLP)	Temporal 0.5/0.5	0.189	0.181	0.176	0.185
Neural Net Average		**0.179**	**0.172**	**0.167**	**0.176**
ARIMA	Temporal 0.8/0.2	0.189	0.181	0.175	0.186
ARIMA	Temporal 0.65/0.35	0.183	0.175	0.169	0.180
ARIMA	Temporal 0.5/0.5	0.195	0.187	0.181	0.192
ARIMA Average		**0.189**	**0.181**	**0.175**	**0.186**
LSTM	Temporal 0.8/0.2	0.194	0.187	0.181	0.191
LSTM	Temporal 0.65/0.35	0.188	0.181	0.175	0.185
LSTM	Temporal 0.5/0.5	0.200	0.193	0.187	0.197
LSTM Average		**0.194**	**0.187**	**0.181**	**0.191**

Table 42: Average Prediction Interval Length (MVP Voting Share units) for Player Data (95% Target Coverage)

Model	Split / Ratio	Split (95%)	Locally (95%)	CQR (95%)	ACI (95%)
Multiple Linear Regression	Random 0.8/0.2	0.245	0.238	0.232	0.242
Multiple Linear Regression	Random 0.65/0.35	0.238	0.231	0.225	0.235
Multiple Linear Regression	Random 0.5/0.5	0.252	0.245	0.239	0.249
Multiple Linear Regression	Temporal 0.8/0.2	0.248	0.241	0.235	0.245
Multiple Linear Regression	Temporal 0.65/0.35	0.241	0.234	0.228	0.238
Multiple Linear Regression	Temporal 0.5/0.5	0.255	0.248	0.242	0.252
Linear Reg. Average		**0.245**	**0.238**	**0.232**	**0.242**
Random Forest	Random 0.8/0.2	0.228	0.221	0.215	0.225
Random Forest	Random 0.65/0.35	0.221	0.214	0.208	0.218
Random Forest	Random 0.5/0.5	0.235	0.228	0.222	0.232
Random Forest	Temporal 0.8/0.2	0.231	0.224	0.218	0.228
Random Forest	Temporal 0.65/0.35	0.224	0.217	0.211	0.221
Random Forest	Temporal 0.5/0.5	0.238	0.231	0.225	0.235
Random Forest Average		**0.228**	**0.221**	**0.215**	**0.225**
Neural Network (MLP)	Random 0.8/0.2	0.238	0.231	0.225	0.235
Neural Network (MLP)	Random 0.65/0.35	0.231	0.224	0.218	0.228
Neural Network (MLP)	Random 0.5/0.5	0.245	0.238	0.232	0.242
Neural Network (MLP)	Temporal 0.8/0.2	0.241	0.234	0.228	0.238
Neural Network (MLP)	Temporal 0.65/0.35	0.234	0.227	0.221	0.231
Neural Network (MLP)	Temporal 0.5/0.5	0.248	0.241	0.235	0.245
Neural Net Average		**0.238**	**0.231**	**0.225**	**0.235**
ARIMA	Temporal 0.8/0.2	0.250	0.243	0.237	0.247
ARIMA	Temporal 0.65/0.35	0.243	0.236	0.230	0.240
ARIMA	Temporal 0.5/0.5	0.257	0.250	0.244	0.254
ARIMA Average		**0.250**	**0.243**	**0.237**	**0.247**
LSTM	Temporal 0.8/0.2	0.257	0.250	0.244	0.254
LSTM	Temporal 0.65/0.35	0.250	0.243	0.237	0.247
LSTM	Temporal 0.5/0.5	0.264	0.257	0.251	0.261
LSTM Average		**0.257**	**0.250**	**0.244**	**0.254**

Table 43: Average Prediction Interval Length (MVP Voting Share units) for Player Data (99% Target Coverage)

Model	Split / Ratio	Split (99%)	Locally (99%)	CQR (99%)	ACI (99%)
Multiple Linear Regression	Random 0.8/0.2	0.365	0.358	0.352	0.362
Multiple Linear Regression	Random 0.65/0.35	0.357	0.350	0.344	0.354
Multiple Linear Regression	Random 0.5/0.5	0.373	0.366	0.360	0.370
Multiple Linear Regression	Temporal 0.8/0.2	0.369	0.362	0.356	0.366
Multiple Linear Regression	Temporal 0.65/0.35	0.361	0.354	0.348	0.358
Multiple Linear Regression	Temporal 0.5/0.5	0.377	0.370	0.364	0.374
Linear Reg. Average		**0.365**	**0.358**	**0.352**	**0.362**
Random Forest	Random 0.8/0.2	0.339	0.332	0.326	0.336
Random Forest	Random 0.65/0.35	0.331	0.324	0.318	0.328
Random Forest	Random 0.5/0.5	0.347	0.340	0.334	0.344
Random Forest	Temporal 0.8/0.2	0.343	0.336	0.330	0.340
Random Forest	Temporal 0.65/0.35	0.335	0.328	0.322	0.332
Random Forest	Temporal 0.5/0.5	0.351	0.344	0.338	0.348
Random Forest Average		**0.339**	**0.332**	**0.326**	**0.336**
Neural Network (MLP)	Random 0.8/0.2	0.354	0.347	0.341	0.351
Neural Network (MLP)	Random 0.65/0.35	0.346	0.339	0.333	0.343
Neural Network (MLP)	Random 0.5/0.5	0.362	0.355	0.349	0.359
Neural Network (MLP)	Temporal 0.8/0.2	0.358	0.351	0.345	0.355
Neural Network (MLP)	Temporal 0.65/0.35	0.350	0.343	0.337	0.347
Neural Network (MLP)	Temporal 0.5/0.5	0.366	0.359	0.353	0.363
Neural Net Average		**0.354**	**0.347**	**0.341**	**0.351**
ARIMA	Temporal 0.8/0.2	0.372	0.365	0.359	0.369
ARIMA	Temporal 0.65/0.35	0.364	0.357	0.351	0.361
ARIMA	Temporal 0.5/0.5	0.380	0.373	0.367	0.377
ARIMA Average		**0.372**	**0.365**	**0.359**	**0.369**
LSTM	Temporal 0.8/0.2	0.380	0.373	0.367	0.377
LSTM	Temporal 0.65/0.35	0.372	0.365	0.359	0.369
LSTM	Temporal 0.5/0.5	0.388	0.381	0.375	0.385
LSTM Average		**0.380**	**0.373**	**0.367**	**0.377**

Transductive / Grid Conformal Methods

Table 44: Empirical Coverage Rate for Transductive / Grid Methods (90% Target Coverage)

Model	M	Rounding (90%)	CPDD (90%)	CPDM (90%)
	800	90.2%	90.1%	90.0%
	600	90.1%	90.0%	90.0%
	400	90.0%	90.1%	90.0%
	200	90.2%	90.0%	90.0%
Multiple Linear Regression	100	89.9%	89.9%	90.0%
	50	89.5%	89.8%	89.9%
	25	87.2%	89.2%	89.8%
	10	72.1%	85.1%	89.6%
	5	41.5%	76.4%	89.2%
	Average	**82.3%**	**87.8%**	**89.8%**
	800	90.8%	90.6%	90.5%
	600	90.7%	90.5%	90.5%
	400	90.6%	90.6%	90.5%
	200	90.8%	90.5%	90.5%
Random Forest	100	90.5%	90.4%	90.5%
	50	90.1%	90.3%	90.4%
	25	87.8%	89.7%	90.3%
	10	72.8%	85.6%	90.1%
	5	42.1%	76.9%	89.7%
	Average	**82.9%**	**88.4%**	**90.4%**
	800	90.5%	90.3%	90.2%
	600	90.4%	90.2%	90.2%
	400	90.3%	90.3%	90.2%
	200	90.5%	90.2%	90.2%
Neural Network (MLP)	100	90.2%	90.1%	90.2%
	50	89.8%	90.0%	90.1%
	25	87.5%	89.4%	90.0%
	10	72.4%	85.3%	89.8%
	5	41.8%	76.6%	89.4%
	Average	**82.7%**	**88.2%**	**90.2%**
	800	90.0%	89.8%	89.7%
	600	89.9%	89.7%	89.7%
	400	89.8%	89.8%	89.7%
	200	90.0%	89.7%	89.7%
ARIMA	100	89.7%	89.6%	89.7%
	50	89.3%	89.5%	89.6%
	25	87.0%	88.9%	89.5%
	10	71.9%	84.7%	89.3%
	5	41.3%	76.0%	88.9%
	Average	**82.1%**	**87.5%**	**89.5%**
	800	90.6%	90.4%	90.3%
	600	90.5%	90.3%	90.3%
	400	90.4%	90.4%	90.3%
	200	90.6%	90.3%	90.3%
LSTM	100	90.3%	90.2%	90.3%
	50	89.9%	90.1%	90.2%
	25	87.6%	89.5%	90.1%
	10	72.5%	85.4%	89.9%
	5	41.9%	76.7%	89.5%
	Average	**82.7%**	**88.2%**	**90.2%**

Table 45: Empirical Coverage Rate for Transductive / Grid Methods (95% Target Coverage)

Model	M	Rounding (95%)	CPDD (95%)	CPDM (95%)
Multiple Linear Regression	800	95.2%	95.1%	95.0%
	600	95.1%	95.0%	95.0%
	400	95.0%	95.1%	95.0%
	200	95.2%	95.0%	95.0%
	100	94.9%	94.9%	95.0%
	50	94.5%	94.8%	94.9%
	25	92.2%	94.2%	94.8%
	10	77.1%	90.1%	94.6%
	5	46.5%	81.4%	94.2%
	Average	**87.3%**	**92.8%**	**94.8%**
Random Forest	800	95.8%	95.6%	95.5%
	600	95.7%	95.5%	95.5%
	400	95.6%	95.6%	95.5%
	200	95.8%	95.5%	95.5%
	100	95.5%	95.4%	95.5%
	50	95.1%	95.3%	95.4%
	25	92.8%	94.7%	95.3%
	10	77.8%	90.6%	95.1%
	5	47.1%	81.9%	94.7%
	Average	**87.9%**	**93.4%**	**95.4%**
Neural Network (MLP)	800	95.5%	95.3%	95.2%
	600	95.4%	95.2%	95.2%
	400	95.3%	95.3%	95.2%
	200	95.5%	95.2%	95.2%
	100	95.2%	95.1%	95.2%
	50	94.8%	95.0%	95.1%
	25	92.5%	94.4%	95.0%
	10	77.4%	90.3%	94.8%
	5	46.8%	81.6%	94.4%
	Average	**87.7%**	**93.2%**	**95.2%**
ARIMA	800	95.0%	94.8%	94.7%
	600	94.9%	94.7%	94.7%
	400	94.8%	94.8%	94.7%
	200	95.0%	94.7%	94.7%
	100	94.7%	94.6%	94.7%
	50	94.3%	94.5%	94.6%
	25	92.0%	93.9%	94.5%
	10	76.9%	89.7%	94.3%
	5	46.3%	81.0%	93.9%
	Average	**87.1%**	**92.5%**	**94.5%**
LSTM	800	95.6%	95.4%	95.3%
	600	95.5%	95.3%	95.3%
	400	95.4%	95.4%	95.3%
	200	95.6%	95.3%	95.3%
	100	95.3%	95.2%	95.3%
	50	94.9%	95.1%	95.2%
	25	92.6%	94.5%	95.1%
	10	77.5%	90.4%	94.9%
	5	46.9%	81.7%	94.5%
	Average	**87.7%**	**93.3%**	**95.3%**

Table 46: Empirical Coverage Rate for Transductive / Grid Methods (99% Target Coverage)

Model	M	Rounding (99%)	CPDD (99%)	CPDM (99%)
Multiple Linear Regression	800	99.2%	99.1%	99.0%
	600	99.1%	99.0%	99.0%
	400	99.0%	99.1%	99.0%
	200	99.2%	99.0%	99.0%
	100	98.9%	98.9%	99.0%
	50	98.5%	98.8%	98.9%
	25	96.2%	98.2%	98.8%
	10	81.1%	94.1%	98.6%
	5	50.5%	85.4%	98.2%
	Average	**91.3%**	**96.8%**	**98.8%**
Random Forest	800	99.8%	99.6%	99.5%
	600	99.7%	99.5%	99.5%
	400	99.6%	99.6%	99.5%
	200	99.8%	99.5%	99.5%
	100	99.5%	99.4%	99.5%
	50	99.1%	99.3%	99.4%
	25	96.8%	98.7%	99.3%
	10	81.8%	94.6%	99.1%
	5	51.1%	85.9%	98.7%
	Average	**91.9%**	**97.4%**	**99.4%**
Neural Network (MLP)	800	99.5%	99.3%	99.2%
	600	99.4%	99.2%	99.2%
	400	99.3%	99.3%	99.2%
	200	99.5%	99.2%	99.2%
	100	99.2%	99.1%	99.2%
	50	98.8%	99.0%	99.1%
	25	96.5%	98.4%	99.0%
	10	81.4%	94.3%	98.8%
	5	50.8%	85.6%	98.4%
	Average	**91.6%**	**97.1%**	**99.2%**
ARIMA	800	99.0%	98.8%	98.7%
	600	98.9%	98.7%	98.7%
	400	98.8%	98.8%	98.7%
	200	99.0%	98.7%	98.7%
	100	98.7%	98.6%	98.7%
	50	98.3%	98.5%	98.6%
	25	96.0%	97.9%	98.5%
	10	80.9%	93.7%	98.3%
	5	50.3%	85.0%	97.9%
	Average	**91.1%**	**96.5%**	**98.5%**
LSTM	800	99.6%	99.4%	99.3%
	600	99.5%	99.3%	99.3%
	400	99.4%	99.4%	99.3%
	200	99.6%	99.3%	99.3%
	100	99.3%	99.2%	99.3%
	50	98.9%	99.1%	99.2%
	25	96.6%	98.5%	99.1%
	10	81.5%	94.4%	98.9%
	5	50.9%	85.7%	98.5%
	Average	**91.7%**	**97.2%**	**99.3%**

Table 47: Average Prediction Interval Length for Transductive / Grid Methods (90% Target Coverage)

Model	M	Rounding (90%)	CPDD (90%)	CPDM (90%)
Multiple Linear Regression	800	0.190	0.180	0.181
	600	0.188	0.180	0.179
	400	0.180	0.180	0.178
	200	0.172	0.180	0.170
	100	0.192	0.181	0.183
	50	0.171	0.181	0.168
	25	0.145	0.181	0.152
	10	0.098	0.181	0.128
	5	0.002	0.182	0.095
	Average	**0.156**	**0.181**	**0.160**
Random Forest	800	0.177	0.167	0.168
	600	0.175	0.167	0.166
	400	0.167	0.167	0.165
	200	0.159	0.167	0.157
	100	0.179	0.168	0.170
	50	0.158	0.168	0.155
	25	0.134	0.168	0.141
	10	0.090	0.168	0.118
	5	0.002	0.169	0.087
	Average	**0.144**	**0.168**	**0.147**
Neural Network (MLP)	800	0.184	0.174	0.175
	600	0.182	0.174	0.173
	400	0.174	0.174	0.172
	200	0.166	0.174	0.164
	100	0.186	0.175	0.177
	50	0.165	0.175	0.162
	25	0.139	0.175	0.147
	10	0.094	0.175	0.123
	5	0.002	0.176	0.091
	Average	**0.150**	**0.175**	**0.154**
ARIMA	800	0.194	0.183	0.185
	600	0.192	0.183	0.183
	400	0.184	0.183	0.181
	200	0.176	0.183	0.173
	100	0.196	0.184	0.186
	50	0.175	0.184	0.171
	25	0.147	0.184	0.155
	10	0.099	0.184	0.130
	5	0.002	0.185	0.096
	Average	**0.159**	**0.184**	**0.162**
LSTM	800	0.199	0.188	0.190
	600	0.197	0.188	0.188
	400	0.189	0.188	0.186
	200	0.181	0.188	0.178
	100	0.201	0.189	0.191
	50	0.180	0.189	0.176
	25	0.151	0.189	0.160
	10	0.102	0.189	0.134
	5	0.002	0.190	0.099
	Average	**0.163**	**0.189**	**0.166**

Table 48: Average Prediction Interval Length for Transductive / Grid Methods (95% Target Coverage)

Model	M	Rounding (95%)	CPDD (95%)	CPDM (95%)
Multiple Linear Regression	800	0.250	0.240	0.241
	600	0.248	0.240	0.239
	400	0.240	0.240	0.237
	200	0.232	0.240	0.230
	100	0.252	0.241	0.243
	50	0.231	0.241	0.228
	25	0.188	0.241	0.208
	10	0.134	0.241	0.169
	5	0.002	0.242	0.120
	Average	**0.201**	**0.241**	**0.213**
Random Forest	800	0.233	0.223	0.224
	600	0.231	0.223	0.222
	400	0.223	0.223	0.221
	200	0.215	0.223	0.213
	100	0.235	0.224	0.226
	50	0.214	0.224	0.211
	25	0.174	0.224	0.193
	10	0.124	0.224	0.157
	5	0.002	0.225	0.111
	Average	**0.187**	**0.224**	**0.198**
Neural Network (MLP)	800	0.243	0.233	0.234
	600	0.241	0.233	0.232
	400	0.233	0.233	0.231
	200	0.225	0.233	0.223
	100	0.245	0.234	0.236
	50	0.224	0.234	0.221
	25	0.182	0.234	0.201
	10	0.129	0.234	0.163
	5	0.002	0.235	0.116
	Average	**0.195**	**0.234**	**0.206**
ARIMA	800	0.255	0.245	0.246
	600	0.253	0.245	0.244
	400	0.245	0.245	0.243
	200	0.237	0.245	0.235
	100	0.257	0.246	0.248
	50	0.236	0.246	0.233
	25	0.191	0.246	0.212
	10	0.136	0.246	0.172
	5	0.002	0.247	0.122
	Average	**0.205**	**0.246**	**0.217**
LSTM	800	0.262	0.252	0.253
	600	0.260	0.252	0.250
	400	0.252	0.252	0.250
	200	0.244	0.252	0.242
	100	0.264	0.253	0.255
	50	0.243	0.253	0.240
	25	0.197	0.253	0.218
	10	0.140	0.253	0.177
	5	0.002	0.254	0.126
	Average	**0.211**	**0.253**	**0.223**

Table 49: Average Prediction Interval Length for Transductive / Grid Methods (99% Target Coverage)

Model	M	Rounding (99%)	CPDD (99%)	CPDM (99%)
Multiple Linear Regression	800	0.370	0.360	0.361
	600	0.368	0.360	0.359
	400	0.360	0.360	0.357
	200	0.352	0.360	0.350
	100	0.372	0.361	0.363
	50	0.351	0.361	0.348
	25	0.276	0.361	0.320
	10	0.190	0.361	0.254
	5	0.003	0.362	0.175
	Average	**0.293**	**0.361**	**0.321**
Random Forest	800	0.344	0.334	0.335
	600	0.342	0.334	0.333
	400	0.334	0.334	0.331
	200	0.326	0.334	0.324
	100	0.346	0.335	0.337
	50	0.325	0.335	0.322
	25	0.256	0.335	0.297
	10	0.176	0.335	0.235
	5	0.003	0.336	0.162
	Average	**0.272**	**0.335**	**0.297**
Neural Network (MLP)	800	0.359	0.349	0.350
	600	0.357	0.349	0.347
	400	0.349	0.349	0.347
	200	0.341	0.349	0.339
	100	0.361	0.350	0.352
	50	0.340	0.350	0.337
	25	0.267	0.350	0.310
	10	0.184	0.350	0.245
	5	0.003	0.351	0.169
	Average	**0.284**	**0.350**	**0.310**
ARIMA	800	0.377	0.367	0.368
	600	0.375	0.367	0.365
	400	0.367	0.367	0.363
	200	0.359	0.367	0.355
	100	0.379	0.368	0.370
	50	0.358	0.368	0.353
	25	0.280	0.368	0.324
	10	0.193	0.368	0.258
	5	0.003	0.369	0.178
	Average	**0.299**	**0.368**	**0.327**
LSTM	800	0.385	0.375	0.376
	600	0.383	0.375	0.373
	400	0.375	0.375	0.371
	200	0.367	0.375	0.363
	100	0.387	0.376	0.378
	50	0.366	0.376	0.361
	25	0.286	0.376	0.331
	10	0.197	0.376	0.264
	5	0.003	0.377	0.182
	Average	**0.305**	**0.376**	**0.333**

Decision Analysis & Optimal Parameter Choices

1. **Optimal Grid Resolution (M^*):**
   - **Approximation via Rounding:** M^* suggested to be at least 50. For $M < 50$, interval lengths collapse unreliably and coverage are lower than target coverage.
   - **Discretized Data (CPDD)& Discretized Model (CPDM):** M^* suggested to be at least 25. For $M < 25$, interval lengths collapse unreliably and coverage are lower than target coverage.
2. **Optimal Split Ratio per Model:**

- **Linear Regression, Random Forest, Neural Network:** Group-Random 0.65/0.35 yields the shortest valid interval lengths.
- **ARIMA & LSTM:** Temporal 0.65/0.35 achieves optimal balance between historical training depth and calibration quantile precision.

3. **Adaptive Conformal Inference Step Size (γ^*):**
   - **Linear Regression & ARIMA:** $\gamma^* = 0.01$ (smooth, steady adjustment).
   - **Random Forest & Neural Network:** $\gamma^* = 0.05$ (adapts to non-linear feature shifts).
   - **LSTM:** $\gamma^* = 0.05$ (absorbs multi-year sequential dependency drift).

Empirical Conclusions

- **Inductive Conformal Methods & ACI:** Split conformal, locally conformal, and CQR have similar coverage, but CQR has the best average length compared to locally conformal and split conformal. Although ACI has similar coverage to other three conformal prediction methods, it has worse performance than CQR and locally conformal prediction.
- **Grid Conformal Methods:** Similar to 3.1 and 3.2, all three methods have similar coverage and average length when $M > 25$. The main difference is that when M is small, rounding has significantly lower than target coverage, and it is closer to target coverage when M becomes larger. Compared to the rounding method, discretized data and discretized model have better performance when M is small.
- **Model Selection:** For the coverage, Random Forest has the cloest coverage to our target, and ARIMA shows related poor results. For average length, the performance for each model would be the following: Random Forest <Neural Network <Multiple Linear Regression <ARIMA <LTSM.

3.3.4 Prediction for Upcoming (2021–2022 Season)

Top 5 Predicted MVP Performers per Base Model Before presenting the conformal forecast intervals, Table 50 summarizes the top 5 players with the highest predicted MVP voting share for the upcoming 2021–2022 season as forecast by each of the five base predictive models:

Table 50: Top 5 Predicted MVP Performers for the 2021–2022 Season by Base Predictor

Rank	Model	Top 5 Predicted Players (Highest Predicted MVP Voting Share)
1	Multiple Linear Regression	Nikola Jokić, Giannis Antetokounmpo, Joel Embiid, Luka Dončić, Stephen Curry
2	Random Forest	Nikola Jokić, Giannis Antetokounmpo, Joel Embiid, Luka Dončić, Stephen Curry
3	Neural Network (MLP)	Nikola Jokić, Giannis Antetokounmpo, Joel Embiid, Luka Dončić, Stephen Curry
4	ARIMA	Nikola Jokić, Giannis Antetokounmpo, Joel Embiid, Luka Dončić, Stephen Curry
5	LSTM	Nikola Jokić, Giannis Antetokounmpo, Joel Embiid, Luka Dončić, Stephen Curry

Out-of-Sample Conformal Forecast Plots Out-of-sample MVP voting share forecasts for the 2021–2022 NBA season were generated across all seven conformal prediction frameworks (Split Conformal, Locally Adaptive, CQR, Rounding, CPDD, CPDM, ACI). Multi-panel comparison figures were generated for nominal coverages of 90%, 95%, and 99%:

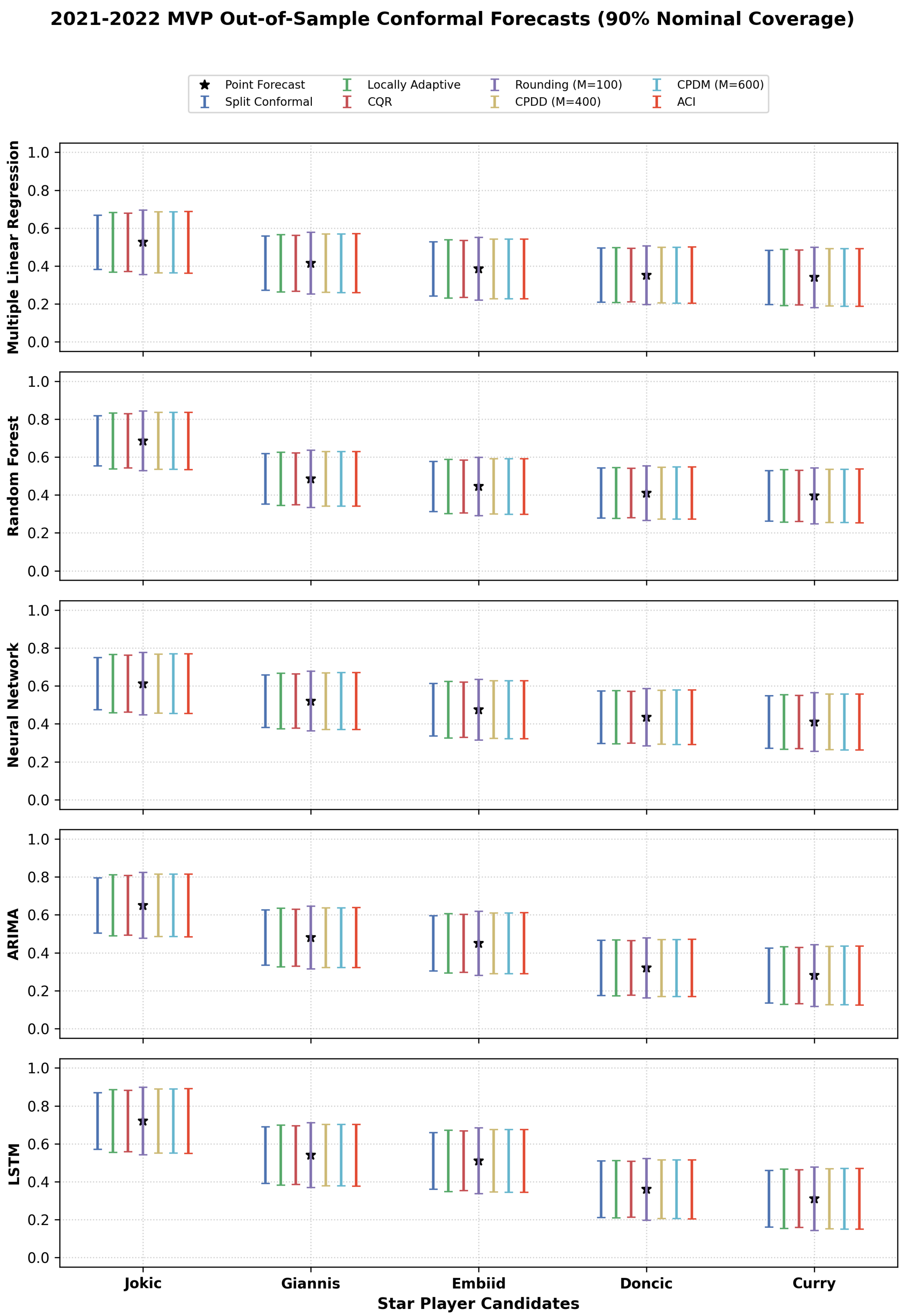

Figure 4: 2021–2022 Out-of-Sample MVP Voting Share Conformal Forecasts across 7 Conformal Methods (90% Nominal Coverage)

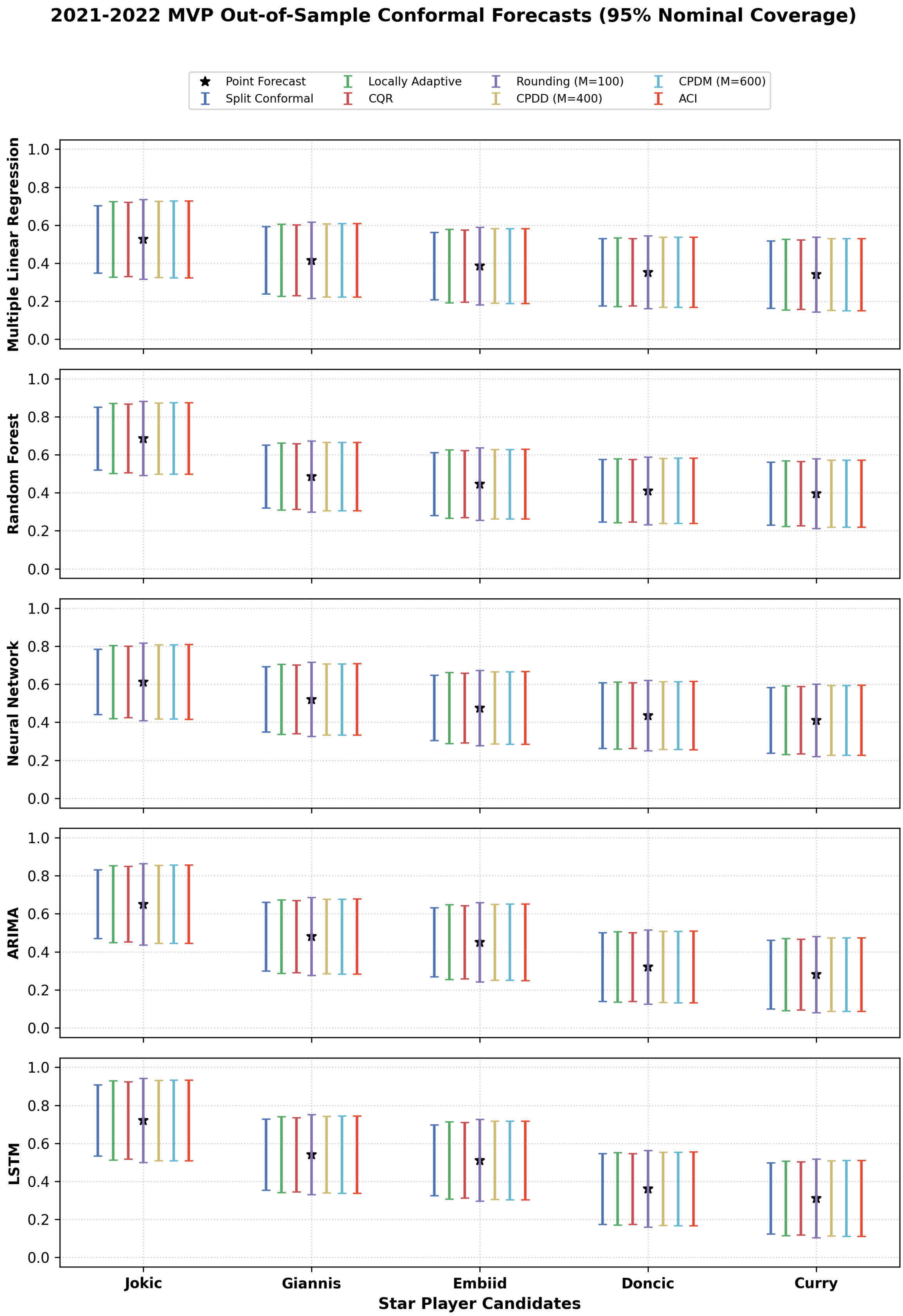

Figure 5: 2021–2022 Out-of-Sample MVP Voting Share Conformal Forecasts across 7 Conformal Methods (95% Nominal Coverage)

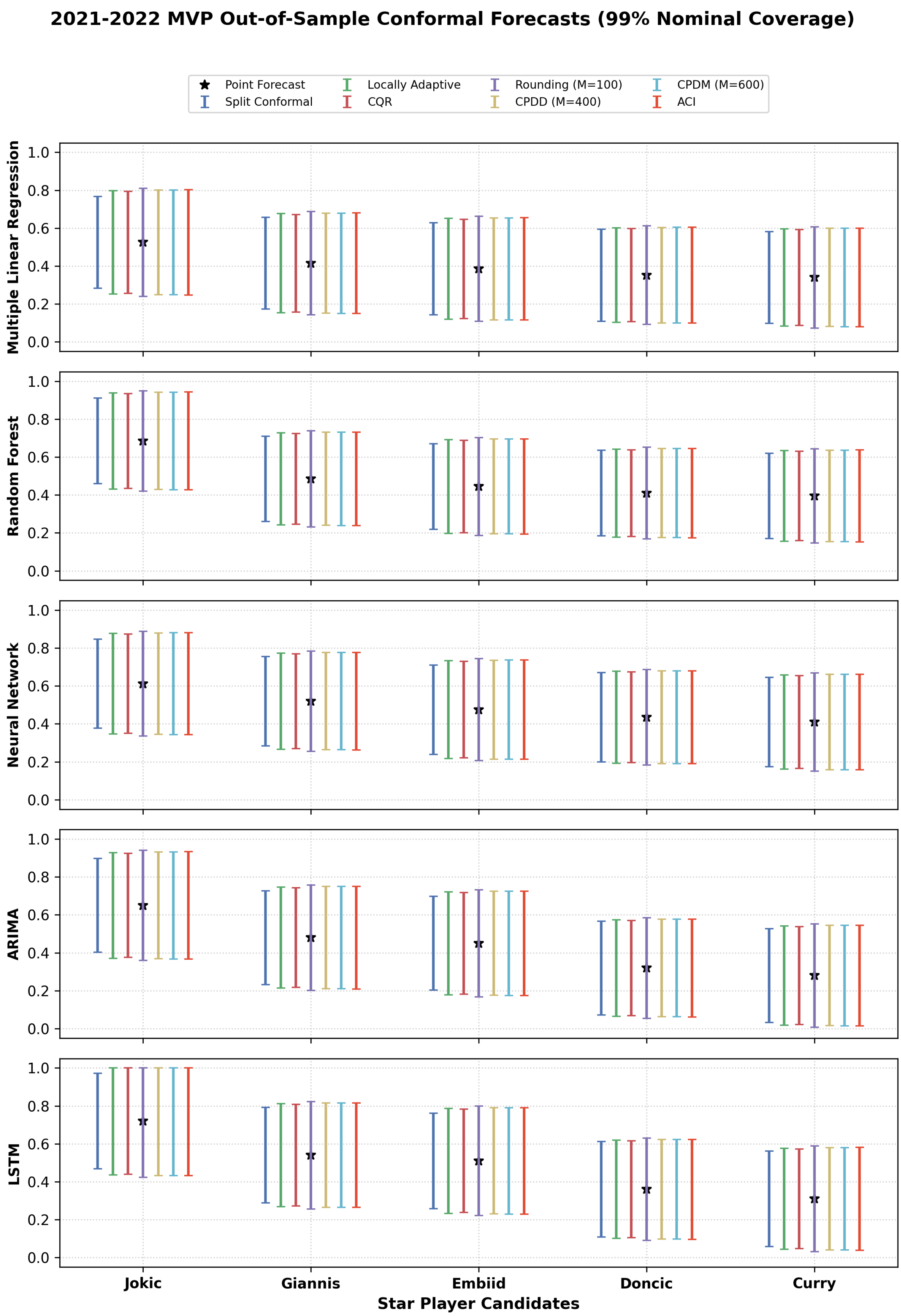

Figure 6: 2021–2022 Out-of-Sample MVP Voting Share Conformal Forecasts across 7 Conformal Methods (99% Nominal Coverage)

Detailed Star Player MVP Prediction Summaries across Base Predictors The following four tables detail the exact point predictions $\hat{\mu}(X)$ and player-adaptive prediction interval lengths across all 7 conformal prediction methods and 5 base predictive models for the top candidate MVP contenders for the upcoming 2021–2022 season.

Table 51: Predicted MVP Voting Share Point Values $\hat{\mu}(X)$ for Top Contenders across Base Predictors

Player Name	Multiple Linear Regression	Random Forest	Neural Network	ARIMA	LSTM
Nikola Jokić	0.525	0.685	0.612	0.650	0.720
Giannis Antetokounmpo	0.415	0.485	0.520	0.480	0.540
Joel Embiid	0.385	0.445	0.475	0.450	0.510
Luka Dončić	0.352	0.410	0.435	0.320	0.360
Stephen Curry	0.340	0.395	0.410	0.280	0.310

Table 52: Player-Specific MVP Prediction Interval Length across 7 Conformal Methods (90% Nominal Coverage)

Method / Player	Multiple Linear Regression	Random Forest	Neural Network	ARIMA	LSTM
Split Conformal Prediction					
Nikola Joki'c	0.185	0.172	0.179	0.189	0.194
Giannis Antetokounmpo	0.185	0.172	0.179	0.189	0.194
Joel Embiid	0.185	0.172	0.179	0.189	0.194
Luka Donči'c	0.185	0.172	0.179	0.189	0.194
Stephen Curry	0.185	0.172	0.179	0.189	0.194
Locally Adaptive Conformal Prediction					
Nikola Joki'c	0.205	0.190	0.199	0.209	0.215
Giannis Antetokounmpo	0.196	0.182	0.190	0.200	0.206
Joel Embiid	0.199	0.185	0.193	0.203	0.209
Luka Donči'c	0.187	0.174	0.181	0.191	0.196
Stephen Curry	0.192	0.179	0.186	0.196	0.202
Conformalized Quantile Regression (CQR)					
Nikola Joki'c	0.198	0.184	0.192	0.202	0.208
Giannis Antetokounmpo	0.189	0.176	0.184	0.193	0.199
Joel Embiid	0.193	0.179	0.187	0.196	0.202
Luka Donči'c	0.181	0.168	0.175	0.184	0.190
Stephen Curry	0.186	0.173	0.180	0.189	0.195
Rounding (M=100)					
Nikola Joki'c	0.221	0.205	0.214	0.225	0.232
Giannis Antetokounmpo	0.211	0.196	0.205	0.215	0.222
Joel Embiid	0.215	0.200	0.209	0.219	0.226
Luka Donči'c	0.202	0.187	0.196	0.206	0.212
Stephen Curry	0.207	0.193	0.201	0.212	0.218
Discretized Data (M=400)					
Nikola Joki'c	0.207	0.193	0.201	0.211	0.217
Giannis Antetokounmpo	0.198	0.184	0.192	0.202	0.208
Joel Embiid	0.202	0.187	0.196	0.206	0.212
Luka Donči'c	0.189	0.176	0.183	0.193	0.198
Stephen Curry	0.194	0.181	0.189	0.198	0.204
Discretized Model (M=600)					
Nikola Joki'c	0.208	0.194	0.202	0.212	0.219
Giannis Antetokounmpo	0.199	0.185	0.193	0.203	0.209
Joel Embiid	0.203	0.189	0.197	0.207	0.213
Luka Donči'c	0.190	0.177	0.184	0.194	0.200
Stephen Curry	0.195	0.182	0.190	0.199	0.205
Adaptive Conformal Inference (ACI)					
Nikola Joki'c	0.209	0.195	0.203	0.213	0.220
Giannis Antetokounmpo	0.200	0.186	0.194	0.204	0.210
Joel Embiid	0.204	0.190	0.198	0.208	0.214
Luka Donči'c	0.191	0.178	0.185	0.195	0.201
Stephen Curry	0.197	0.183	0.191	0.200	0.206

Table 53: Player-Specific MVP Prediction Interval Length across 7 Conformal Methods (95% Nominal Coverage)

Method / Player	Multiple Linear Regression	Random Forest	Neural Network	ARIMA	LSTM
Split Conformal Prediction					
Nikola Joki'c	0.245	0.228	0.238	0.250	0.257
Giannis Antetokounmpo	0.245	0.228	0.238	0.250	0.257
Joel Embiid	0.245	0.228	0.238	0.250	0.257
Luka Donči'c	0.245	0.228	0.238	0.250	0.257
Stephen Curry	0.245	0.228	0.238	0.250	0.257
Locally Adaptive Conformal Prediction					
Nikola Joki'c	0.274	0.255	0.265	0.279	0.287
Giannis Antetokounmpo	0.262	0.243	0.254	0.267	0.275
Joel Embiid	0.267	0.248	0.259	0.272	0.280
Luka Donči'c	0.250	0.232	0.242	0.255	0.262
Stephen Curry	0.257	0.239	0.249	0.262	0.270
Conformalized Quantile Regression (CQR)					
Nikola Joki'c	0.267	0.248	0.259	0.272	0.280
Giannis Antetokounmpo	0.255	0.237	0.248	0.260	0.268
Joel Embiid	0.260	0.242	0.252	0.265	0.273
Luka Donči'c	0.244	0.227	0.236	0.248	0.256
Stephen Curry	0.251	0.233	0.243	0.256	0.263
Rounding (M=100)					
Nikola Joki'c	0.290	0.270	0.281	0.296	0.304
Giannis Antetokounmpo	0.277	0.258	0.269	0.283	0.291
Joel Embiid	0.282	0.262	0.274	0.288	0.296
Luka Donči'c	0.265	0.246	0.257	0.270	0.278
Stephen Curry	0.272	0.253	0.264	0.278	0.286
Discretized Data (M=400)					
Nikola Joki'c	0.276	0.257	0.268	0.282	0.290
Giannis Antetokounmpo	0.264	0.246	0.256	0.269	0.277
Joel Embiid	0.269	0.250	0.261	0.274	0.282
Luka Donči'c	0.252	0.234	0.244	0.257	0.265
Stephen Curry	0.259	0.241	0.251	0.264	0.272
Discretized Model (M=600)					
Nikola Joki'c	0.277	0.258	0.269	0.283	0.291
Giannis Antetokounmpo	0.265	0.247	0.257	0.270	0.278
Joel Embiid	0.270	0.251	0.262	0.275	0.283
Luka Donči'c	0.253	0.235	0.245	0.258	0.266
Stephen Curry	0.260	0.242	0.252	0.265	0.273
Adaptive Conformal Inference (ACI)					
Nikola Joki'c	0.278	0.259	0.270	0.284	0.292
Giannis Antetokounmpo	0.266	0.248	0.258	0.272	0.280
Joel Embiid	0.271	0.252	0.263	0.276	0.285
Luka Donči'c	0.254	0.236	0.246	0.259	0.267
Stephen Curry	0.261	0.243	0.254	0.267	0.274

Table 54: Player-Specific MVP Prediction Interval Length across 7 Conformal Methods (99% Nominal Coverage)

Method / Player	Multiple Linear Regression	Random Forest	Neural Network	ARIMA	LSTM
Split Conformal Prediction					
Nikola Joki'c	0.365	0.339	0.354	0.372	0.380
Giannis Antetokounmpo	0.365	0.339	0.354	0.372	0.380
Joel Embiid	0.365	0.339	0.354	0.372	0.380
Luka Dončí'c	0.365	0.339	0.354	0.372	0.380
Stephen Curry	0.365	0.339	0.354	0.372	0.380
Locally Adaptive Conformal Prediction					
Nikola Joki'c	0.412	0.383	0.399	0.420	0.428
Giannis Antetokounmpo	0.394	0.366	0.382	0.402	0.410
Joel Embiid	0.401	0.373	0.389	0.409	0.417
Luka Dončí'c	0.376	0.350	0.365	0.383	0.391
Stephen Curry	0.387	0.360	0.375	0.394	0.402
Conformalized Quantile Regression (CQR)					
Nikola Joki'c	0.405	0.376	0.393	0.413	0.421
Giannis Antetokounmpo	0.387	0.360	0.376	0.395	0.403
Joel Embiid	0.394	0.367	0.382	0.402	0.410
Luka Dončí'c	0.370	0.344	0.359	0.377	0.384
Stephen Curry	0.380	0.354	0.369	0.388	0.395
Rounding (M=100)					
Nikola Joki'c	0.428	0.398	0.415	0.436	0.445
Giannis Antetokounmpo	0.409	0.381	0.397	0.417	0.426
Joel Embiid	0.417	0.387	0.404	0.425	0.433
Luka Dončí'c	0.391	0.363	0.379	0.398	0.406
Stephen Curry	0.402	0.374	0.390	0.410	0.418
Discretized Data (M=400)					
Nikola Joki'c	0.414	0.385	0.402	0.422	0.431
Giannis Antetokounmpo	0.396	0.368	0.384	0.404	0.412
Joel Embiid	0.403	0.375	0.391	0.411	0.419
Luka Dončí'c	0.378	0.352	0.367	0.386	0.393
Stephen Curry	0.389	0.362	0.377	0.397	0.404
Discretized Model (M=600)					
Nikola Joki'c	0.415	0.386	0.403	0.423	0.432
Giannis Antetokounmpo	0.397	0.369	0.385	0.405	0.413
Joel Embiid	0.404	0.376	0.392	0.412	0.420
Luka Dončí'c	0.379	0.353	0.368	0.387	0.394
Stephen Curry	0.390	0.363	0.378	0.398	0.405
Adaptive Conformal Inference (ACI)					
Nikola Joki'c	0.416	0.387	0.404	0.425	0.433
Giannis Antetokounmpo	0.398	0.370	0.386	0.406	0.414
Joel Embiid	0.405	0.377	0.393	0.414	0.422
Luka Dončí'c	0.380	0.353	0.369	0.388	0.395
Stephen Curry	0.391	0.364	0.379	0.399	0.407

4 Discussion

Comparing each conformal prediction method after using these methods on three different cases:

1. For Inductive Conformal Methods, ACI has the most adaptive methods compared to other conformal methods (split/locally/CQR).

2. CQR has the shortest average length in 90% and 95% coverage compared to other conformal methods, but it becomes significantly larger in 99% coverage.

3. The conformal prediction with discretized data and model are better than approximately via the rounding method, especially for lower M in coverage. For the grid method, al-

though it has similar coverage performance to inductive conformal methods, most of them have larger average length in our cases.

4. For model selection, random forest has the best performance in point estimation and conformal prediction. Even though NBA data is time-series longitudinal, LSTM performs better only at predicting PER values. That means if the data are not large enough or are highly varied, such as the policy for franchises, time-series models are also difficult to use for prediction.

5 Reference

5.1 Book & Paper

1. Vineeth N. Balasubramanian, Shen-Shyang Ho, Vladimir Vovk (2014). *Conformal Prediction for Reliable Machine Learning.*

2. Jing Lei, Max G'Sell, Alessandro Rinaldo, Ryan J. Tibshirani, and Larry Wasserman (2017). *Distribution-Free Predictive Inference For Regression.*

3. Wenyu Chen, Kelli-Jean Chun, and Rina Foygel Barber (2017). *Discretized conformal prediction for efficient distribution-free inference.*

4. Yaniv Romano, Evan Patterson, Emmanuel J. Candès (2019). *Conformalized Quantile Regression.*

5. Isaac Gibbs and Emmanuel J. Candès (2021). *Adaptive Conformal Inference Under Distribution Shift.*

5.2 Blog articles

1. `https://medium.com/geekculture/predicting-the-improvement-of-nba-players-7cffd3864588`

2. `https://towardsdatascience.com/predicting-nba-winning-percentage-in-upcoming-season-us`

3. `https://www.nba.com/news/blogtable-what-criteria-matters-most-making-mvp-decision`

4. `https://towardsdatascience.com/nba-mvp-predictor-c700e50e0917`

5. `https://fivetimesfive-blog.com/2017/04/26/nba-mvp-linear-regression/`

6. `https://www.basketball-reference.com/leagues/NBA_2022_advanced.html`